

Phenix Experimental Phasing

5 documentation pages

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Experimental Phasing

xray-experimental-phasing.html

Why

One of the biggest challenges in crystallography is the phase problem. We measure amplitudes for structure factors, but cannot directly measure the phases. Unfortunately, these phases are necessary to generate the electron density of the molecules in the crystal.

Phases can be determined by experimental phasing, which relies on a few atoms in the macromolecule with known special properties, such as a large number of electrons and/or anomalous scattering. These properties are exploited to determine the location of the special atoms (substructure), and then knowledge of the substructure in one or more crystals is used to deduce phase information for the entire macromolecule. Thus, experimental phasing solves the phase problem by using the differences in amplitudes from different diffraction experiments to determine the structure factor phases. In modern crystallography, it is most common to use single or multiple anomalous diffraction experiments (SAD/MAD).

How

In Phenix, the primary program for performing experimental phasing is `phenix.autosol`. This is a comprehensive, automatic tool that performs experimental phasing with the MAD, MIR, SIR, or SAD methods. The program locates the substructure, estimates phases, performs density modification, identifies noncrystallographic symmetry, and builds and refines a preliminary model.

To run `phenix.autosol`, you need to provide diffraction data, typically one or more anomalous data sets, or a native data set plus one or more derivative data sets (if you are performing SIR/MIR phasing). You also need to provide the sequence of your macromolecule, and information about the anomalous scatterers and/or heavy atoms. The `phenix.autosol` results include a model file containing the anomalous scatterers, an MTZ file containing the experimentally determined phases, the optimized phases from density modification, and, if the maps are of sufficient quality, an atomic model.

How to use the `phenix.autosol` GUI: [Click here](#)

Phenix reference manual for `phenix.autosol`

Common issues

- Anomalous substructure not determined: There can be several reasons for this. Likely candidates are a lack of anomalous signal or an incorrect space group assignment. Use `phenix.xtriage` or `phenix.anomalous_signal` to estimate the resolution of the data for anomalous substructure determination. If this resolution is very low, it might not be possible to locate the substructure (note, the experimental sigmas are used in this calculation so it is important that these are well estimated). If there is uncertainty about the space group, it might be necessary to try the alternatives.
- An interpretable model isn't built: The most likely reason is that the electron density map is of insufficient quality to be automatically interpreted.
- Frequently asked questions about experimental phasing

Related programs

- `phenix.plan_sad_experiment`: This tool allows you to plan your SAD data collection strategy by estimating the anomalous signal and the probability of solving your structure with SAD based on the anomalously-scattering atoms, wavelength, size of your molecule, and overall $I/\sigma(I)$ for your dataset.

- `phenix.scale_and_merge`: This tool scales any number of datasets together to obtain a merged, scaled anomalous dataset. It also produces two half-datasets of scaled anomalous data that can be used by `phenix.anomalous_signal` to accurately evaluate the anomalous signal in your data.
- `phenix.anomalous_signal`: This tool is like `phenix.plan_sad_experiment` except that it uses your measured data to give much more accurate estimates. Specifically, it uses your scaled data, two half-dataset files of your scaled data, the name of the anomalously-scattering atom, and the number of sites in the substructure in order to estimate the anomalous signal in your data and the probability of solving your structure by SAD. You can create the necessary scaled half-datasets using `phenix.scale_and_merge`.
- `phenix.hyss`: This hybrid substructure search program is used by `phenix.autosol` to locate the anomalous substructure or heavy atoms. It is possible to run this program separately (for anomalous data only) if you want to only find the substructure. In general, it is recommended to use `phenix.autosol` for this as it makes optimal use of `phenix.hyss` to locate sites.
- Phaser EP: Phaser can directly perform SAD phasing, given an initial anomalous substructure. Phases can be calculated for the substructure's given hand (enantiomer) or both hands. Phaser attempts to complete the substructure automatically. Note, Phaser can also combine information from a molecular replacement solution in the phase calculation process (in MR-SAD mode).
- `phenix.emma`: This Euclidian model matching program can compare different substructure solutions by superimposing sites taking into account crystallographic symmetry, possible origin shifts, and polar axes. This can be useful if comparing solutions from different programs.

Analyzing the anomalous signal in a SAD dataset with anomalous_signal

anomalous_signal.html

Author(s)

- anomalous_signal: Tom Terwilliger

Purpose

anomalous_signal is a tool for estimating the anomalous signal in a SAD experiment and for predicting whether this signal is sufficient to solve the structure. This tool normally is used in combination with plan_sad_experiment (planning the experiment) and scale_and_merge (scaling the unmerged data and also obtaining two half-datasets).

Usage

How anomalous_signal works:

- You supply anomalous_signal with scaled anomalous data, two half-dataset files with scaled anomalous data, and the number of sites or a sequence file and name of the anomalously-scattering atom.
- anomalous_signal will calculate the anomalous signal in your dataset from (1) the half-dataset anomalous correlation, (2) the skew of the anomalous difference Patterson map, and (3) the estimated measurement error in your data.
- anomalous_signal will then estimate the probability that you can solve this dataset using likelihood-based HySS (standard run) and will estimate the figure of merit of phasing that you should obtain.
- The way that plan_sad_experiment and anomalous_signal estimate the probability that you can solve this dataset is to compare the anomalous signal in this dataset with the anomalous signal in other datasets at the same resolution. Then the fraction of similar datasets that can be solved by HySS is used as the probability that your dataset will also be solved.

Output from anomalous_signal

anomalous_signal provides a summary of the skew, measurement error, and half-dataset correlations in your dataset and the estimated anomalous signal, probability of solving the structure and estimated figure of merit of phasing. Here is an example:

```
-----Summary of signal in this dataset -----  
  
Shell  
  
CCano Nrefl Nrefl  
Resolution Esqr I/sigI half anom half  
48.2- 6.0 0.09 33.09 0.85 2090 2032  
6.0- 5.5 0.34 23.31 0.53 667 659  
5.5- 5.0 0.34 22.84 0.49 956 947  
5.0- 4.5 0.28 29.23 0.38 1468 1447  
4.5- 4.0 0.42 24.00 0.16 2305 2276  
4.0- 3.5 0.57 14.02 0.05 3832 3751
```

```
3.5- 3.0 0.84 5.91 0.01 6869 6494
3.0- 2.9 1.52 2.39 0.00 2035 1734
```

Cumulative

```
-----Data quality----- Best guess of expected
results of finding sites
----- and phasing-----
```

```
CCano Nrefl P(Substr)
Resolution Skew Esqr half anom CC* Signal +/- (%) FOM* +/-
48.2- 6.0 0.03 0.08 0.85 2090 0.71 6.3 0.4 23 0.3 0.1
48.2- 5.5 0.02 0.10 0.82 2757 0.76 7.8 0.3 40 0.3 0.1
48.2- 5.0 0.01 0.12 0.78 3713 0.70 8.3 0.6 47 0.3 0.1
48.2- 4.5 0.03 0.14 0.72 5181 0.68 9.5 0.7 61 0.3 0.1
48.2- 4.0 0.03 0.17 0.62 7486 0.72 12.1 0.6 76 0.4 0.0
48.2- 3.5 0.02 0.25 0.43 11318 0.66 13.6 1.4 83 0.4 0.0
48.2- 3.0 0.00 0.47 0.24 18187 0.48 12.7 2.0 79 0.3 0.1
48.2- 2.9 0.00 0.59 0.22 20222 0.42 11.8 1.7 74 0.3 0.1
```

Notes:

Skew is skew of origin-removed anomalous difference Patterson
Esqr is the squared ratio of sigmas to differences: $\frac{\text{SigAno}^2}{\text{Dano}^2}$;
I/sigI is the mean value of (Ihkl/sigIhkl)
CCano half is the correlation between half-dataset anomalous differences
Nrefl half is the number of anomalous differences in both half-datasets
Nrefl anom is the number of anomalous differences in the entire dataset
CC* is estimate of anomalous correlation to ideal data
Signal +/- sigma is estimate of anomalous signal (peak height at coordinates of
anomalously-scattering atoms in difference Fourier phased with model
phases.)

Note that the anomalous signal increases with the square root of the
number of reflections and decreases with the square root of the number
of sites.

P(Substr) is estimate of probability that the sub-structure can be found
with LLG-based HySS and this data.
FOM* +/- sigma is estimate of phasing FOM if sub-structure is solved

Possible Problems

anomalous_signal assumes that your crystal is similar to others that have been solved. If your crystal has serious decay or other factors that reduce signal to noise, the estimates provided by plan_sad_experiment may be too optimistic.

Literature

List of all available keywords

input_filesdata = None Anomalous data (I+ and I- or F+ and F-). You must supply unmerged or merged data plus two half-dataset datafiles (usually these half-dataset files are obtained with phenix.scale_and_merge)
Any standard format is fine.data_labels = None Optional label specifying which columns of anomalous data to use. Not necessary if your input file has only one set of anomalous data. .short_caption = Data
labelshalf_dataset_a = None Data for half-dataset A. Normally required if the main dataset is merged. This half-dataset normally comes from running phenix.scale_and_merge on unmerged data to produce half_datasets A and B. Any standard format is fine.half_dataset_a_labels = None Optional label specifying which columns of anomalous data to use. Not necessary if your input file has only one set of anomalous data. .short_caption = Half-dataset A labelshalf_dataset_b = None Data for half-dataset B. Normally required if the

main dataset is merged. This half-dataset normally comes from running `phenix.scale_and_merge` on unmerged data to produce half_datasets A and B. Any standard format is fine. `half_dataset_b_labels = None` Optional label specifying which columns of anomalous data to use. Not necessary if your input file has only one set of anomalous data. `.short_caption = Half-dataset B labels` `output_filescaling_log = scale.log` Output log file for scaling (if applied) `output_file = scaled_data.mtz` Output data file name for merged scaled data. Output files are produced if half-dataset files are not supplied and unmerged input data is scaled and divided into half datasets half-dataset files are not supplied `output_half_dataset_a = half_dataset_a.mtz` Output data file name for half dataset A. Output files are produced if half-dataset files are not supplied and unmerged input data is scaled and divided into half datasets half-dataset files are not supplied `output_half_dataset_b = half_dataset_b.mtz` Output data file name for half dataset B. Output files are produced if half-dataset files are not supplied and unmerged input data is scaled and divided into half datasets half-dataset files are not supplied `directoriestemp_dir = None` Optional temporary work directory. If specified, it will be used for temporary files. If `clean_up=True` and it did not exist before, it will be removed at the end of the run. If it did exist, it will be left. `output_dir = ""` Output directory where files are to be written `gui_output_dir = None` GUI use only - does not apply to command line version `crystal_inforesolution = None` high-resolution limit `low_resolution = None` low-resolution limit `space_group = None` Space group (normally read from the data file) `unit_cell = None` Unit Cell (normally read from the data file) `seq_file = None` Optional sequence file (1-letter code). Separate chains with a blank line or line starting with > If not supplied, you will need to supply the number of sites with `sites=xxx`. `chain_type = *PROTEIN RNA DNA` Chain type (PROTEIN RNA DNA). This is used to estimate the number of atoms from the number of residues. If not supplied, you will need to supply the number of sites with `sites=xxx`. `atom_type = None` Name of anomalously-scattering atom. If not supplied, you will need to supply the number of sites with `sites=xxx`. `sites = None` The number of anomalously-scattering atoms in the asymmetric unit. `b_value_anomalous = None` Estimated Wilson B-value for the anomalously-scattering atoms. Normally leave as None and it will be estimated from Wilson `Banalysis` `half_dataset_cc = True` Analyze half-dataset anomalous correlation. Requires unmerged data file or `half_dataset_a` and `half_dataset_b` files, each with half of data from an unmerged dataset. The half-dataset files normally come from running `phenix.scale_and_merge` on unmerged data to produce half_datasets A and B. `skip_scaling = False` Skip scaling and use `merging_statistics` to get half-dataset `CC.resolution_dependent_predictors = True` Use Bayesian predictors that are resolution-dependent `planned_number_of_data...`

- `input_filesdata = None` Anomalous data (I+ and I- or F+ and F-). You must supply unmerged or merged data plus two half-dataset datafiles (usually these half-dataset files are obtained with `phenix.scale_and_merge`) Any standard format is fine. `data_labels = None` Optional label specifying which columns of anomalous data to use. Not necessary if your input file has only one set of anomalous data. `.short_caption = Data label` `half_dataset_a = None` Data for half-dataset A. Normally required if the main dataset is merged. This half-dataset normally comes from running `phenix.scale_and_merge` on unmerged data to produce half_datasets A and B. Any standard format is fine. `half_dataset_a_labels = None` Optional label specifying which columns of anomalous data to use. Not necessary if your input file has only one set of anomalous data. `.short_caption = Half-dataset A label` `half_dataset_b = None` Data for half-dataset B. Normally required if the main dataset is merged. This half-dataset normally comes from running `phenix.scale_and_merge` on unmerged data to produce half_datasets A and B. Any standard format is fine. `half_dataset_b_labels = None` Optional label specifying which columns of anomalous data to use. Not necessary if your input file has only one set of anomalous data. `.short_caption = Half-dataset B labels`
- `data = None` Anomalous data (I+ and I- or F+ and F-). You must supply unmerged or merged data plus two half-dataset datafiles (usually these half-dataset files are obtained with `phenix.scale_and_merge`) Any standard format is fine.

- `data_labels = None` Optional label specifying which columns of anomalous data to use. Not necessary if your input file has only one set of anomalous data. `.short_caption = Data labels`
- `half_dataset_a = None` Data for half-dataset A. Normally required if the main dataset is merged. This half-dataset normally comes from running `phenix.scale_and_merge` on unmerged data to produce half_datasets A and B. Any standard format is fine.
- `half_dataset_a_labels = None` Optional label specifying which columns of anomalous data to use. Not necessary if your input file has only one set of anomalous data. `.short_caption = Half-dataset A labels`
- `half_dataset_b = None` Data for half-dataset B. Normally required if the main dataset is merged. This half-dataset normally comes from running `phenix.scale_and_merge` on unmerged data to produce half_datasets A and B. Any standard format is fine.
- `half_dataset_b_labels = None` Optional label specifying which columns of anomalous data to use. Not necessary if your input file has only one set of anomalous data. `.short_caption = Half-dataset B labels`
- `output_filescaling_log = scale.log` Output log file for scaling (if applied)
`output_file = scaled_data.mtz` Output data file name for merged scaled data. Output files are produced if half-dataset files are not supplied and unmerged input data is scaled and divided into half datasets
half-dataset files are not supplied
`output_half_dataset_a = half_dataset_a.mtz` Output data file name for half dataset A. Output files are produced if half-dataset files are not supplied and unmerged input data is scaled and divided into half datasets
half-dataset files are not supplied
`output_half_dataset_b = half_dataset_b.mtz` Output data file name for half dataset B. Output files are produced if half-dataset files are not supplied and unmerged input data is scaled and divided into half datasets
half-dataset files are not supplied
- `scaling_log = scale.log` Output log file for scaling (if applied)
- `output_file = scaled_data.mtz` Output data file name for merged scaled data. Output files are produced if half-dataset files are not supplied and unmerged input data is scaled and divided into half datasets
half-dataset files are not supplied
- `output_half_dataset_a = half_dataset_a.mtz` Output data file name for half dataset A. Output files are produced if half-dataset files are not supplied and unmerged input data is scaled and divided into half datasets
half-dataset files are not supplied
- `output_half_dataset_b = half_dataset_b.mtz` Output data file name for half dataset B. Output files are produced if half-dataset files are not supplied and unmerged input data is scaled and divided into half datasets
half-dataset files are not supplied
- `directoriestemp_dir = None` Optional temporary work directory. If specified, it will be used for temporary files. If `clean_up=True` and it did not exist before, it will be removed at the end of the run. If it did exist, it will be left.
`output_dir = ""` Output directory where files are to be written
`gui_output_dir = None` GUI use only - does not apply to command line version
- `temp_dir = None` Optional temporary work directory. If specified, it will be used for temporary files. If `clean_up=True` and it did not exist before, it will be removed at the end of the run. If it did exist, it will be left.
- `output_dir = ""` Output directory where files are to be written
- `gui_output_dir = None` GUI use only - does not apply to command line version
- `crystal_inforesolution = None` high-resolution limit
`low_resolution = None` low-resolution limit
`space_group = None` Space group (normally read from the data file)
`unit_cell = None` Unit Cell (normally read from the data file)
`seq_file = None` Optional sequence file (1-letter code). Separate chains with a blank line or line starting with > If not supplied, you will need to supply the number of sites with `sites=xxx`.
`chain_type = *PROTEIN RNA DNA` Chain type (PROTEIN RNA DNA). This is used to estimate

the number of atoms from the number of residues. If not supplied, you will need to supply the number of sites with sites=xxx. atom_type = None Name of anomalously-scattering atom. If not supplied, you will need to supply the number of sites with sites=xxx. sites = None The number of anomalously-scattering atoms in the asymmetric unit. b_value_anomalous = None Estimated Wilson B-value for the anomalously-scattering atoms. Normally leave as None and it will be estimated from Wilson B

- resolution = None high-resolution limit
- low_resolution = None low-resolution limit
- space_group = None Space group (normally read from the data file)
- unit_cell = None Unit Cell (normally read from the data file)
- seq_file = None Optional sequence file (1-letter code). Separate chains with a blank line or line starting with > If not supplied, you will need to supply the number of sites with sites=xxx.
- chain_type = *PROTEIN RNA DNA Chain type (PROTEIN RNA DNA). This is used to estimate the number of atoms from the number of residues. If not supplied, you will need to supply the number of sites with sites=xxx.
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- sites = None The number of anomalously-scattering atoms in the asymmetric unit.
- b_value_anomalous = None Estimated Wilson B-value for the anomalously-scattering atoms. Normally leave as None and it will be estimated from Wilson B
- analysishalf_dataset_cc = True Analyze half-dataset anomalous correlation. Requires unmerged data file or half_dataset_a and half_dataset_b files, each with half of data from an unmerged dataset. The half-dataset files normally come from running phenix.scale_and_merge on unmerged data to produce half_datasets A and B. skip_scaling = False Skip scaling and use merging_statistics to get half-dataset CC. resolution_dependent_predictors = True Use Bayesian predictors that are resolution-dependent. planned_number_of_datasets = None You can predict the useful anomalous correlation, anomalous signal, and figure of merit of phasing that you would expect if you collected a total of N datasets like this one. See also ideal_cc_anom which is the estimated maximum anomalous correlation achievable with these crystals. ideal_cc_anom = 0.75 Estimate of useful anomalous correlation for perfectly measured data. Used as maximum anomalous correlation achievable when planned_number_of_datasets is set.
- half_dataset_cc = True Analyze half-dataset anomalous correlation. Requires unmerged data file or half_dataset_a and half_dataset_b files, each with half of data from an unmerged dataset. The half-dataset files normally come from running phenix.scale_and_merge on unmerged data to produce half_datasets A and B.
- skip_scaling = False Skip scaling and use merging_statistics to get half-dataset CC.
- resolution_dependent_predictors = True Use Bayesian predictors that are resolution-dependent
- planned_number_of_datasets = None You can predict the useful anomalous correlation, anomalous signal, and figure of merit of phasing that you would expect if you collected a total of N datasets like this one. See also ideal_cc_anom which is the estimated maximum anomalous correlation achievable with these crystals.
- ideal_cc_anom = 0.75 Estimate of useful anomalous correlation for perfectly measured data. Used as maximum anomalous correlation achievable when planned_number_of_datasets is set.

- `controlverbose = False` Verbose output
`clean_up = True` Clean up and remove temp directory after finishing
`job_title = None` Job title in PHENIX GUI, not used on command line
- `verbose = False` Verbose output
- `clean_up = True` Clean up and remove temp directory after finishing
- `job_title = None` Job title in PHENIX GUI, not used on command line

Automated Structure Solution using AutoSol

[autosol.html](#)

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Authors

AutoSol Wizard: Tom Terwilliger
 Phaser: Gabor Bunkoczi, Airlie McCoy, Randy Read
 Hyss: Ralf Grosse-Kunstleve, Tom Terwilliger, Gabor Bunkoczi, Randy Read
 Xtriage: Peter Zwart
 PHENIX GUI: Nathaniel Echols

Purpose

The AutoSol Wizard uses HYSS, SOLVE, Phaser, RESOLVE, xtriage and phenix.refine to solve a macromolecular crystallographic structure and generate experimental phases with the MAD, MIR, SIR, or SAD methods. The Wizard begins with datafiles (.sca, .hkl, etc) containing amplitudes (or intensities) of structure factors, identifies heavy-atom sites, calculates phases, carries out density modification and NCS identification, and builds and refines a preliminary model.

Usage

The AutoSol Wizard can be run from the PHENIX GUI, from the command-line, and from parameters files. All three versions are identical except in the way that they take commands from the user. See Using the PHENIX Wizards for details of how to run a Wizard. The command-line version will be described here, except for MIR and multiple datasets, which can only be run with the GUI or with a parameters file. Nearly all the parameters described here as command-line keyword=value pairs are parameters that can be set in the GUI, so nearly everything described here can be done in the GUI. The GUI is documented separately.

How the AutoSol Wizard works

The basic steps that the AutoSol Wizard carries out are described below. They are: Setting up inputs, Analyzing and scaling the data, Finding heavy-atom (anomalously-scattering atom) sites, Scoring of heavy-atom solutions, Phasing, Density modification (including NCS averaging), and Preliminary model-building and refinement. The data for structure solution are grouped into Datasets and solutions are

stored in Solution objects.

Setting up inputs

The AutoSol Wizard expects the following basic information:

- a datafile name (w1.sca or data=w1.sca)
- a sequence file (seq.dat or seq_file=seq.dat)
- how many sites to look for (2 or sites=2)
- what the anomalously-scattering atom is (Se or atom_type=Se)
- The wavelength has to be given for SAD phasing; for other phasing methods it is helpful to provide the wavelength as well as f_prime and f_double_prime for each wavelength or derivative that you have

You can also specify many other parameters, including resolution, number of sites, whether to search in a thorough or quick fashion, how thoroughly to build a model, etc. If you have a heavy-atom solution from a previous run or another approach, you can read it in directly as well.

Your parameters can be specified on the command-line, using a GUI, or by editing a parameters file (examples below).

Datafile formats in AutoSol

AutoSol will accept the following formats of data:

- scalepack unmerged original index: I, SIGI
- scalepack premerged: I+, SIGI+, I-, SIGI-
- mtz unmerged: I, SIGI, M_ISYM
- mtz premerged: I(+), SIGI(+), I(-), SIGI(-)
- d*trek
- CNS

The data from any of these formats will be converted to amplitudes (F+ , sigF+, and F-, sigF-) internally.

If you have multiple data files for a single dataset, or if you have original index data with multiple observations of each reflection, you might want to run phenix.scale_and_merge on your data to scale and merge the datasets before you run AutoSol. This is particularly useful for weak SAD data (such as S-SAD datasets).

Additionally, within AutoSol if you supply all scalepack unmerged original index files or all mtz unmerged files then the information in these files can be used to improve scaling. In most cases this is not necessary however. If all the files are scalepack unmerged original index or all the files are mtz unmerged and no anisotropy correction is applied, then SOLVE local scaling will be applied to the data prior to merging and averaging equivalent reflections. In all other cases equivalent reflections will be averaged prior to scaling, so that the scaling may not be as effective at removing systematic errors due to absorption or other effects.

Sequence file format

The sequence file for autosol (and autobuild) is the one-letter code sequence of each chain in the structure to be solved, where chains are separated by blank lines or lines starting with a > character. Here is a simple example with two different chains:

```
> sequence of chain A follows. This line not required
LVLKWMSTKYVEAGELKEGSYVVIDGEPCRVVEIEKSKTGKHGSAKARIVAVGVFDGGKRTLSPVDAQVEVPPIIEKFT
```

> sequence of chain B follows. This line could be blank to indicate new chain
AQILSVSGDVIQLMDMRDYKTIEVPMKYVEEEAKGRLAPGAEEVWQILDYKIIRVKG

Usually the chain type (RNA, DNA, PROTEIN) is guessed from the sequence file. You can also specify it directly with a command such as `chain_type=PROTEIN`.

If there are multiple copies of a chain (ncs) then you can put in a single copy and it will be used for all of them. If there are multiple copies of a set of chains (A,A,A, B,B,B would be 3 copies of chains A and B) then you can put in the unique set (A and B). If there are different numbers of copies of different chains, then put in the unique set, and be sure to set the `solvent_fraction`.

If you have more than one type of chain (RNA, DNA, PROTEIN) then just put in the sequence for the one that is the biggest, and be sure to specify `solvent_fraction=xxx` so that the correct solvent fraction is used.

NOTE 1: Characters such as numbers and non-printing characters in the sequence file are ignored.

NOTE 2: Be sure that your sequence file does not have any blank lines in the middle of your sequence, as these are interpreted as the beginning of another chain.

Datasets and Solutions in AutoSol

AutoSol breaks down the data for a structure solution into datasets, where a dataset is a set of data that corresponds to a single set of heavy-atom sites. An entire MAD dataset is a single dataset. An MIR structure solution consists of several datasets (one for each native-derivative combination). A MAD + SIR structure has one dataset for the MAD data and a second dataset for the SIR data. The heavy-atom sites for each dataset are found separately (but using difference Fouriers from any previously-solved datasets to help). In the phasing step all the information from all datasets is merged into a single set of phases.

The AutoSol wizard uses a "Solution" object to keep track of heavy-atom solutions and the phased datasets that go with them. There are two types of Solutions: those which consist of a single dataset (Primary Solutions) and those that are combinations of datasets (Composite Solutions). "Primary" Solutions have information on the datafiles that were part of the dataset and on the heavy-atom sites for this dataset. Composite Solutions are simply sets of Primary Solutions, with associated origin shifts. The hand of the heavy-atom or anomalously-scattering atom substructure is part of a Solution, so if you have two datasets, each with two Solutions related by inversion, then AutoSol would normally construct four different Composite Solutions from these and score each one as described below.

Analyzing and scaling the data

The AutoSol Wizard analyzes input datasets with `phenix.xtriage` to identify twinning and other conditions that may require special care. The data is scaled with `SOLVE`. For MAD data, FA values are calculated as well.

Note on anisotropy corrections:

The AutoSol wizard will apply an anisotropy correction and B-factor sharpening to all the raw experimental data by default (controlled by the keyword `remove_aniso=True`). The target overall Wilson B factor can be set with the keyword `b_iso`, as in `b_iso=25`. By default the target Wilson B will be 10 times the resolution of the data (e.g., if the resolution is 3 Å then `b_iso=30`), or the actual Wilson B of the data, whichever is lower.

If an anisotropy correction is applied then the entire AutoSol run will be carried out with anisotropy-corrected and sharpened data. At the very end of the run the final model will be re-refined against the uncorrected refinement data and this re-refined model and the uncorrected refinement data (with freeR flags) will be written out. For the top solution this will be `overall_best.pdb` and `overall_best_refine_data.mtz`; for all other solutions the files will be listed at the end of the log file.

Finding heavy-atom (anomalously-scattering atom) sites

The AutoSol Wizard uses HYSS to find heavy-atom sites. The result of this step is a list of possible heavy-atom solutions for a dataset. For SIR or SAD data, the isomorphous or anomalous differences, respectively are used as input to HYSS. For MAD data, the anomalous differences at each wavelength, and the FA estimates of complete heavy-atom structure factors from SOLVE are each used as separate inputs to HYSS. Each heavy-atom substructure obtained from HYSS corresponds to a potential solution. In space groups where the heavy-atom structure can be either hand, a pair of enantiomorphic solutions is saved for each run of HYSS.

For SAD and MAD data (except for FA estimates) the Phaser LLG completion approach is used to find the heavy-atom sites. This can be quite a bit more powerful than direct methods completion.

Running AutoSol separately in related space groups

AutoSol will check for the opposite hand of the heavy-atom solution, and at the same time it will check for the opposite hand of your space group (It will invert the heavy-atom solution from HYSS and invert the hand of the space group at the same time). Therefore you do not need to run AutoSol twice for space groups that are chiral (for example P41). The corresponding inverse space groups will be checked automatically (P43). If there are possibilities for your space group other than the inverse hand of the space group, then you should test them all, one at a time. For example if you were not able to measure 00l reflections in a hexagonal space group, your space group might be P6, P61, P62, P63, P64 or P65. In this case you would have to run it in P6, P61 P62 and P63 (and then P65 and P64 will be done automatically as the inverses of P61 and P62). Normally only one of these will give a plausible solution.

Scoring of heavy-atom solutions

Potential heavy-atom solutions are scored based on a set of criteria (SKEW, CORR_RMS, CC_DENMOD, RFACTOR, NCS_OVERLAP, TRUNCATE, REGIONS, CONTRAST, FOM, FLATNESS, described below), using either a Bayesian estimate or a Z-score system to put all the scores on a common scale and to combine them into a single overall score. The overall scoring method chosen (BAYES-CC or Z-SCORE) is determined by the value of the keyword `overall_score_method`. The default is BAYES-CC. Note that for all scoring methods, the map that is being evaluated, and the estimates of map-perfect-model correlation, refer to the experimental electron density map, not the density-modified map.

Bayesian CC scores (BAYES-CC). Bayesian estimates of the quality of experimental electron density maps are obtained using data from a set of previously-solved datasets. The standard scoring criteria were evaluated for 1905 potential solutions in a set of 246 MAD, SAD, and MIR datasets. As each dataset had previously been solved, the correlation between the refined model and each experimental map (CC_PERFECT) could be calculated for each solution (after offsetting the maps to account for origin differences).

Histograms have been tabulated of the number of instances that a scoring criterion (e.g., SKEW) had various possible values, as a function of the CC_PERFECT of the corresponding experimental map to the refined model. These histograms yield the relative probability of measuring a particular value of that scoring criterion (SKEW), given the value of CC_PERFECT. Using Bayes' rule, these probabilities can be used to estimate the relative probabilities of values of CC_PERFECT given the value of each scoring criterion for a particular electron density map. The mean estimate (BAYES-CC) is reported (multiplied x 100), with a +/-2SD estimate of the uncertainty in this estimate of CC_PERFECT. The BAYES-CC values are estimated independently for each scoring criterion used, and also from all those selected with the keyword `score_type_list` and not selected with the keyword `skip_score_list`.

Z-scores (Z-SCORE). The Z-score for one criterion for a particular solution is given by,

$Z = (\text{Score} - \text{mean_random_solution_score}) / (\text{SD_of_random_solution_scores})$ where Score is the score for this solution, mean_random_solution_score is the mean score for a solution with randomized phases, and SD_of_random_solution_scores is the standard deviation of the scores of solutions with randomized phases. To create a total score based on Z-scores, the Z-scores for each criterion are simply summed.

The principal scoring criteria are:

- The skew (SKEW; third moment or normalized $\langle \rho^3 \rangle$) of the density in an electron density map is a good measure of its quality, because a random map has a skew of zero (density histograms look like a Gaussian), while a good map has a very positive skew (density histograms very strong near zero, but many points with very high density). This criterion is used in scoring by default. Correlation of local rms density (CORR_RMS). The presence of contiguous flat solvent regions in a map was detected using the correlation coefficient of the smoothed squared electron density calculated as described above, with the same quantity calculated using half the value of the smoothing radius, yielding the correlation of rms density, r2RMS. In this way the local value of the rms density within a small local region (typically within a radius of 3 Å) is compared with the local rms density in a larger local region (typically within a radius of 6 Å). If there were a large, contiguous solvent region and another large contiguous region containing the macromolecule, the local rms density in the small region would be expected to be highly correlated with the rms density in the larger region. On the other hand, if the solvent region were broken up into many small flat regions, then this correlation would be expected to be smaller.
- Correlation of map-phased electron density map with experimentally- phased map (CC_DENMOD). The statistical density modification in RESOLVE allows the calculation of map-based phases that are (mostly) independent of the experimental phases. The phase information in statistical density modification comes from two sources: your experimental phases and maximization of the agreement of the map with expectations (such as a flat solvent region). Normally the phase probabilities from these two sources are merged together, yielding your density-modified phases. This score is calculated based on the correlation of the phase information from these two sources before combining them, and is a good indication of the quality of the experimental phases. This criterion is used in scoring by default.
- The R-factor for density modification (RFACTOR). Statistical density modification provides an estimate of structure factors that is (mostly) independent of the measured structure factors, so the R-factor between FC and Fobs is a good measure of the quality of experimental phases. This criterion is used in scoring by default. Note that this is different from a refinement R value.
- Non-crystallographic symmetry (NCS_OVERLAP). The presence of NCS in a map is a nearly-positive indication that the map is good, or has some correct features. The AutoSol Wizard uses symmetry in heavy-atom sites to suggest NCS, and RESOLVE identifies the actual correlation of NCS-related density for the NCS overlap score. This score is used by default if NCS is present in the Z-score method of scoring.
- Figure of merit (FOM). The figure of merit of phasing is a good indicator of the internal consistency of a solution. This score is not normalized by the SD of randomized phase sets (as that has no meaning; rather a standard SD=0.05 is used). This score is used by default if NCS is present in the Z-score method of scoring and in the Bayesian CC estimate method.
- Map correlation after truncation (TRUNCATION). Dummy atoms (the same number as estimated non-hydrogen atoms in the structure) are placed in positions of high density of the map, and a new map is calculated based on these atomic positions. The correlation of these maps is calculated after adjusting an overall B-value for the dummy atoms to maximize the correlation. A good map will show a high correlation of these maps. This score is by default not used.

- Number of contiguous regions per 100 Å³ comprising top 5% of density in map (REGIONS). The top 5% of points in the map are marked, and the number of contiguous regions that result are counted, and divided by the volume of the asymmetric unit, then multiplied by 100. A good map will have just a few contiguous regions at a high contour level, a poor map will have many isolated peaks. This score is by default not used.
- Contrast, or standard deviation of local rms density (CONTRAST). The local rms density in the map is calculated using a smoothing radius of 3 times the high-resolution cutoff (or 6 Å, if less than 6Å). Then the standard deviation of the local rms, normalized to the mean value of the local rms, is reported. This criteria will be high if there are regions of high local rms (the macromolecule) and separate regions of low local rms (the solvent) and low if the map is random. This score is by default not used.

Phasing

The AutoSol Wizard uses Phaser to calculate experimental phases from SAD data, and SOLVE to calculate phases from MIR, MAD, and multiple-dataset cases.

Density modification (including NCS averaging)

The AutoSol Wizard uses RESOLVE to carry out density modification. It identifies NCS from symmetries in heavy-atom sites with RESOLVE and applies this NCS if it is present in the electron density map.

Preliminary model-building and refinement

The AutoSol Wizard carries out one cycle of model-building and refinement after obtaining density-modified phases. The model-building is done with RESOLVE. The refinement is carried out with phenix.refine.

Resolution limits in AutoSol

There are several resolution limits used in AutoSol. You can leave them all to default, or you can set any of them individually. Here is a list of these limits and how their default values are set:

- resolution: This is the overall resolution for a dataset. By default it is based on the highest resolution for any datafile in this dataset. For multiple datasets, it is the highest resolution for any dataset.
- refinement_resolution: This is the resolution for refinement
- resolution_build: This is the resolution for model-building
- res_phase: This is the resolution for phasing for a dataset. If phase_full_resolution is True then this is the same as "resolution". Otherwise, the value of "recommended_resolution" based on an analysis of signal-to-noise in the dataset is used to define it.
- res_eval: This is the resolution for evaluation of solution quality. The default is the value of "resolution" or 2.5 Å, whichever is lower resolution.

Output files from AutoSol

When you run AutoSol the output files will be in a subdirectory with your run number:

```
AutoSol_run_1_/
```

The key output files that are produced are:

- A log file describing everything in the run and the files produced: AutoSol_run_1_1.log # overall log file

- A summary file listing the results of the run and the other files produced: AutoSol_summary.dat # overall summary
- A warnings file listing any warnings about the run: AutoSol_warnings.dat # any warnings
- Density-modified map coefficients (NOTE: These files will be aniso-corrected and sharpened if remove_aniso=True): overall_best_denmod_map_coeffs.mtz # map coefficients (density modified phases)
- Current preliminary model: overall_best.pdb # model produced for top solution

NOTE: If there are multiple chains or multiple ncs copies, each chain will be given its own chainID (A B C D...). Segments that are not assigned to a chain are given a separate chainID and are given a segid of "UNK" to indicate that their assignment is unknown. The chainID for solvent molecules is normally S, and the chainID for heavy-atoms is normally Z.

- An mtz file for use in refinement NOTE 1: not aniso corrected and not sharpened. NOTE 2: Two sets of HL coefficients may be present. Normally use HLA HLB etc . However, if you supplied a model with input_partpdb_file=my_model.pdb then use instead HlanomA HlanomB etc. The reason is that HL coeffs contain phase information from my_model.pdb in this case and you do not want that information passed to your refinement program.
- overall_best_refine_data.mtz # F Sigma HL coeffs, freeR-flags for refinement NOTE: if this is a SAD or MAD dataset then the overall_best_refine_data.mtz file will normally have your original anomalous data. For MAD data this will be from the wavelength of data with the highest-resolution data present.
- Heavy atom sites in PDB format: overall_best_ha_pdb.pdb # ha file for top solution
- NCS information (if any): overall_best_ncs_file.ncs_spec # NCS information for top solution
- Experimental phases and HL coefficients (NOTE: These files are aniso-corrected and sharpened if remove_aniso=True): overall_best_hklout_phased.mtz # phases and HL coeffs for top solution
- Log file for experimental phasing: overall_best_log_phased.log # experimental phasing log file for top solution
- Log file for scaling: overall_best_log_phased.log # experimental phasing log file for top solution
- Log file for heavy-atom substructure search: overall_best_log_hyss.log # ha search log file for top solution

How to run the AutoSol Wizard

Running the AutoSol Wizard is easy. From the command-line you can type:

```
phenix.autosol w1.sca seq.dat 2 Se f_prime=-8 f_double_prime=4.5
```

From the GUI you load in these files and specify the number of anomalously-scattering atoms and the atom type and the scattering factors on the main GUI page.

The AutoSol Wizard will assume that w1.sca is a datafile (because it ends in .sca and is a file) and that seq.dat is a sequence file, that there are 2 heavy-atom sites, and that the heavy-atom is Se. The f_prime and f_double_prime values are set explicitly

You can also specify each of these things directly:

```
phenix.autosol data=w1.sca seq_file=seq.dat sites=2 \
atom_type=Se f_prime=-8 f_double_prime=4.5
```

You can specify many more parameters as well. See the list of keywords, defaults and descriptions at the end of this page and also general information about running Wizards at Using the PHENIX Wizards for how to do

this. Some of the most common parameters are:

```
sites=3 # 3 sites
sites_file=sites.pdb # ha sites in PDB or fractional xyz format
atom_type=Se # Se is the heavy-atom
seq_file=seq.dat # sequence file (1-aa code, separate chains with &gt;&gt;&gt;&gt;)
quick=True # try to find sites quickly
data=w1.sca # input datafile
lambda=0.9798 # wavelength for SAD
```

Running from a parameters file

You can run phenix.autosol from a parameters file. This is often convenient because you can generate a default one with:

```
phenix.autosol --show_defaults &gt; my_autosol.eff
```

and then you can just edit this file to match your needs and run it with:

```
phenix.autosol my_autosol.eff
```

NOTE: the autosol parameters file my_autosol.eff will have just one blank native, derivative, and wavelength. You can cut and paste them to put in as many as you want to have.

NEW: AutoSol is optimized for weak SAD data from version 1.8.5

If you want to use a very weak anomalous signal in AutoSol you will want to turn on enable_extreme_dm. This allows AutoSol to turn on the features below if the figure of merit of phasing is low.

The new AutoSol is specifically engineered to be able to solve structures at low or high resolution with a very weak anomalous signal. One feature you may notice right away is that the new AutoSol will try to optimize several choices on the fly. AutoSol will use the Bayesian estimates of map quality and the R-value in density modification to decide which choices lead to the best phasing. AutoSol will try using sharpened data for substructure identification as well as unscaled data as input to AutoSol and pick the one leading to the best map. AutoSol will also try several smoothing radii for identification of the solvent boundary and pick the one that gives the best density modification R-value.

You'll also notice that AutoSol uses the new parallel HySS and that it can find substructures with SAD data that are very weak or that only have signal to low resolution. You can use any number of processors on your machine in the HySS step (so far the parallelization is only for HySS, not the other steps in AutoSol, but those are planned as well). The biggest change in AutoSol is that it now uses iterative Phaser LLG completion to improve the anomalously-scattering substructure for SAD phasing.

The key idea is to use the density-modified map (and later, the model built by AutoSol) to iterate the identification of the substructure. This feature is amazingly powerful in cases where only some of the sites can be identified at the start by HySS and by initial Phaser completion. Phaser LLG completion is more powerful if an estimate of part of the structure (from the map or from a model) is available. The new AutoSol may take a little longer than the old one due to the heavy-atom iteration, but you may find that it gives a much improved map and model.

What to do next after AutoSol

The next step after AutoSol is usually rebuilding your model with AutoBuild.

After you have run AutoSol, have a close look at the output files and summary statistics. You are hoping to find that AutoSol has found the heavy atoms or anomalously-scattering atoms, carried out density

modification, found non-crystallographic symmetry if any, and has built a partial or mostly-complete model.

Note any non-crystallographic symmetry found. If you know something about your molecule, this may match or not match what you expect.

Evaluate the plausibility of the number of sites found by HySS (if it is a SeMet structure you know about how many to find, for example). Note whether the sites show non-crystallographic symmetry if it was found.

Note the R-factor for density modification. These R values are not the same as those for refinement. A good R-factor for density modification will be in the range of 0.3-0.4, and an ok one from 0.4-0.5. Note that if AutoBuild is used to carry out density modification you may see a line like: Cycle 2 R/Rfree=999.90/999.90 Built=0 Placed=0. This just means that cycle 2 did not involve any refinement in maps_only mode.

If you have high-resolution data (better than 2 Å), you should obtain a nearly-complete model containing waters, with R/Rfree values near 0.2-0.25. For lower-resolution data (2-3 Å), expect a partial model with R/Rfree values in the range of 0.25-0.3, and for low resolution data (worse than 3 Å), expect a partial model without waters and with R/Rfree values in the range of 0.3-0.35. Have a look at your model and the overall best density-modified map. These should agree closely, with perhaps some parts of the map that show density not yet accounted for by the model (either not built or representing ligands).

If your results are in expected ranges and your map looks good, the next step is usually to run AutoBuild to try and obtain a more complete model. In the GUI you can just hit the Run AutoBuild button in the AutoSol output panel.

If results are not ok, recheck your Xtriage analysis and see if you have considered the right space group. You might try varying the resolution of data considered, either overall or just for HySS (finding sites). You can try running with thorough defaults or even try the extreme parameters.

If you have MIR data, try different combinations of derivatives. If you have native data and SAD data, try both together.

If attempts at experimental structure solution fail, try molecular replacement with a homology model or an AlphaFold model.

Examples

SAD dataset

You can use AutoSol on SAD data specifying just a few key items. You can load a sequence file and a data file in to the GUI and specify the atom type and wavelength and that is sufficient. You can do this on the command line with:

```
phenix.autosol w1.sca seq.dat 2 Se lambda=0.9798
```

The sequence file is used to estimate the solvent content of the crystal and for model-building. The wavelength (lambda) is used to look up values for f_prime and f_double_prime from a table, but if measured values are available from a fluorescence scan, these should be given in addition to the wavelength.

SAD dataset specifying solvent fraction

You can set the solvent fraction in the AutoSol GUI main page or on the command line:

```
phenix.autosol w1.sca seq.dat 2 Se lambda=0.9798 \  
solvent_fraction=0.45
```

This will force the solvent fraction to be 0.45. This illustrates a general feature of the Wizards: they will try to estimate values of parameters, but if you input them directly, they will use your input values.

SAD dataset without model-building

To skip model_building you can set build to False in the GUI or on the command_line:

```
phenix.autosol w1.sca seq.dat 2 Se lambda=0.9798 build=False
```

This will carry out the usual structure solution, but will skip model-building

SAD dataset, building RNA instead of protein

You can specify the chain type (RNA, DNA, PROTEIN):

```
phenix.autosol w1.sca seq.dat 2 Se lambda=0.9798 \
chain_type=RNA
```

This will carry out the usual structure solution, but will build an RNA chain. For DNA, specify chain_type=DNA. You can only build one type of chain at a time in the AutoSol Wizard. To build protein and DNA, use the AutoBuild Wizard and run it first with chain_type=PROTEIN, then run it again specifying the protein model as input_lig_file_list=proteinmodel.pdb and with chain_type=DNA.

SAD dataset, selecting a particular dataset from an MTZ file

If you have an input MTZ file with more than one anomalous dataset, you can type something like:

```
phenix.autosol w1.mtz seq.dat 2 Se lambda=0.9798 \
labels='F+ SIGF+ F- SIGF-'
```

This will carry out the usual structure solution, but will choose the input data columns based on the labels: 'F+ SIGF+ F- SIGF-' NOTE: to specify anomalous data with F+ SIGF+ F- SIGF- like this, these 4 columns must be adjacent to each other in the MTZ file with no other columns in between. FURTHER NOTE: to instead use a FAVG SIGFAVG DANO SIGDANO array in AutoSol, the data file or an input refinement file MUST also contain a separate array for FP SIGFP or I SIGI or equivalent. This is because FAVG DANO arrays are ONLY allowed as anomalous information, not as amplitudes or intensities. You can use F+ SIGF+ F- SIGF- arrays as a source of both anomalous differences and amplitudes if you want, however. If you run the AutoSol Wizard with SAD data and an MTZ file containing more than one anomalous dataset and don't tell it which one to use, all possible values of labels are printed out for you so that you can just paste the one you want in.

You can also find out all the possible label strings to use by typing:

```
phenix.autosol display_labels=w1.mtz # display all labels for w1.mtz
```

MRSAD -- SAD dataset with an MR model; Phaser SAD phasing including the model

If you are carrying out SAD phasing with Phaser, you can carry out a combination of molecular replacement phasing and SAD phasing (MRSAD) by adding a single new keyword to your AutoSol run:

```
input_partpdb_file=MR.pdb
```

NOTE: You must run molecular replacement separately first before you run autosol. MR.pdb is the result from your molecular replacement run.

You can optionally also specify an estimate of the RMSD between your model and the true structure with a command like:

```
partpdb_rms=1.5
```

In this case the MR.pdb file will be used as a partial model in a maximum-likelihood SAD phasing calculation with Phaser to calculate phases and identify sites in Phaser, and the combined MR+SAD phases will be written out.

Notes on model bias and MRSAD.

There are a number of factors that influence how much model bias there is in a MR-SAD experiment. Additionally the model bias is different at different steps in the procedure. You also have several choices that affect how much model bias there is:

Options for reducing model bias in MR-SAD initial phasing

- Considering just the initial phasing step, you have two options in Phenix:
- You can find sites with phaser LLG maximization using your input model as part of the total model, then phase using the sites that you find and exclude the model in the phasing step. This is the keyword "phaser_sites_then_phase=True" and it has essentially no model bias.
- You can find sites and phase in one step with Phaser, in which case your phases will contain information from the model and there can be model bias. You get two different sets of HL coefficients, one is HLA etc, the other is HLANOM etc. The first contains the same information in the phases, and may be biased by the model. The second does not contain phase information from the model, only from the anomalous differences.

Options for reducing model bias in MR-SAD in density modification and refinement

- Once you have obtained calculated phases and phase probabilities with and without the model (HL and HLANOM), you have additional choices as to how to use this information. In all cases the starting phases are those from step 1, and include model information unless you set phaser_sites_then_phase=True. The choices in this stage consist of whether to use the phase probabilities with model information (HL) or without (HLANOM).
- You can specify whether to use phase probabilities with or without the model in density modification and refinement. If you set use_hl_anom_in_denmod=False (default) then model information will be used in density modification. If you set use_hl_anom_in_denmod=True then HLANOM coefficients not including model information will be used.
- You can separately specify whether to use model information in refinement. If you set use_hl_anom_in_refinement=True (default if a partial model has been supplied as in MR-SAD), then phase probabilities without the model will be used.

Avoiding model bias in MR-SAD

You can avoid model bias in MR-SAD completely with phaser_sites_then_phase=True. You can also have reduced model bias if you set use_hl_anom_in_denmod=True and use_hl_anom_in_refinement=True. If you do not reduce model bias in one of these ways, the amount of bias is going to depend very much on the strength of the anomalous signal, the amount of solvent, and the resolution, as all of these factors contribute to the phase information coming from somewhere other than the model.

A good way to check on model bias in this or any other case is to remove a part of your starting model (say, a helix) that you are worried about, and use that omit model in the whole procedure. Then if the density shows up in the end for this feature you can be sure it is not due to model bias. You can do this after the fact — you solve your structure, you are worried that something in the map is model bias, you run the whole thing over

starting with a model that does not have this feature, it comes back in your maps, and you know it is ok. You can show the resulting map to anyone and they will know that it is not biased as well.

Using placed density from a Phaser MR density search to find sites and as a source of phase information (method #2 for MRSAD)

You can carry out a procedure almost like using `input_partpdb_file` except that the model information comes as a set of map coefficients for density. These map coefficients can be from Phaser or any other source of phase information. In this case the keyword to use is `input_part_map_coeffs_file` instead of `input_partpdb_file`. You can also identify the labels with `input_part_map_coeffs_labels`.

SAD dataset, reading heavy-atom sites from a PDB file written by phenix.hyss

You can type from the command_line:

```
phenix.autosol 11 Pb data=deriv.sca seq_file=seq.dat \
  sites_file=deriv_hyss_consensus_model.pdb lambda=0.95
```

This will carry out the usual structure solution process, but will read sites from `deriv_hyss_consensus_model.pdb`, try both hands, and carry on from there. If you know the hand of the substructure, you can fix it with `have_hand=True`.

MAD dataset

The inputs for a MAD dataset need to specify `f_prime` and `f_double_prime` for each wavelength. You can use a parameters file "mad.eff" to input MAD data. You run it with "phenix.autosol mad.eff". Here is an example of a parameters file for a MAD dataset. You can set many additional parameters as well (see the list at the end of this document).

```
autosol {
  seq_file = seq.dat
  sites = 2
  atom_type = Se
  wavelength {
    data = peak.sca
    lambda = .9798
    f_prime = -8.0
    f_double_prime = 4.5
  }
  wavelength {
    data = inf.sca
    lambda = .9792
    f_prime = -9.0
    f_double_prime = 1.5
  }
}
```

MAD dataset, selecting particular datasets from an MTZ file

This is similar to the case for running a SAD analysis, selecting particular columns of data from an MTZ file. If you have an input MTZ file with more than one anomalous dataset, you can use a parameters file like the one above for MAD data, but adding information on the labels in the MTZ file that are to be chosen for each wavelength:

```
autosol {
  seq_file = seq.dat
  sites = 2
  atom_type = Se
```

```

wavelength {
data = mad.mtz
lambda = .9798
f_prime = -8.0
f_double_prime = 4.5
labels='peak(+) SIGpeak(+) peak(-) SIGpeak(-)'
}
wavelength {
data = mad.mtz
lambda = .9792
f_prime = -9.0
f_double_prime = 1.5
labels='infl(+) SIGinfl(+) infl(-) SIGinfl(-)'
}
}

```

This will carry out the usual structure solution, but will choose the input peak data columns based on the label keywords. As in the SAD case, you can find out all the possible label strings to use by typing:

```
phenix.autosol display_labels=w1.mtz # display all labels for w1.mtz
```

SIR dataset

The standard inputs for a SIR dataset are the native and derivative, the sequence file, the heavy-atom type, and the number of sites, as well as whether to use anomalous differences (or just isomorphous differences). From the command line you can say:

```
phenix.autosol native.data=native.sca deriv.data=deriv.sca \
atom_type=I sites=2 inano=inano
```

This will set the heavy-atom type to Iodine, look for 2 sites, and include anomalous differences. You can also specify many more parameters using a parameters file. This parameters file shows some of them:

```

autosol {
  seq_file = seq.dat
  native {
    data = native.sca
  }
  deriv {
    data = pt.sca
    lambda = 1.4
    atom_type = Pt
    f_prime = -3.0
    f_double_prime = 3.5
    sites = 3
  }
}

```

SAD with more than one anomalously-scattering atom

You can tell the AutoSol wizard to look for more than one anomalously-scattering atom. Specify one atom type (Se) in the usual way. Then specify any additional ones in the GUI window or like this if you are running AutoSol from the command line:

```
mad_ha_add_list="Br Pt"
```

Optionally, you can add `f_prime` and `f_double_prime` values for the additional atom types with commands like `mad_ha_add_f_prime_list=" -7 -10"` `mad_ha_add_f_double_prime_list=" 4.2 12"` but the values from table lookup should be fine. Note that there must be the same number of entries in each of these three keyword lists, if given. During phasing Phaser will try to add whichever atom types best fit the scattering from each new site. This option is available for SAD phasing only and only for a single dataset (not with SAD+MIR etc).

Note: if you are running SAD or MRSAD and your wavelength is longer than 3 Å, you will need to supply f' and f'' for all atoms that are anomalous scatterers (i.e., any atoms in your model plus any atoms that are added). You can use the `mad_ha_add_list` to do this.

A particularly useful way to use this feature is to add in S atoms in a protein structure that has SeMet (specifying S with `mad_ha_add_list` will then, with luck, find the Cys sulfurs), or to add in P atoms in a nucleic acid structure phased by some other atom such as I. This works especially well at high resolution.

MIR dataset

You can run an MIR dataset from the GUI or using a parameters file such as "mir.eff" which you then run with "phenix.autosol mir.eff". Here is an example parameters file for MIR:

```
autosol {
  seq_file = seq.dat
  native {
    data = native.sca
  }
  deriv {
    data = pt.sca
    lambda = 1.4
    atom_type = Pt
  }
  deriv {
    data = ki.sca
    lambda = 1.5
    atom_type = I
  }
}
```

You can enter as many derivatives as you want. If you specify a wavelength and heavy atom type then scattering factors are calculated from a table for that heavy-atom. You can instead enter scattering factors with the keywords "`f_prime = -3.0`" "`f_double_prime = 5.0`" if you want.

SIR + SAD datasets

A combination of SIR and SAD datasets (or of SAD+SAD or MIR+SAD+SAD or any other combination) is easy with a parameters file. You tell the wizard which grouping each wavelength, native, or derivative goes with with a keyword such as "`group=1`":

```
autosol {
  seq_file = seq.dat
  native {
    group = 1
    data = native.sca
  }
  deriv {
    group = 1
    data = pt.sca
    lambda = 1.4
    atom_type = Pt
  }
  wavelength {
    group = 2
    data = w1.sca
    lambda = .9798
    atom_type = Se
    f_prime = -7.
    f_double_prime = 4.5
  }
}
```

The SIR and SAD datasets will be solved separately (but whichever one is solved first will use difference or anomalous difference Fouriers to locate sites for the other). Then phases will be combined by addition of Hendrickson-Lattman coefficients and the combined phases will be density modified.

AutoSol with an extremely weak signal

You can use AutoSol in cases with a very weak anomalous signal. The big challenges in this kind of situation are finding the anomalously-scattering atoms and density modification. For finding the sites, you may need to try everything available, including running HySS with various resolution cutoffs for the data and trying other software for finding sites as well. You may then want to start with the sites found using one of these approaches and provide those to phenix.autosol with a command like,:

```
sites_file=my_sites.pdb
```

For the density modification step, there is a keyword (`extreme_dm`) that may be helpful in cases where the starting phases are very poor:

```
extreme_dm = True
```

This keyword works with the keyword `fom_for_extreme_dm` to determine whether a set of defaults for density modification with weak phases is appropriate. If so, a very large radius for identification of the solvent boundary (20 Å) is used, and the number of density modification cycles is reduced. This can make a big difference in getting started with density modification in such a case.

AutoSol with a cluster compound

You can run SAD phasing in AutoSol with a cluster compound (not MIR or MAD yet). Normally you should supply a PDB file with an example of the cluster with the keyword `cluster_pdb_file=my_cluster_compound` and a unique residue name XX (yes, really the two letters XX, not XX replaced by some other name). Set the keyword `atom_type=XX` as well. If your cluster is Ta₆Br₁₂ then you can simply put `atom_type=TX` and skip the `cluster_pdb_file`. For MAD/MIR, cluster compounds are not currently supported. Instead just use a standard atom.

R-free flags and test set

In Phenix the parameter `test_flag_value` sets the value of the test set that is to be free. Normally Phenix sets up test sets with values of 0 and 1 with 1 as the free set. The CCP4 convention is values of 0 through 19 with 0 as the free set. Either of these is recognized by default in Phenix and you do not need to do anything special. If you have any other convention (for example values of 0 to 19 and test set is 1) then you can specify this with:

```
test_flag_value=1
```

Specific limitations and problems

- The size of the asymmetric unit in the SOLVE/RESOLVE portion of the AutoSol wizard is limited by the memory in your computer and the binaries used. The Wizard is supplied with regular-size ("", size=6), giant ("_giant", size=12), huge ("_huge", size=18) and extra_huge ("_extra_huge", size=36). Larger-size versions can be obtained on request.
- The keywords "cell" and "sg" have been replaced with "unit_cell" and "space_group" to make the keywords the same as in other phenix applications.

- The keywords for running MIR and SIR and MAD datasets from parameter files and the command line have been changed to make the inputs more consistent and suitable for a static GUI.
- The AutoSol Wizard can take a maximum of 6 derivatives for MIR.
- The AutoSol Wizard can take most settings of most space groups, however it can only use the hexagonal setting of rhombohedral space groups (eg., #146 R3:H or #155 R32:H), and it cannot use space groups 114-119 (not found in macromolecular crystallography) even in the standard setting due to difficulties with the use of asuset in the version of ccp4 libraries used in PHENIX for these settings and space groups.

Literature

Decision-making in structure solution using Bayesian estimates of map quality: the PHENIX AutoSol wizard. T.C. Terwilliger, P.D. Adams, R.J. Read, A.J. McCoy, N.W. Moriarty, R.W. Grosse-Kunstleve, P.V. Afonine, P.H. Zwart, and L.W. Hung. *Acta Crystallogr D Biol Crystallogr* 65, 582-601 (2009).

- Decision-making in structure solution using Bayesian estimates of map quality: the PHENIX AutoSol wizard. T.C. Terwilliger, P.D. Adams, R.J. Read, A.J. McCoy, N.W. Moriarty, R.W. Grosse-Kunstleve, P.V. Afonine, P.H. Zwart, and L.W. Hung. *Acta Crystallogr D Biol Crystallogr* 65, 582-601 (2009).

Simple algorithm for a maximum-likelihood SAD function. A.J. McCoy, L.C. Storoni, and R.J. Read. *Acta Crystallogr D Biol Crystallogr* 60, 1220-8 (2004).

- Simple algorithm for a maximum-likelihood SAD function. A.J. McCoy, L.C. Storoni, and R.J. Read. *Acta Crystallogr D Biol Crystallogr* 60, 1220-8 (2004).

Substructure search procedures for macromolecular structures. R.W. Grosse-Kunstleve, and P.D. Adams. *Acta Cryst. D* 59, 1966-1973. (2003).

- Substructure search procedures for macromolecular structures. R.W. Grosse-Kunstleve, and P.D. Adams. *Acta Cryst. D* 59, 1966-1973. (2003).

MAD phasing: Bayesian estimates of F(A). T.C. Terwilliger. *Acta Crystallogr D Biol Crystallogr* 50, 11-6 (1994).

- MAD phasing: Bayesian estimates of F(A). T.C. Terwilliger. *Acta Crystallogr D Biol Crystallogr* 50, 11-6 (1994).

Rapid automatic NCS identification using heavy-atom substructures. T.C. Terwilliger. *Acta Crystallogr D Biol Crystallogr* 58, 2213-5 (2002).

- Rapid automatic NCS identification using heavy-atom substructures. T.C. Terwilliger. *Acta Crystallogr D Biol Crystallogr* 58, 2213-5 (2002).

Maximum-likelihood density modification. T.C. Terwilliger. *Acta Crystallogr D Biol Crystallogr* 56, 965-72 (2000).

- Maximum-likelihood density modification. T.C. Terwilliger. *Acta Crystallogr D Biol Crystallogr* 56, 965-72 (2000).

Statistical density modification with non-crystallographic symmetry. T.C. Terwilliger. *Acta Crystallogr D Biol Crystallogr* 58, 2082-6 (2002).

- Statistical density modification with non-crystallographic symmetry. T.C. Terwilliger. *Acta Crystallogr D Biol Crystallogr* 58, 2082-6 (2002).

Automated side-chain model building and sequence assignment by template matching. T.C. Terwilliger. *Acta Crystallogr D Biol Crystallogr* 59, 45-9 (2003).

- Automated side-chain model building and sequence assignment by template matching. T.C. Terwilliger. Acta Crystallogr D Biol Crystallogr 59, 45-9 (2003).

Model morphing and sequence assignment after molecular replacement. T.C. Terwilliger, R.J. Read, P.D. Adams, A.T. Brunger, P.V. Afonine, and L.W. Hung. Acta Crystallogr D Biol Crystallogr 69, 2244-50 (2013).

- Model morphing and sequence assignment after molecular replacement. T.C. Terwilliger, R.J. Read, P.D. Adams, A.T. Brunger, P.V. Afonine, and L.W. Hung. Acta Crystallogr D Biol Crystallogr 69, 2244-50 (2013).

Automated main-chain model building by template matching and iterative fragment extension. T.C. Terwilliger. Acta Crystallogr D Biol Crystallogr 59, 38-44 (2003).

- Automated main-chain model building by template matching and iterative fragment extension. T.C. Terwilliger. Acta Crystallogr D Biol Crystallogr 59, 38-44 (2003).

List of all available keywords

job_title = None Job title in PHENIX GUI, not used on command line

autosolatom_type = None Anomalous-scattering atom type. This sets the atom_type in all derivatives and wavelengths. Normally it is used as a shortcut for SAD or SIR cases. NOTE: For SAD only: if you have a cluster compound, normally you should supply a PDB file with an example of the cluster with the keyword

cluster_pdb_file=my_cluster_compound and a unique residue name XX (yes, XX). Set the keyword atom_type=XX as well. If your cluster is Ta6Br12 then you can simply put atom_type=TX and skip the cluster_pdb_file. For MAD/MIR, cluster compounds are not currently supported. Instead just use a standard atom. NOTE: if you specify keep_atom_type and supply a sites_file then atom_type will still apply to added sites.

lambda = None Wavelength (Å). This sets the wavelength value in all derivatives and wavelengths. Normally it is used as a shortcut for SAD or SIR cases.

f_prime = None F-prime value. This sets the f_prime value in all derivatives and wavelengths. Normally it is used as a shortcut for SAD or SIR cases.

f_double_prime = None F-double-prime value. This sets the f_double_prime value in all derivatives and wavelengths. Normally it is used as a shortcut for SAD or SIR cases.

wavelength_name = peak inf high low remote Optional name of wavelength for SAD data. This sets the name in all wavelengths. Normally it is used as a shortcut for SAD cases.

sites = None Number of heavy-atom sites. This sets the number of sites in all derivatives and wavelengths. Normally it is used as a shortcut for SAD or SIR cases.

sites_file = None PDB or plain-text file with ha sites. This sets the sites in all derivatives and wavelengths. Normally it is used as a shortcut for SAD or SIR cases.

If you want to keep the atom types specify keep_atom_types=True

seq_file = Auto Text file with 1-letter code of protein sequence. Separate chains with a blank line or line starting with > Normally you should include one copy of each unique chain. NOTE: if 1 copy of each unique chain is provided it is assumed that there are ncs_copies (could be 1) of each unique chain. If more than one copy of any chain is provided it is assumed that the asymmetric unit contains the number of copies of each chain that are given, multiplied by ncs_copies. So if the sequence file has two copies of the sequence for chain A and one of chain B, the cell contents are assumed to be ncs_copies*2 of chain A and ncs_copies of chain B. If there are unequal numbers of copies of chains, be sure to set solvent_fraction.

ADDITIONAL NOTES: 1. lines starting with > are ignored and separate chains 2. FASTA format is fine 3. If you enter a PDB file for rebuilding and it has the sequence you want, then the sequence file is not necessary. NOTE: You can also enter the name of a PDB file that contains SEQRES records, and the sequence from the SEQRES records will be read, written to seq_from_seqres_records.dat, and used as your input sequence. If you have a duplex DNA, enter each strand as a separate chain. 4. Characters such as numbers and non-printing characters in the sequence file are ignored. 5 Be sure that your sequence file does not have any blank lines in the middle of your sequence, as these are interpreted as the beginning of another chain.

quick = None Run everything quickly (Same as thoroughness=quick)

data = None Shortcut for name of datafile (SAD data only. For SIR use

"native.data=native.sca" and "deriv.data=deriv.sca",. For MIR and MAD use a parameters file and specify data under "native" and "deriv" or for "wavelength".) NOTE: For command_line input it is easiest if each wavelength of data is in a separate data file with obvious data columns. File types that are easy to read include Scalepack sca files , CNS hkl files, mtz files with just one wavelength of data, or just native or just derivative. In this case the Wizard can ...

- job_title = None Job title in PHENIX GUI, not used on command line
- autosolatom_type = None Anomalous-scattering atom type. This sets the atom_type in all derivatives and wavelengths. Normally it is used as a shortcut for SAD or SIR cases. NOTE: For SAD only: if you have a cluster compound, normally you should supply a PDB file with an example of the cluster with the keyword cluster_pdb_file=my_cluster_compound and a unique residue name XX (yes, XX). Set the keyword atom_type=XX as well . If your cluster is Ta6Br12 then you can simply put atom_type=TX and skip the cluster_pdb_file. For MAD/MIR, cluster compounds are not currently supported. Instead just use a standard atom. NOTE: if you specify keep_atom_type and supply a sites_file then atom_type will still apply to added sites.lambda = None Wavelength (Å). This sets the wavelength value in all derivatives and wavelengths. Normally it is used as a shortcut for SAD or SIR cases.f_prime = None F-prime value. This sets the f_prime value in all derivatives and wavelengths. Normally it is used as a shortcut for SAD or SIR cases.f_double_prime = None F-double-prime value. This sets the f_double_prime value in all derivatives and wavelengths. Normally it is used as a shortcut for SAD or SIR cases.wavelength_name = peak inf high low remote Optional name of wavelength for SAD data. This sets the name in all wavelengths. Normally it is used as a shortcut for SAD cases.sites = None Number of heavy-atom sites. This sets the number of sites in all derivatives and wavelengths. Normally it is used as a shortcut for SAD or SIR cases.sites_file = None PDB or plain-text file with ha sites. This sets the sites in all derivatives and wavelengths. Normally it is used as a shortcut for SAD or SIR cases. If you want to keep the atom types specify keep_atom_types=True.seq_file = Auto Text file with 1-letter code of protein sequence. Separate chains with a blank line or line starting with > Normally you should include one copy of each unique chain. NOTE: if 1 copy of each unique chain is provided it is assumed that there are ncs_copies (could be 1) of each unique chain. If more than one copy of any chain is provided it is assumed that the asymmetric unit contains the number of copies of each chain that are given, multiplied by ncs_copies. So if the sequence file has two copies of the sequence for chain A and one of chain B, the cell contents are assumed to be ncs_copies*2 of chain A and ncs_copies of chain B. If there are unequal numbers of copies of chains, be sure to set solvent_fraction. ADDITIONAL NOTES: 1. lines starting with > are ignored and separate chains 2. FASTA format is fine 3. If you enter a PDB file for rebuilding and it has the sequence you want, then the sequence file is not necessary. NOTE: You can also enter the name of a PDB file that contains SEQRES records, and the sequence from the SEQRES records will be read, written to seq_from_seqres_records.dat, and used as your input sequence. If you have a duplex DNA, enter each strand as a separate chain. 4. Characters such as numbers and non-printing characters in the sequence file are ignored. 5 Be sure that your sequence file does not have any blank lines in the middle of your sequence, as these are interpreted as the beginning of another chain.quick = None Run everything quickly (Same as thoroughness=quick)data = None Shortcut for name of datafile (SAD data only. For SIR use "native.data=native.sca" and "deriv.data=deriv.sca",. For MIR and MAD use a parameters file and specify data under "native" and "deriv" or for "wavelength".) NOTE: For command_line input it is easiest if each wavelength of data is in a separate data file with obvious data columns. File types that are easy to read include Scalepack sca files , CNS hkl files, mtz files with just one wavelength of data, or just native or just derivative. In this case the Wizard can read your data without further information. <p> If you have ...

- `atom_type = None` Anomalous-scattering atom type. This sets the `atom_type` in all derivatives and wavelengths. Normally it is used as a shortcut for SAD or SIR cases. NOTE: For SAD only: if you have a cluster compound, normally you should supply a PDB file with an example of the cluster with the keyword `cluster_pdb_file=my_cluster_compound` and a unique residue name XX (yes, XX). Set the keyword `atom_type=XX` as well. If your cluster is Ta6Br12 then you can simply put `atom_type=TX` and skip the `cluster_pdb_file`. For MAD/MIR, cluster compounds are not currently supported. Instead just use a standard atom. NOTE: if you specify `keep_atom_type` and supply a `sites_file` then `atom_type` will still apply to added sites.
- `lambda = None` Wavelength (Å). This sets the wavelength value in all derivatives and wavelengths. Normally it is used as a shortcut for SAD or SIR cases.
- `f_prime = None` F-prime value. This sets the `f_prime` value in all derivatives and wavelengths. Normally it is used as a shortcut for SAD or SIR cases.
- `f_double_prime = None` F-double-prime value. This sets the `f_double_prime` value in all derivatives and wavelengths. Normally it is used as a shortcut for SAD or SIR cases.
- `wavelength_name = peak inf high low remote` Optional name of wavelength for SAD data. This sets the name in all wavelengths. Normally it is used as a shortcut for SAD cases.
- `sites = None` Number of heavy-atom sites. This sets the number of sites in all derivatives and wavelengths. Normally it is used as a shortcut for SAD or SIR cases.
- `sites_file = None` PDB or plain-text file with ha sites. This sets the sites in all derivatives and wavelengths. Normally it is used as a shortcut for SAD or SIR cases. If you want to keep the atom types specify `keep_atom_types=True`
- `seq_file = Auto` Text file with 1-letter code of protein sequence. Separate chains with a blank line or line starting with `>`; Normally you should include one copy of each unique chain. NOTE: if 1 copy of each unique chain is provided it is assumed that there are `ncs_copies` (could be 1) of each unique chain. If more than one copy of any chain is provided it is assumed that the asymmetric unit contains the number of copies of each chain that are given, multiplied by `ncs_copies`. So if the sequence file has two copies of the sequence for chain A and one of chain B, the cell contents are assumed to be `ncs_copies*2` of chain A and `ncs_copies` of chain B. If there are unequal numbers of copies of chains, be sure to set `solvent_fraction`. ADDITIONAL NOTES: 1. lines starting with `>`; are ignored and separate chains 2. FASTA format is fine 3. If you enter a PDB file for rebuilding and it has the sequence you want, then the sequence file is not necessary. NOTE: You can also enter the name of a PDB file that contains SEQRES records, and the sequence from the SEQRES records will be read, written to `seq_from_seqres_records.dat`, and used as your input sequence. If you have a duplex DNA, enter each strand as a separate chain. 4. Characters such as numbers and non-printing characters in the sequence file are ignored. 5 Be sure that your sequence file does not have any blank lines in the middle of your sequence, as these are interpreted as the beginning of another chain.
- `quick = None` Run everything quickly (Same as `thoroughness=quick`)
- `data = None` Shortcut for name of datafile (SAD data only. For SIR use `"native.data=native.sca"` and `"deriv.data=deriv.sca"`; For MIR and MAD use a parameters file and specify data under `"native"` and `"deriv"` or for `"wavelength"`;) NOTE: For command_line input it is easiest if each wavelength of data is in a separate data file with obvious data columns. File types that are easy to read include Scalepack sca files, CNS hkl files, mtz files with just one wavelength of data, or just native or just derivative. In this case the Wizard can read your data without further information. <p> If you have a datafile with many columns, you can use the `"labels"` keyword to specify which data columns to read. (It may be easier in some

cases to use the GUI or to split it with phenix.reflection_file_converter first, however.)

- labels = None Shortcut for specification string for data labels (SAD data only). Only necessary if the wizard does not automatically choose the correct set of data from your file For SIR use "native.labels" and "deriv.labels". For MIR and MAD use a parameters file and specify labels under "native" and "deriv"; NOTE: To find out what the appropriate strings are, type "phenix.autosol display_labels=your-datafile-here.mtz";
- derscale = None shortcut for derivative scale
- crystal_infounit_cell = None Enter cell parameter (a b c alpha beta gamma)space_group = None Space Group symbol (i.e., C2221 or C 2 2 21)solvent_fraction = None Solvent fraction in crystals (0 to 1). This is normally set automatically from the number of NCS copies and the sequence. If your has unequal numbers of different chains, then be sure to set the solvent fraction.chain_type = *Auto PROTEIN DNA RNA You can specify whether to build protein, DNA, or RNA chains. At present you can only build one of these in a single run. If you have both DNA and protein, build one first, then run AutoBuild again, supplying the prebuilt model in the "input_lig_file_list" and build the other. NOTE: default for this keyword is Auto, which means "carry out normal process to guess this keyword". The process is to look at the sequence file and/or input pdb file to see what the chain type is. If there are more than one type, the type with the larger number of residues is guessed. If you want to force the chain_type, then set it to PROTEIN RNA or DNA. resolution = 0 High-resolution limit. Used as resolution limit for density modification and as general default high-resolution limit. If resolution_build or refinement_resolution are set then they override this for model-building or refinement. If overall_resolution is set then data beyond that resolution is ignored completely. Zero means keep everything.change_sg = False You can change the space group. In AutoSol the Wizard will use ImportRawData and let you specify the sg and cell. In AutoMR the wizard will give you an entry form to specify them. NOTE: This only applies when reading in new datasets. It does nothing when changed after datasets are read in. residues = None Number of amino acid residues in the au (or equivalent) sequence = None Plain text containing 1-letter code of protein sequence Same as seq_file except the sequence is read directly, not from a file. If both are given, seq_file is ignored.
- unit_cell = None Enter cell parameter (a b c alpha beta gamma)
- space_group = None Space Group symbol (i.e., C2221 or C 2 2 21)
- solvent_fraction = None Solvent fraction in crystals (0 to 1). This is normally set automatically from the number of NCS copies and the sequence. If your has unequal numbers of different chains, then be sure to set the solvent fraction.
- chain_type = *Auto PROTEIN DNA RNA You can specify whether to build protein, DNA, or RNA chains. At present you can only build one of these in a single run. If you have both DNA and protein, build one first, then run AutoBuild again, supplying the prebuilt model in the "input_lig_file_list" and build the other. NOTE: default for this keyword is Auto, which means "carry out normal process to guess this keyword". The process is to look at the sequence file and/or input pdb file to see what the chain type is. If there are more than one type, the type with the larger number of residues is guessed. If you want to force the chain_type, then set it to PROTEIN RNA or DNA.
- resolution = 0 High-resolution limit. Used as resolution limit for density modification and as general default high-resolution limit. If resolution_build or refinement_resolution are set then they override this for model-building or refinement. If overall_resolution is set then data beyond that resolution is ignored completely. Zero means keep everything.
- change_sg = False You can change the space group. In AutoSol the Wizard will use ImportRawData and let you specify the sg and cell. In AutoMR the wizard will give you an entry form to specify them.

NOTE: This only applies when reading in new datasets. It does nothing when changed after datasets are read in.

- residues = None Number of amino acid residues in the au (or equivalent)
- sequence = None Plain text containing 1-letter code of protein sequence Same as seq_file except the sequence is read directly, not from a file. If both are given, seq_file is ignored.
- output_filetarget_output_format = *None pdb mmCIF Desired output format (if possible). Choices are None (try to use input format), pdb, mmCIF. If output model does not fit in pdb format, mmCIF will be used. Default is pdb.
- target_output_format = *None pdb mmCIF Desired output format (if possible). Choices are None (try to use input format), pdb, mmCIF. If output model does not fit in pdb format, mmCIF will be used. Default is pdb.
- input_filesCIF_def_file_list = None You can enter any number of CIF definition files. These are normally used to tell phenix.refine about the geometry of a ligand or unusual residue. You usually will use these in combination with "PDB file with metals/ligands" (keyword "input_lig_file_list") which allows you to attach the contents of any PDB file you like to your model just before it gets refined. You can use phenix.elbow to generate these if you do not have a CIF file and one is requested by phenix.refine group_labels_list = None For command-line and script running of AutoSol, you may wish to use keywords to specify which set of data columns to be used from an MTZ or other file type with multiple datasets. (From the GUI, it is easy because you are prompted with the column labels). You can do this by specifying a string that identifies which dataset to include. All allowed values of this identification string will be written out any time AutoSol is run on this dataset like this: NOTE: To specify a particular set of data you can specify one of the following (this example is for MAD data, specifying data for peak wavelength): ...: peak.labels='F SIGF DANO SIGDANO' peak.labels='F(+) SIGF(+) F(-) SIGF(-)' You can then use one of the above commands on the command-line to identify the dataset of interest. If you want to use a script instead, you can specify N files in your input_data_file_list, and then specify N values for group_labels_list like this: group_labels_list 'F,SIGF,DANO,SIGDANO' 'F(+),SIGF(+),F(-),SIGF(-)' This will take 'F,SIGF,DANO,SIGDANO' as the data for datafile 1 and 'F(+),SIGF(+),F(-),SIGF(-)' for datafile 2 You can identify one dataset from each input file in this way. If you want more than one, then please use phenix.reflection_file_converter to split your input file, or else use the GUI version of AutoSol in which you can select any subset of the data that you wish. input_file_list = None Normally not used. Use "data=" or "wavelength.data=" or "native.data=" or "deriv.data=" instead. input_phase_is_part_map_coeffs = True If input_phase_is_part_map_coeffs is true then both input_phase_file and input_part_map_coeffs_file will be the same file (whichever is specified) input_phase_file = None MTZ data file with FC PHIC or equivalent to use for finding heavy-atom sites with difference Fourier methods. NOTE: compare with input_part_map_coeffs_file which is a file with map coefficients for use with Phaser completion to find anomalously-scattering atoms. NOTE 2: You can supply both input_phase_file and input_part_map_coeffs_file. NOTE 3: if input_phase_is_part_map_coeffs is true then both input_part_map_coeffs_file or phases from partpdb_file will be copied to input_phase_file input_phase_labels = None Labels for FC and PHIC for data file with FC PHIC or equivalent to use for finding heavy-atom sites with difference Fourier methods. input_refinement_file = None Data file to use for refinement. The data in this file should not be corrected for anisotropy. It will be combined with experimental phase information for refinement. If you leave this blank, then the output of phasing will be used in refinement (see below). If no anisotropy correction is applied to the data you do not need to specify a datafile for refinement. If an anisotropy correction is applied to the data files, then you must enter a datafile for refinement if you want to refine your model. (See "remove_aniso" for specifying whether an anisotropy correction is applied. In most cases it is not.) If an anisotropy

correction is applied and no refinement datafile is supplied, then no refinement will be carried out in the model-building step. You can choose any of your datafiles to be the refinement file, or a native that is not part of the datasets for structure solution. If there are more than one dataset you will be asked each time for...

- `cif_def_file_list = None` You can enter any number of CIF definition files. These are normally used to tell phenix.refine about the geometry of a ligand or unusual residue. You usually will use these in combination with `"PDB file with metals/ligands"` (keyword `"input_lig_file_list"`;) which allows you to attach the contents of any PDB file you like to your model just before it gets refined. You can use phenix.elbow to generate these if you do not have a CIF file and one is requested by phenix.refine

- `group_labels_list = None` For command-line and script running of AutoSol, you may wish to use keywords to specify which set of data columns to be used from an MTZ or other file type with multiple datasets. (From the GUI, it is easy because you are prompted with the column labels). You can do this by specifying a string that identifies which dataset to include. All allowed values of this identification string will be written out any time AutoSol is run on this dataset like this: NOTE: To specify a particular set of data you can specify one of the following (this example is for MAD data, specifying data for peak wavelength): ...: `peak.labels='F SIGF DANO SIGDANO'` `peak.labels='F(+) SIGF(+) F(-) SIGF(-)'` You can then use one of the above commands on the command-line to identify the dataset of interest. If you want to use a script instead, you can specify N files in your `input_data_file_list`, and then specify N values for `group_labels_list` like this: `group_labels_list 'F,SIGF,DANO,SIGDANO' 'F(+),SIGF(+),F(-),SIGF(-)'` This will take 'F,SIGF,DANO,SIGDANO' as the data for datafile 1 and 'F(+),SIGF(+),F(-),SIGF(-)' for datafile 2. You can identify one dataset from each input file in this way. If you want more than one, then please use phenix.reflection_file_converter to split your input file, or else use the GUI version of AutoSol in which you can select any subset of the data that you wish.

- `input_file_list = None` Normally not used. Use `"data="` or `"wavelength.data="` or `"native.data="` or `"deriv.data="` instead.

- `input_phase_is_part_map_coeffs = True` If `input_phase_is_part_map_coeffs` is true then both `input_phase_file` and `input_part_map_coeffs_file` will be the same file (whichever is specified)

- `input_phase_file = None` MTZ data file with FC PHIC or equivalent to use for finding heavy-atom sites with difference Fourier methods. NOTE: compare with `input_part_map_coeffs_file` which is a file with map coefficients for use with Phaser completion to find anomalously-scattering atoms. NOTE 2: You can supply both `input_phase_file` and `input_part_map_coeffs_file`. NOTE 3: if `input_phase_is_part_map_coeffs` is true then both `input_part_map_coeffs_file` or phases from `partpdb_file` will be copied to `input_phase_file`

- `input_phase_labels = None` Labels for FC and PHIC for data file with FC PHIC or equivalent to use for finding heavy-atom sites with difference Fourier methods.

- `input_refinement_file = None` Data file to use for refinement. The data in this file should not be corrected for anisotropy. It will be combined with experimental phase information for refinement. If you leave this blank, then the output of phasing will be used in refinement (see below). If no anisotropy correction is applied to the data you do not need to specify a datafile for refinement. If an anisotropy correction is applied to the data files, then you must enter a datafile for refinement if you want to refine your model. (See `"remove_aniso"` for specifying whether an anisotropy correction is applied. In most cases it is not.) If an anisotropy correction is applied and no refinement datafile is supplied, then no refinement will be carried out in the model-building step. You can choose any of your datafiles to be the refinement file, or a native that is not part of the datasets for structure solution. If there are more than one dataset you will be asked each time for a refinement file, but only the last one will be used. Any standard format is

fine; normally only F and sigF will be used. Bijvoet pairs and duplicates will be averaged. If an mtz file is provided then a free R flag can be read in as well. If you do not provide a refinement file then the structure factors from the phasing step will be used in refinement. This is normally satisfactory for SAD data and MIR data. For MAD data you may wish to supply a refinement file because the structure factors from phasing are a combination of data from different wavelengths of data. It is better if you choose your best wavelength of data for refinement.

- `input_refinement_labels` = None Labels for input refinement file columns (FP SIGFP FreeR_flag)
- `input_seq_file` = Auto Normally not used. Use instead `"seq_file"`;
- `refine_eff_file_list` = None You can enter any number of refinement parameter files. These are normally used to tell phenix.refine defaults to apply, as well as creating specialized definitions such as unusual amino acid residues and linkages. These parameters override the normal phenix.refine defaults. They themselves can be overridden by parameters set by the Wizard and by you, controlling the Wizard. NOTE: Any parameters set by AutoBuild directly (such as `number_of_macro_cycles`, `high_resolution`, etc...) will not be taken from this parameters file. This is useful only for adding extra parameters not normally set by AutoBuild.
- `wavelength` Enter a SAD or MAD dataset by filling in information for one or more wavelengths. You can cut and paste an entire wavelength section and enter as many as you like. If you have multiple datasets (i.e., MIR+MAD) then group them using the `"group"` keyword. `wavelength_name` = peak inf high low remote Optionally indicate if this is the peak, inflection point, high energy remote or low energy remote or remote `data` = None Datafile for this wavelength. `labels` = None Specification string for data labels for peak wavelength. Only necessary if the wizard does not automatically choose the correct set of data from your file To find out what the appropriate strings are, type `"phenix.autosol display_labels=your-datafile-here.mtz"`; `atom_type` = None Anomalous-scattering atom type. You only need to specify this for one of the wavelengths in MAD datasets. NOTE: if you want Phaser to add additional heavy-atoms of other types, you can specify them with `mad_ha_add_list`. `lambda` = None wavelength (Å). If you supply an `atom_type` and `lambda` then if you do not supply `f_prime` and `f_double_prime` a guess will be made for them from a table. `res_hyss` = None resolution for running HYSS for this wavelength/deriv `res_eval` = None resolution for evaluation of solutions for this wavelength/deriv `f_prime` = None F-prime value for this wavelength. It is best to supply it if you know it. `f_double_prime` = None F-double_prime value for this wavelength. It is best to supply it if you know it. `sites` = None Number of anomalously-scattering sites for this wavelength You only need to specify this for one wavelength. If you have only MAD data you can also just specify `"sites=2"`; `sites_file` = None PDB or plain-text file with heavy-atom sites. The sites will be taken from this file if supplied. `scale` = None derivative scale factor `group` = 1 Phasing group(s) this wavelength is associated with (Relevant in cases where you have 2 MAD datasets or MAD+SAD or MAD+MIR etc...) `added_wavelength` = False Used internally to flag if this wavelength was added automatically `ignore` = False Ignore this wavelength of data
- `wavelength_name` = peak inf high low remote Optionally indicate if this is the peak, inflection point, high energy remote or low energy remote or remote
- `data` = None Datafile for this wavelength.
- `labels` = None Specification string for data labels for peak wavelength. Only necessary if the wizard does not automatically choose the correct set of data from your file To find out what the appropriate strings are, type `"phenix.autosol display_labels=your-datafile-here.mtz"`;
- `atom_type` = None Anomalous-scattering atom type. You only need to specify this for one of the wavelengths in MAD datasets. NOTE: if you want Phaser to add additional heavy-atoms of other types, you can specify them with `mad_ha_add_list`.

- `lambda = None` wavelength (Å). If you supply an `atom_type` and `lambda` then if you do not supply `f_prime` and `f_double_prime` a guess will be made for them from a table.
- `res_hyss = None` resolution for running HYSS for this wavelength/deriv
- `res_eval = None` resolution for evaluation of solutions for this wavelength/deriv
- `f_prime = None` F-prime value for this wavelength. It is best to supply it if you know it.
- `f_double_prime = None` F-double_prime value for this wavelength. It is best to supply it if you know it.
- `sites = None` Number of anomalously-scattering sites for this wavelength You only need to specify this for one wavelength. If you have only MAD data you can also just specify `"sites=2"`;
- `sites_file = None` PDB or plain-text file with heavy-atom sites. The sites will be taken from this file if supplied
- `derscale = None` derivative scale factor
- `group = 1` Phasing group(s) this wavelength is associated with (Relevant in cases where you have 2 MAD datasets or MAD+SAD or MAD+MIR etc...)
- `added_wavelength = False` Used internally to flag if this wavelength was added automatically
- `ignore = False` Ignore this wavelength of data
- `native` Enter an MIR or SIR dataset by filling in information for a native and one or more derivatives. You can cut and paste these sections and enter as many as you like. If you have multiple datasets (i.e., MIR+MAD) then group them using the `"group"` keyword.`data = None` Datafile for native
`labels = None` Specification string for data labels for native. Only necessary if the wizard does not automatically choose the correct set of data from your file To find out what the appropriate strings are, type `"phenix.autosol display_labels=your-datafile-here.mtz"`
`lambda = None` wavelength (Å) (Not used, for your reference only).`group = 1` Phasing group(s) this native is associated with (Relevant in cases where you have more than one group of native+derivs or you have MIR + MAD or SAD)`added_native = False` Used internally to flag if this native was added automatically
`ignore = False` Ignore this native data
- `data = None` Datafile for native
- `labels = None` Specification string for data labels for native. Only necessary if the wizard does not automatically choose the correct set of data from your file To find out what the appropriate strings are, type `"phenix.autosol display_labels=your-datafile-here.mtz"`;
- `lambda = None` wavelength (Å) (Not used, for your reference only).
- `group = 1` Phasing group(s) this native is associated with (Relevant in cases where you have more than one group of native+derivs or you have MIR + MAD or SAD)
- `added_native = False` Used internally to flag if this native was added automatically
- `ignore = False` Ignore this native data
- `deriv` Enter an MIR or SIR dataset by filling in information for a native and one or more derivatives. You can cut and paste these sections and enter as many as you like. If you have multiple datasets (i.e., MIR+MAD) then group them using the `"group"` keyword.`data = None` Datafile for this derivative
`labels = None` Specification string for data labels for deriv. Only necessary if the wizard does not automatically choose the correct set of data from your file To find out what the appropriate strings are, type `"phenix.autosol display_labels=datafile.mtz"`
`atom_type = None` Heavy-atom type for deriv . `sites = None` Number of heavy-atom sites for deriv . `sites_file = None` PDB or plain-text file with heavy-atom sites. The sites will be taken from this file if supplied
`res_hyss = None` resolution for running HYSS for this wavelength/deriv
`res_eval = None` resolution for evaluation of solutions for this

wavelength/derivinano = noinano *inano anoonly Use anomalous differences for deriv . noinano means do not use anomalous differences. inano means use anomalous differences and isomorphous differences. anoonly means use anomalous differences and not iso differences.f_prime = None F-prime value for this derivative. f_double_prime = None F-double_prime value for this derivative. lambda = None wavelength (A). Used with atom_type to calculate f_prime and f_double_prime if they are not suppliedderscale = None derivative scale factorgroup = 1 Phasing group(s) this derivative is associated with (Relevant in cases where you have more than one group of native+derivs or you have MIR + MAD or SAD)added_deriv = False Used internally to flag if this derivative was added automaticallyignore = False Ignore this deriv data

- data = None Datafile for this derivative
- labels = None Specification string for data labels for deriv. Only necessary if the wizard does not automatically choose the correct set of data from your file To find out what the appropriate strings are, type "phenix.autosol display_labels=datafile.mtz "
- atom_type = None Heavy-atom type for deriv .
- sites = None Number of heavy-atom sites for deriv .
- sites_file = None PDB or plain-text file with heavy-atom sites. The sites will be taken from this file if supplied
- res_hyss = None resolution for running HYSS for this wavelength/deriv
- res_eval = None resolution for evaluation of solutions for this wavelength/deriv
- inano = noinano *inano anoonly Use anomalous differences for deriv . noinano means do not use anomalous differences. inano means use anomalous differences and isomorphous differences. anoonly means use anomalous differences and not iso differences.
- f_prime = None F-prime value for this derivative.
- f_double_prime = None F-double_prime value for this derivative.
- lambda = None wavelength (A). Used with atom_type to calculate f_prime and f_double_prime if they are not supplied
- derscale = None derivative scale factor
- group = 1 Phasing group(s) this derivative is associated with (Relevant in cases where you have more than one group of native+derivs or you have MIR + MAD or SAD)
- added_deriv = False Used internally to flag if this derivative was added automatically
- ignore = False Ignore this deriv data
- decision_makingalways_include_peak = True Choose True to add PEAK dataset on for HYSS if not automatically chosenadd_extra_if_fa = True Choose True to try an extra file for HYSS if FA values are used. This may be useful to solve cases where FA values are poor but their sigmas are small. If True then the anomalous differences will be used for HYSS as well. create_scoring_table = None Choose whether you want a scoring table for solutions A scoring table is slower but better desired_coverage = None Choose what probability you want to have that the correct solution is in your current list of top solutions. A good value is 0.80. If you set a low value (0.01) then only one solution will be kept at any time; if you set a high value, then many solutions will be kept (and it will take longer).
- skip_hyss_other_derivs_if_quick = True You can skip HYSS and instead use difference Fourier's only to find sites for all other derivatives once a good HYSS solution is found. Only used if thoroughness=quick
- self_diff_fourier = True Choose whether, in cases where there are multiple derivatives or multiple datasets, you want to use difference Fourier analysis on the same derivative(s) used in phasing (True), or

instead (False) only phasing other derivatives `combine_siblings = True` You can specify that in MIR or multiple-dataset solutions the solutions to combine must all be ultimately derived by difference fourier from the same parent. Compare with `combine_same_parent_only` where any solutions must have the same immediate parent (unless one is a composite solution). `max_cc_extra_unique_solutions = 0.5` Specify the maximum value of CC between experimental maps for two solutions to consider them substantially different. Solutions that are within the range for consideration based on `desired_coverage`, but are outside of the number of allowed `max_choices`, will be considered, up to `max_extra_unique_solutions`, if they have a correlation of no more than `max_cc_extra_unique_solutions` with all other solutions to be tested. `max_choices = None` Number of choices for solutions to consider. Set automatically with quick: 1 and thorough: 3 `max_composite_choices = 8` Number of choices for composite solutions to consider `max_extra_unique_solutions = None` Specify the maximum number of solutions to consider based on their uniqueness as well as their high scores. Solutions that are within the range for consideration based on `desired_coverage`, but are outside of the number of allowed `max_choices`, will be considered, up to `max_extra_unique_solutions`, if they have a correlation of no more than `max_cc_extra_unique_solutions` with all other solutions to be tested. Set automatically with quick: 0 ; thorough: 2 `max_range_to_keep = 4` The range of solutions to be kept is `range_to_keep * SD` of the group of solutions. This sets the maximum of `range_to_keep` `min_fom = 0.05` Minimum fom of a solution to keep it at all `low_fom = 0.20` If best FOM is less than `low_fom`, double `range_to_keep` `minimum_merge_cc = 0.25` Minimum ratio of CC of solutions to expected in merge_mir keep at all `min_fom_for_dm = 0` Minimum fom of a solution to density modify (otherwise just copy over phases). This is useful in cases where the phasing is so weak that density modification does nothing or makes the phases worse. `extreme_dm = False` Turns on extreme density modification if True or if Auto and FOM is up to `fom_for_extreme_dm`. Use `extreme_dm=True` if your phasing is really weak and density modification is not working well. NOTE: if left default, `extreme_dm` is set to Auto if quick=False. `fom_for_extreme_dm = 0.35` If `extreme_dm` is on and FOM of phasing is up to `fom_for_extreme_dm` then defaults for density modification become: `mask_type=wang` `wang_radius=20` `mask_cycles=1` `minor_cycles=4` `minimum_ncs_cc = 0.30` Minimum NCS correlation to keep, except in case of `extreme_dm` `min_phased_each_deriv = 1` You can require that the wizard phase at least this number of solutions from each derivative, even if they are poor solutions. Usually at...

- `always_include_peak = True` Choose True to add PEAK dataset on for HYSS if not automatically chosen
- `add_extra_if_fa = True` Choose True to try an extra file for HYSS if FA values are used. This may be useful to solve cases where FA values are poor but their sigmas are small. If True then the anomalous differences will be used for HYSS as well.
- `create_scoring_table = None` Choose whether you want a scoring table for solutions A scoring table is slower but better
- `desired_coverage = None` Choose what probability you want to have that the correct solution is in your current list of top solutions. A good value is 0.80. If you set a low value (0.01) then only one solution will be kept at any time; if you set a high value, then many solutions will be kept (and it will take longer).
- `skip_hyss_other_derivs_if_quick = True` You can skip HYSS and instead use difference Fouriers only to find sites for all other derivatives once a good HYSS solution is found. Only used if thoroughness=quick .
- `self_diff_fourier = True` Choose whether, in cases where there are multiple derivatives or multiple datasets, you want to use difference Fourier analysis on the same derivative(s) used in phasing (True), or instead (False) only phasing other derivatives
- `combine_siblings = True` You can specify that in MIR or multiple-dataset solutions the solutions to combine must all be ultimately derived by difference fourier from the same parent. Compare with

combine_same_parent_only where any solutions must have the same immediate parent (unless one is a composite solution).

- max_cc_extra_unique_solutions = 0.5 Specify the maximum value of CC between experimental maps for two solutions to consider them substantially different. Solutions that are within the range for consideration based on desired_coverage, but are outside of the number of allowed max_choices, will be considered, up to max_extra_unique_solutions, if they have a correlation of no more than max_cc_extra_unique_solutions with all other solutions to be tested.
- max_choices = None Number of choices for solutions to consider. Set automatically with quick: 1 and thorough:3
- max_composite_choices = 8 Number of choices for composite solutions to consider
- max_extra_unique_solutions = None Specify the maximum number of solutions to consider based on their uniqueness as well as their high scores. Solutions that are within the range for consideration based on desired_coverage, but are outside of the number of allowed max_choices, will be considered, up to max_extra_unique_solutions, if they have a correlation of no more than max_cc_extra_unique_solutions with all other solutions to be tested. Set automatically with quick:0 ; thorough:2
- max_range_to_keep = 4 The range of solutions to be kept is range_to_keep * SD of the group of solutions. This sets the maximum of range_to_keep
- min_fom = 0.05 Minimum fom of a solution to keep it at all
- low_fom = 0.20 If best FOM is less than low_fom, double range_to_keep
- minimum_merge_cc = 0.25 Minimum ratio of CC of solutions to expected in merge_mir keep at all
- min_fom_for_dm = 0 Minimum fom of a solution to density modify (otherwise just copy over phases). This is useful in cases where the phasing is so weak that density modification does nothing or makes the phases worse.
- extreme_dm = False Turns on extreme density modification if True or if Auto and FOM is up to fom_for_extreme_dm. Use extreme_dm=True if your phasing is really weak and density modification is not working well. NOTE: if left default, extreme_dm is set to Auto if quick=False.
- fom_for_extreme_dm = 0.35 If extreme_dm is on and FOM of phasing is up to fom_for_extreme_dm then defaults for density modification become: mask_type=wang wang_radius=20 mask_cycles=1 minor_cycles=4
- minimum_ncs_cc = 0.30 Minimum NCS correlation to keep, except in case of extreme_dm
- min_phased_each_deriv = 1 You can require that the wizard phase at least this number of solutions from each derivative, even if they are poor solutions. Usually at least 1 is a good idea so that one derivative does not dominate the solutions.
- n_random = 6 Number of random solutions to generate when setting up scoring table
- res_eval = 0 Resolution for running resolve evaluation (usually 2.5 Å) It will be set automatically if you do not set it
- score_individual_offset_list = None Offsets for individual scores in CC-scoring. Each score will be multiplied by the score_individual_scale_list value, then score_individual_offset_list value is added, to estimate the CC**2 value using this score by itself. The uncertainty in the CC**2 value is given by score_individual_sd_list. NOTE: These scores are not used in calculation of the overall score. They are for information only
- score_individual_scale_list = None Scale factors for individual scores in CC-scoring. Each score will be multiplied by the score_individual_scale_list value, then score_individual_offset_list value is added, to

estimate the CC**2 value using this score by itself. The uncertainty in the CC**2 value is given by score_individual_sd_list. NOTE: These scores are not used in calculation of the overall score. They are for information only

- score_individual_sd_list = None Uncertainties for individual scores in CC-scoring. Each score will be multiplied by the score_individual_scale_list value, then score_individual_offset_list value is added, to estimate the CC**2 value using this score by itself. The uncertainty in the CC**2 value is given by score_individual_sd_list. NOTE: These scores are not used in calculation of the overall score. They are for information only
- score_overall_offset = None Overall offset for scores in CC-scoring. The weighted scores will be summed, then all multiplied by score_overall_scale, then score_overall_offset will be added.
- score_overall_scale = None Overall scale factor for scores in CC-scoring. The weighted scores will be summed, then all multiplied by score_overall_scale, then score_overall_offset will be added.
- score_overall_sd = None Overall SD of CC**2 estimate for scores in CC-scoring. The weighted scores will be summed, then all multiplied by score_overall_scale, then score_overall_offset will be added. This is an estimate of CC**2, with uncertainty about score_overall_sd. Then the square root is taken to estimate CC and SD(CC), where SD(CC) now depends on CC due to the square root.
- score_type_list = SKEW CORR_RMS You can choose what scoring methods to include in scoring of solutions in AutoSol. (The choices available are: CC_DENMOD RFACTOR SKEW NCS_COPIES NCS_IN_GROUP TRUNCATE FLATNESS CORR_RMS REGIONS CONTRAST FOM) NOTE: If you are using Z-SCORE or BAYES-CC scoring, The default is CC_RMS RFACTOR SKEW FOM (and NCS_OVERLAP if ncs_copies is at least equal to ncs_copies_min_for_overlap.
- score_weight_list = None Weights on scores for CC-scoring. Enter the weight on each score in score_type_list. The weighted scores will be summed, then all multiplied by score_overall_scale, then score_overall_offset will be added.
- skip_score_list = None You can evaluate some scores but not use them. Include the ones you do not want to use in the final score in skip_score_list.
- ncs_copies_min_for_overlap = 2 Minimum number of ncs copies (set automatically from composition and cell or with ncs_copies=xx) to use NCS_OVERLAP in scoring
- rho_overlap_min = 0.3 Sets minimum average overlap of NCS-related density to keep NCS. Cutoff of overlap will be rho_overlap_min for 2 ncs copies, and proportionally smaller ($\text{rho_overlap_min} \times 2/N$) for N ncs copies.
- rho_overlap_min_scoring = 0.5 Once NCS is found, rho_overlap_min_scoring sets threshold for whether the NCS is used in scoring. Cutoff of overlap will be rho_overlap_min_scoring for 2 ncs copies, and proportionally smaller ($\text{rho_overlap_min_scoring} \times 2/N$) for N ncs copies. (Compare with rho_overlap_min, which sets cutoff for finding NCS, not scoring with it)
- hyss_scoringmodel_ha_iteration = None You can iterate heavy-atom location using a model in the process. This takes a lot longer than simple ha_iteration but can be more powerful for finding weak sites. NOTE: only available for Phaser SAD phasing. Default is to use it if extreme_dm applies. Note: not available on Windows. acceptable_r = 0.35 Used to decide whether the model is acceptable enough to not do heavy-atom iteration. A good value is 0.35 max_model_ha_iterations = None Max number of iterations of model-building and finding heavy-atom sites. max_models_in_ha_iteration = None Max number of models to include in iterations of model-building and finding heavy-atom sites. ha_iteration = None Choose whether you want to iterate the heavy-atom search. With iteration, sites are found with HYSS, then used to phase and carry out quick density-modification, then either Phaser LLG completion (for SAD data) or difference Fourier or anomalous difference Fourier analysis is used to find sites. Default

is to use `ha_iteration` if `extreme_dm` applies. Note: not available on Windows. `max_ha_iterations = None` Number of iterations of phasing/density modification searching for heavy-atom sites.

`minimum_improvement = 0` Minimum improvement in score to continue ha iteration

- `model_ha_iteration = None` You can iterate heavy-atom location using a model in the process. This takes a lot longer than simple `ha_iteration` but can be more powerful for finding weak sites. NOTE: only available for Phaser SAD phasing. Default is to use it if `extreme_dm` applies. Note: not available on Windows.
- `acceptable_r = 0.35` Used to decide whether the model is acceptable enough to not do heavy-atom iteration. A good value is 0.35
- `max_model_ha_iterations = None` Max number of iterations of model-building and finding heavy-atom sites.
- `max_models_in_ha_iteration = None` Max number of models to include in iterations of model-building and finding heavy-atom sites.
- `ha_iteration = None` Choose whether you want to iterate the heavy-atom search. With iteration, sites are found with HYSS, then used to phase and carry out quick density-modification, then either Phaser LLG completion (for SAD data) or difference Fourier or anomalous difference Fourier analysis is used to find sites. Default is to use `ha_iteration` if `extreme_dm` applies. Note: not available on Windows.
- `max_ha_iterations = None` Number of iterations of phasing/density modification searching for heavy-atom sites.
- `minimum_improvement = 0` Minimum improvement in score to continue ha iteration
- `build_scoringoverall_score_method = *BAYES-CC Z-SCORE` You have 2 choices for an overall scoring method: (1) Sum of individual Z-scores (Z-SCORE) (2) Bayesian estimate of CC of map to perfect model (BAYES-CC) You can specify which scoring criteria to include with `score_type_list` (default is SKEW CORR_RMS for BAYES-CC and CC RFACTOR SKEW FOM for Z-SCORE. Additionally, if NCS is present, NCS_OVERLAP is used by default in the Z-SCORE method). `r_switch = 0.4` R-value criteria for deciding whether to use R-value or residues built. A good value is 0.40 `acceptable_quality = 40` You can specify the minimum overall quality of a model (as defined by `overall_score_method`) to be considered acceptable `acceptable_secondary_structure_cc = 0.35` You can specify the minimum correlation of density from a secondary structure model to be considered acceptable `trace_chain = False` You can build a CA-only model right after density modification using `trace_chain` `trace_chain_score = False` You can score density-modified maps with the number of residues built with regular secondary-structure using `trace_chain.include_dm_score = None` You can score density-modified maps with the R-factor Normally only included if `map_model_cc` is not clear and best dm R is less than `max_useful_dm_ruse_map_model_cc = None` You can score models with CC to map Only included if CC is higher then `quality_cc_min` `quality_cc_min = 0.50` Minimum map-model CC value to use CC in scoring `max_useful_dm_r = 0.40` Maximum R-value in density modification to use in `dm_score`
- `overall_score_method = *BAYES-CC Z-SCORE` You have 2 choices for an overall scoring method: (1) Sum of individual Z-scores (Z-SCORE) (2) Bayesian estimate of CC of map to perfect model (BAYES-CC) You can specify which scoring criteria to include with `score_type_list` (default is SKEW CORR_RMS for BAYES-CC and CC RFACTOR SKEW FOM for Z-SCORE. Additionally, if NCS is present, NCS_OVERLAP is used by default in the Z-SCORE method).
- `r_switch = 0.4` R-value criteria for deciding whether to use R-value or residues built. A good value is 0.40
- `acceptable_quality = 40` You can specify the minimum overall quality of a model (as defined by `overall_score_method`) to be considered acceptable

- `acceptable_secondary_structure_cc = 0.35` You can specify the minimum correlation of density from a secondary structure model to be considered acceptable
- `trace_chain = False` You can build a CA-only model right after density modification using `trace_chain`
- `trace_chain_score = False` You can score density-modified maps with the number of residues built with regular secondary-structure using `trace_chain`.
- `include_dm_score = None` You can score density-modified maps with the R-factor Normally only included if `map_model_cc` is not clear and best dm R is less than `max_useful_dm_r`
- `use_map_model_cc = None` You can score models with CC to map Only included if CC is higher then `quality_cc_min`
- `quality_cc_min = 0.50` Minimum map-model CC value to use CC in scoring
- `max_useful_dm_r = 0.40` Maximum R-value in density modification to use in `dm_score`
- `dev_scoringrandom_scoring = False` For testing purposes you can generate random scores `use_perfect = False` You can use the CC between each solution and `hklperfect` in scoring. This is only for methods development purposes.`hklperfect = None` You can supply an mtz file with idealized coefficients for a map. This will be compared with all maps calculated during structure solution `perfect_labels = None` Labels for input data columns for `hklperfect` if present. Typical value: `"FP PHIC FOM"`;
- `random_scoring = False` For testing purposes you can generate random scores
- `use_perfect = False` You can use the CC between each solution and `hklperfect` in scoring. This is only for methods development purposes.
- `hklperfect = None` You can supply an mtz file with idealized coefficients for a map. This will be compared with all maps calculated during structure solution
- `perfect_labels = None` Labels for input data columns for `hklperfect` if present. Typical value: `"FP PHIC FOM"`;
- `scalingremove_aniso = Auto` *True False Choose if you want to apply a correction for anisotropy to the data. True means always apply correction, No means never apply it, Auto means apply it if the data is severely anisotropic (recommended=True). If you set `remove_aniso=Auto` then if the range of anisotropic B-factors is greater than `delta_b_for_auto_remove_aniso` and the ratio of the largest to the smallest less than `ratio_b_for_auto_remove_aniso` then the correction will be applied. Anisotropy correction will be applied to all input data before scaling. If used, the default overall target B factor is is minimum of (`max_b_iso`, lowest B of datasets, `target_b_ratio*resolution`)`b_iso = None` Target overall B value for anisotropy correction. Ignored if `remove_aniso = False`. If None, default is minimum of (`max_b_iso`, lowest B of datasets, `target_b_ratio*resolution`)`max_b_iso = 40`. Default maximum overall B value for anisotropy correction. Ignored if `remove_aniso = False`. Ignored if `b_iso` is set. If used, default is minimum of (`max_b_iso`, lowest B of datasets, `target_b_ratio*resolution`)`target_b_ratio = 10`. Default ratio of target B value to resolution for anisotropy correction. Ignored if `remove_aniso = False`. Ignored if `b_iso` is set. If used, default is minimum of (`max_b_iso`, lowest B of datasets, `target_b_ratio*resolution`)`try_orig_sad_data_in_hyss = Auto` You can try original (unscaled) data in Hyss in addition to scaled data. Applies to SAD datasets only. Default is False if `quick=True`.`localscale_before Phaser = True` You can apply SOLVE localscaling to SAD data before passing it to Phaser for SAD phasing`delta_b_for_auto_remove_aniso = 20` Choose what range of aniso B values is so big that you want to correct for anisotropy by default. Both `ratio_b` and `delta_b` must be large to correct. See also `ratio_b_for_auto_remove_aniso`. See also `"remove_aniso"` which overrides this default if set to `"True"`;`ratio_b_for_auto_remove_aniso = 1.0` Choose what ratio aniso B values is so big that you want to correct for anisotropy by default. Both `ratio_b` and `delta_b` must be large to correct. see also `delta_b_for_auto_remove_aniso` See also `"remove_aniso"` which

overrides this default if set to "True"; test_remove_aniso = True Choose whether you want to try applying or not applying an anisotropy correction if the run fails. First your original selection for applying or not will be tried, and then the opposite will be tried if the run fails. use_sca_as_is = True Choose True to allow use of sca files (and mtz files) without conversion even if the space group is changed. If False, then original index files will always be converted to premerged if the space group is changed. derscale_list = None List of deriv scale factors. Not normally used. Use derscale for deriv or wavelength. scale_only = False Just scale data and stop phase_only = False Just scale data, phase and stop

- remove_aniso = Auto *True False Choose if you want to apply a correction for anisotropy to the data. True means always apply correction, No means never apply it, Auto means apply it if the data is severely anisotropic (recommended=True). If you set remove_aniso=Auto then if the range of anisotropic B-factors is greater than delta_b_for_auto_remove_aniso and the ratio of the largest to the smallest less than ratio_b_for_auto_remove_aniso then the correction will be applied. Anisotropy correction will be applied to all input data before scaling. If used, the default overall target B factor is minimum of (max_b_iso, lowest B of datasets, target_b_ratio*resolution)
- b_iso = None Target overall B value for anisotropy correction. Ignored if remove_aniso = False. If None, default is minimum of (max_b_iso, lowest B of datasets, target_b_ratio*resolution)
- max_b_iso = 40. Default maximum overall B value for anisotropy correction. Ignored if remove_aniso = False. Ignored if b_iso is set. If used, default is minimum of (max_b_iso, lowest B of datasets, target_b_ratio*resolution)
- target_b_ratio = 10. Default ratio of target B value to resolution for anisotropy correction. Ignored if remove_aniso = False. Ignored if b_iso is set. If used, default is minimum of (max_b_iso, lowest B of datasets, target_b_ratio*resolution)
- try_orig_sad_data_in_hyss = Auto You can try original (unscaled) data in Hyss in addition to scaled data. Applies to SAD datasets only. Default is False if quick=True.
- localscale_before Phaser = True You can apply SOLVE local scaling to SAD data before passing it to Phaser for SAD phasing
- delta_b_for_auto_remove_aniso = 20 Choose what range of aniso B values is so big that you want to correct for anisotropy by default. Both ratio_b and delta_b must be large to correct. See also ratio_b_for_auto_remove_aniso. See also "remove_aniso"; which overrides this default if set to "True";
- ratio_b_for_auto_remove_aniso = 1.0 Choose what ratio aniso B values is so big that you want to correct for anisotropy by default. Both ratio_b and delta_b must be large to correct. see also delta_b_for_auto_remove_aniso See also "remove_aniso"; which overrides this default if set to "True";
- test_remove_aniso = True Choose whether you want to try applying or not applying an anisotropy correction if the run fails. First your original selection for applying or not will be tried, and then the opposite will be tried if the run fails.
- use_sca_as_is = True Choose True to allow use of sca files (and mtz files) without conversion even if the space group is changed. If False, then original index files will always be converted to premerged if the space group is changed
- derscale_list = None List of deriv scale factors. Not normally used. Use derscale for deriv or wavelength.
- scale_only = False Just scale data and stop
- phase_only = False Just scale data, phase and stop

- `heavy_atom_searchmin_hyss_cc = 0.05` Minimum CC of a heavy-atom solution in HYSS to keep it at all acceptable.
`acceptable_cc_hyss = 0.2` Solutions with CC better than `acceptable_cc_hyss` will not be rescored.
`good_cc_hyss = 0.3` Hyss will be run up to `best_of_n_hyss_always` times at a given resolution. If the best CC value is greater than `good_cc_hyss` and the number of sites found is at least `min_fraction_of_sites_found` times the number expected and Hyss was tried at least `best_of_n_hyss` times, then the search is ended. Also if `thoroughness=quick` and a solution with CC at least as high as `good_cc_hyss` is found, no more searches will be done at all.
`n_add_res_max = 2` Hyss will be run up to `n_add_res_max+1` resolutions starting with `res_hyss` and adding increments of `add_res_max/n_add_res_max`. If the best CC value is greater than `good_cc_hyss` then no more resolutions are tried.
`add_res_max = 2` Hyss will be run at up to `n_add_res_max+1` resolutions starting with `res_hyss` and adding increments of `add_res_max/n_add_res_max`. If the best CC value is greater than `good_cc_hyss` then no more resolutions are tried.
`try_recommended_resolution_for_hyss = True` If yes, then hyss will be run at recommended_resolution based on anomalous signal in addition to default resolution if CC at default resolution is less than `good_cc_hyss` and recommended_resolution is more than 0.1 Å.
`less_than_default_hyss_runs_min = 2` If there are multiple derivatives or candidate wavelengths for HYSS, run at least `hyss_runs_min` of these.
`best_of_n_hyss = 1` Hyss will be run up to `best_of_n_hyss_always` times at a given resolution. If the best CC value is greater than `good_cc_hyss` and the number of sites found is at least `min_fraction_of_sites_found` times the number expected and Hyss was tried at least `best_of_n_hyss` times, then the search is ended if `hyss_runs_min` data files have been attempted.
`best_of_n_hyss_always = 10` Hyss will be run up to `best_of_n_hyss_always` times at a given resolution. If the best CC value is greater than `good_cc_hyss` and the number of sites found is at least `min_fraction_of_sites_found` times the number expected and Hyss was tried at least `best_of_n_hyss` times, then the search is ended if `hyss_runs_min` data files have been attempted.
`min_fraction_of_sites_found = 0.667` Hyss will be run up to `best_of_n_hyss_always` times at a given resolution. If the best CC value is greater than `good_cc_hyss` and the number of sites found is at least `min_fraction_of_sites_found` times the number expected and Hyss was tried at least `best_of_n_hyss` times, then the search is ended if `hyss_runs_min` data files have been attempted.
`max_single_sites = 5` In `sites_from_denmod` a core set of sites that are strong is identified. If the hand of the solution is known then additional sites are added all at once up to the expected number of sites. Otherwise sites are added one at a time, up to a maximum number of tries of `max_single_sites`.
`hyss_enable_early_termination = True` You can specify whether to stop HYSS as soon as it finds a convincing solution (True, default) or to keep trying...
`hyss_general_positions_only = True` Select True if you want HYSS only to consider general positions and ignore sites on special positions. This is appropriate for SeMet or S-Met solutions, not so appropriate for heavy-atom soaks.
`hyss_min_distance = 3.5` Enter the minimum distance between heavy-atom sites to keep them in HYSS.
`hyss_n_fragments = 3` Enter the number of fragments in HYSS.
`hyss_n_patterson_vectors = 33` Enter the number of Patterson vectors to consider in HYSS.
`hyss_random_seed = 792341` Enter an integer as random seed for HYSS.
`res_hyss = None` Overall resolution for running HYSS (usually default is fine).
`try_low_res_for_cys = True` Use low-resolution hyss (4.5 Å) if number of cys is greater than 2x the number of met and the anomalously-scattering atom is S.
`direct_methods_only = False` Use only direct methods (no phaser completion) in Hyss (applies to SAD/MAD data only).
`use_measurability = True` Use measurability (fro...

- `min_hyss_cc = 0.05` Minimum CC of a heavy-atom solution in HYSS to keep it at all

- `acceptable_cc_hyss = 0.2` Solutions with CC better than `acceptable_cc_hyss` will not be rescored.

- `good_cc_hyss = 0.3` Hyss will be run up to `best_of_n_hyss_always` times at a given resolution. If the best CC value is greater than `good_cc_hyss` and the number of sites found is at least `min_fraction_of_sites_found` times the number expected and Hyss was tried at least `best_of_n_hyss` times, then the search is ended. Also if `thoroughness=quick` and a solution with CC at least as high as

good_cc_hyss is found, no more searches will be done at all

- `n_add_res_max = 2` Hyss will be run at up to `n_add_res_max+1` resolutions starting with `res_hyss` and adding increments of `add_res_max/n_add_res_max`. If the best CC value is greater than `good_cc_hyss` then no more resolutions are tried.
- `add_res_max = 2` Hyss will be run at up to `n_add_res_max+1` resolutions starting with `res_hyss` and adding increments of `add_res_max/n_add_res_max`. If the best CC value is greater than `good_cc_hyss` then no more resolutions are tried.
- `try_recommended_resolution_for_hyss = True` If yes, then hyss will be run at `recommended_resolution` based on anomalous signal in addition to default resolution if CC at default resolution is less than `good_cc_hyss` and `recommended_resolution` is more than 0.1 Å less than default
- `hyss_runs_min = 2` If there are multiple derivatives or candidate wavelengths for HYSS, run at least `hyss_runs_min` of these.
- `best_of_n_hyss = 1` Hyss will be run up to `best_of_n_hyss_always` times at a given resolution. If the best CC value is greater than `good_cc_hyss` and the number of sites found is at least `min_fraction_of_sites_found` times the number expected and Hyss was tried at least `best_of_n_hyss` times, then the search is ended if `hyss_runs_min` data files have been attempted.
- `best_of_n_hyss_always = 10` Hyss will be run up to `best_of_n_hyss_always` times at a given resolution. If the best CC value is greater than `good_cc_hyss` and the number of sites found is at least `min_fraction_of_sites_found` times the number expected and Hyss was tried at least `best_of_n_hyss` times, then the search is ended if `hyss_runs_min` data files have been attempted.
- `min_fraction_of_sites_found = 0.667` Hyss will be run up to `best_of_n_hyss_always` times at a given resolution. If the best CC value is greater than `good_cc_hyss` and the number of sites found is at least `min_fraction_of_sites_found` times the number expected and Hyss was tried at least `best_of_n_hyss` times, then the search is ended if `hyss_runs_min` data files have been attempted.
- `max_single_sites = 5` In `sites_from_denmod` a core set of sites that are strong is identified. If the hand of the solution is known then additional sites are added all at once up to the expected number of sites. Otherwise sites are added one at a time, up to a maximum number of tries of `max_single_sites`
- `hyss_enable_early_termination = True` You can specify whether to stop HYSS as soon as it finds a convincing solution (True, default) or to keep trying...
- `hyss_general_positions_only = True` Select True if you want HYSS only to consider general positions and ignore sites on special positions. This is appropriate for SeMet or S-Met solutions, not so appropriate for heavy-atom soaks
- `hyss_min_distance = 3.5` Enter the minimum distance between heavy-atom sites to keep them in HYSS
- `hyss_n_fragments = 3` Enter the number of fragments in HYSS
- `hyss_n_patterson_vectors = 33` Enter the number of Patterson vectors to consider in HYSS
- `hyss_random_seed = 792341` Enter an integer as random seed for HYSS
- `res_hyss = None` Overall resolution for running HYSS (usually default is fine)
- `try_low_res_for_cys = True` Use low-resolution hyss (4.5 Å) if number of cys is greater than 2x the number of met and the anomalously-scattering atom is S
- `direct_methods_only = False` Use only direct methods (no phaser completion) in Hyss (applies to SAD/MAD data only)
- `use_measurability = True` Use measurability (from xtriage) to estimate recommended resolution for HYSS and for initial phasing. Only applies to MAD/SAD phasing. Alternative is to use signal-to-noise from

Solve scaling.

- `use Phaser rescoring = False` Run Phaser rescoring for HYSS heavy-atom search (only SAD data) if initial try fails
- `use Automatic hyss = True` Use automated hyss with direct methods/Phaser rescoring combination for SAD/MAD data
- `mad_ha_n = None` Normally not used. Use instead `"sites"` for a wavelength. Number of anomalously-scattering atoms in the au
- `mad_ha_type = "Se"` Normally not used. Use instead `"atom_type"` for a wavelength. Anomalously-scattering or heavy atom type. For example, Se or Au. NOTE: if you want Phaser to add additional heavy-atoms of other types, you can specify them with `mad_ha_add_list`.
- `phasingkeep_atom_types = False` Allows you to keep the atom types in your `sites_file` `madbst = True` Choose whether you want to carry out FA calculation Skipping it speeds up MAD phasing but may reduce the ability to find the sites with HYSS `overallscale = False` You can choose to have only an overall scale factor for this dataset (no local scaling applied). Use this if your data is already fully scaled. `res_phase = 0` Enter the high-resolution limit for phasing (0= use all) `phase_full_resolution = True` You can choose to use the full resolution of the data in phasing, instead of using the `recommended_resolution`. This is always a good idea with Phaser phases. `fixscattfactors = None` For SOLVE phasing and MAD data you can choose whether scattering factors are to be fixed by choosing True to fix them or False to refine them. Normally choose True (fix) if the data are weak and False (refine) if the data are strong. `fixscattfactors_in_phasing = False` Fix scattering factors in phasing step. For SOLVE phasing and MAD data you can choose whether scattering factors are to be fixed by choosing True to fix them or False to refine them. Normally False. This command only applies to the phasing step and not initial heavy-atom refinement. It does not apply to Phaser SAD phasing. `fix_xyz_in_phasing = None` Fix coordinates in phasing step. For SOLVE phasing and MAD data you can choose whether ha coordinates are to be fixed by choosing True to fix them or False to refine them. May be useful in maintaining the coordinates of the solutions that were tested in initial phasing steps. If None, then it will be set to True if the resolution of final phasing step is higher than the highest resolution of test phasing runs This command only applies to the phasing step and not initial heavy-atom refinement. It does not apply to Phaser SAD `phasinghave_hand = False` Normally you will not know the hand of the heavy-atom substructure, so `have_hand=False`. However if you do know it (you got the sites from a difference Fourier or you know the answer another way) you can specify that the hand is known. `id_scale_ref = None` By default the datafile with the highest resolution is used for the first step in scaling of MAD data. You can choose to use any of the datafiles in your MAD dataset. NOTE: not applicable for multi-dataset analyses `ratio_out = 10`. You can choose the ratio of del ano or del iso to the rms in the shell for rejection of a reflection. Default = 10. `ratmin = 0`. Reflections with $I/\sigma I$ less than `ratmin` will be ignored when read in. `require_nat = True` Choose yes to skip any reflection with no native (for SIR) or no data (MAD/SAD) or where anom difference is very large. This keyword (default=True) allows the routines in SOLVE to remove reflections with an implausibly large anomalous difference (greater than `ratio_out` times the rms anomalous difference). `ikeepflag = 1` You can choose to keep all reflections in merging steps. This is separate from rejecting reflections with high iso or ano diffs. Default=1 (keep them) `phasing_method = SOLVE` *PHASER You can choose to phase with SOLVE or with Phaser. (Only applies to SAD phasing at present) `cluster_pdb_file = None` For SAD only: if you have a cluster compound, normally you should supply a PDB file with an example of the cluster with the keyword `cluster_pdb_file=my_cluster_compound` and a unique residue name XX (yes, XX). Set the keyword `atom_type=XX` as well . If your cluster is Ta6Br12 then you can simply put `atom_type=TX` and skip the `cluster_pdb_file`. For MAD/MIR, cluster compounds are not currently supported. Instead just use a standard atom. `input_partpdb_file = None` You can enter a PDB file (usually from molecular replacement)

for use in identifying heavy-atom sites and phasing. NOTE 1: This procedure works best if the model is refined. NOTE 2: This file is only used in SAD phasing with Phaser on a single dataset. In all other cases it is ignored. NOTE 3: The output phases in phaser_xx.mtz will contain both SAD and mod...

- `keep_atom_types = False` Allows you to keep the atom types in your `sites_file`
- `do_madbst = True` Choose whether you want to carry out FA calculation. Skipping it speeds up MAD phasing but may reduce the ability to find the sites with HYSS
- `overallscale = False` You can choose to have only an overall scale factor for this dataset (no local scaling applied). Use this if your data is already fully scaled.
- `res_phase = 0` Enter the high-resolution limit for phasing (0= use all)
- `phase_full_resolution = True` You can choose to use the full resolution of the data in phasing, instead of using the recommended_resolution. This is always a good idea with Phaser phases.
- `fixscattfactors = None` For SOLVE phasing and MAD data you can choose whether scattering factors are to be fixed by choosing True to fix them or False to refine them. Normally choose True (fix) if the data are weak and False (refine) if the data are strong.
- `fixscattfactors_in_phasing = False` Fix scattering factors in phasing step. For SOLVE phasing and MAD data you can choose whether scattering factors are to be fixed by choosing True to fix them or False to refine them. Normally False. This command only applies to the phasing step and not initial heavy-atom refinement. It does not apply to Phaser SAD phasing.
- `fix_xyz_in_phasing = None` Fix coordinates in phasing step. For SOLVE phasing and MAD data you can choose whether ha coordinates are to be fixed by choosing True to fix them or False to refine them. May be useful in maintaining the coordinates of the solutions that were tested in initial phasing steps. If None, then it will be set to True if the resolution of final phasing step is higher than the highest resolution of test phasing runs. This command only applies to the phasing step and not initial heavy-atom refinement. It does not apply to Phaser SAD phasing
- `have_hand = False` Normally you will not know the hand of the heavy-atom substructure, so `have_hand=False`. However if you do know it (you got the sites from a difference Fourier or you know the answer another way) you can specify that the hand is known.
- `id_scale_ref = None` By default the datafile with the highest resolution is used for the first step in scaling of MAD data. You can choose to use any of the datafiles in your MAD dataset. NOTE: not applicable for multi-dataset analyses
- `ratio_out = 10`. You can choose the ratio of $\Delta I/\sigma$ or $\Delta I/\sigma_{iso}$ to the rms in the shell for rejection of a reflection. Default = 10.
- `ratmin = 0`. Reflections with I/σ less than `ratmin` will be ignored when read in.
- `require_nat = True` Choose yes to skip any reflection with no native (for SIR) or no data (MAD/SAD) or where anom difference is very large. This keyword (default=True) allows the routines in SOLVE to remove reflections with an implausibly large anomalous difference (greater than `ratio_out` times the rms anomalous difference).
- `ikeepflag = 1` You can choose to keep all reflections in merging steps. This is separate from rejecting reflections with high iso or ano diffs. Default=1 (keep them)
- `phasing_method = SOLVE *PHASER` You can choose to phase with SOLVE or with Phaser. (Only applies to SAD phasing at present)
- `cluster_pdb_file = None` For SAD only: if you have a cluster compound, normally you should supply a PDB file with an example of the cluster with the keyword `cluster_pdb_file=my_cluster_compound` and a

unique residue name XX (yes, XX). Set the keyword `atom_type=XX` as well. If your cluster is Ta6Br12 then you can simply put `atom_type=TX` and skip the `cluster_pdb_file`. For MAD/MIR, cluster compounds are not currently supported. Instead just use a standard atom.

- `input_partpdb_file = None` You can enter a PDB file (usually from molecular replacement) for use in identifying heavy-atom sites and phasing. NOTE 1: This procedure works best if the model is refined. NOTE 2: This file is only used in SAD phasing with Phaser on a single dataset. In all other cases it is ignored. NOTE 3: The output phases in `phaser_xx.mtz` will contain both SAD and model information. They are not completely suitable for use with AutoBuild or other iterative model-building procedures because the phases are not entirely experimental (but they may work). NOTE 4: if `input_phase_is_part_map_coeffs` is true then `partpdb_file` will also be used to generate an `input_phase_file` which will be used to find sites as well. NOTE 5: you can choose if this file is used to find sites only and then phase with those sites (`phaser_sites_then_phase=True`) or to phase with the model and also the sites. If you use `phaser_sites_then_phase=True` then you do not need to worry about bias from the model, but the phasing may not be as good. NOTE 6: if `include_partpdb_in_building=True` then this model will be used to start model-building
- `include_partpdb_in_building = True` If an input `partpdb_file` is supplied, use it to start model-building.
- `ncs_from_partpdb = True` If input `partpdb_file` is supplied and `ncs_from_partpdb` is set and `find_ncs` is set, then try to find NCS from the model in `partpdb_file`.
- `input_part_map_ha_fraction = 0.75` Use effective $f_o + f'$ value of `input_part_map_ha_fraction` for real part of anomalously-scattering atoms when `part_map_coeffs` are used. (Lower than 1.0 because the map will have some contribution corresponding to these atoms)
- `phaser_sites_then_phase = False` When using `part_map_coeffs` or `partpdb_file` or iterating heavy-atom searches, find sites with this partial model, then calculate phases without the partial model. This approach removes the bias from the model and keeps the real part of the scattering from the heavy atoms. If you use `phaser_sites_then_phase=True` you can use HLA HLB HLC HLD and not worry about HLANOM values.
- `partpdb_rms = None` Estimate of rmsd of partial PDB file from correct structure. Default is 1A for `partpdb_file` and 1.5 for `part_map_coeffs_file`
- `input_part_map_coeffs_file = None` If you have done MR on density instead of on a model you can enter the output map coefficients file from Phaser MR instead of an input `partpdb_file`. It will be used in the same way as an input `partpdb_file`. These map coefficients must correspond to the placed density. The map coefficients must also be in the same unit cell as the input data. Note that `partpdb_rms` also applies to these map coefficients. It is an estimate of how close this density matches the true model. Note that you can choose if this file is used to find sites only and then phase with those sites (`phaser_sites_then_phase=True`) or to phase with the model and also the sites. If you use `phaser_sites_then_phase=True` then you do not need to worry about bias from the model, but the phasing may not be as good. NOTE: labels in this mtz file must be FC,PHIC unless they are set with `input_part_map_coeffs_labels` Note also that there is a similar keyword: `input_phase_file` that has a different result. The `input_phase_file` keyword uses a difference Fourier to find sites. Also if you use `input_phase_file=xxx` and you use `ha_iteration=True` then during iteration of density modification and finding sites the sites will be found with a difference Fourier. NOTE 2: You can supply both `input_phase_file` and `input_part_map_coeffs_file`. NOTE 3: if `input_phase_is_part_map_coeffs` is true then both `input_part_map_coeffs_file` or phases from `partpdb_file` will be copied to `input_phase_file`
- `input_part_map_coeffs_labels = FC,PHIC` Labels for input `part_map_coeffs`. Normally FC,PHIC.
- `llgc_sigma = None`

- `phaser_completion = True` You can choose to use phaser log-likelihood gradients to complete your heavy-atom sites. This can be used with or without the `ha_iteration` option.
- `use_phaser_hklstart = True` You can choose to start density modification with FWT PHWT from Phaser (Only applies to SAD phasing at present)
- `combine_same_parent_only = False` You can choose to only combine solutions with the same parent (and that have a parent) in MIR, unless one solution is a composite. Compare with `combine_siblings` in which case the solutions do not have to have the same immediate parents, but can be derived from the same ultimate parent through several difference fourier steps.
- `skip_extra_phasing = *Auto True False` You can choose to skip an extra phasing step to speed up the process. If the extra step is used then the evaluation of solutions is done with data to `res_eval` (2.5 Å) and then all the data are used in an extra phasing step. Only applicable to Phaser SAD phasing.
- `read_sites = False` Choose if you want to enter ha sites from a file The name of the file will be requested after scaling is finished. The file can have sites in fractional coordinates or be a PDB file. Normally you do not need to set this. Set automatically if you specify a `sites_file`
- `f_double_prime_list = None` f-double-prime for the heavy-atom for this dataset Normally not used. Use `f_double_prime` for wavelength or deriv
- `f_prime_list = None` f-prime for the heavy-atom for this dataset Normally not used. Use `f_prime` for wavelength or deriv
- `mad_ha_add_f_double_prime_list = None` F-double_prime values of additional heavy-atom types. You must specify the same number of entries of `mad_ha_add_f_double_prime_list` as you do for `mad_ha_add_f_prime_list` and for `mad_ha_add_list`. Only use for Phaser SAD phasing with a single dataset
- `mad_ha_add_f_prime_list = None` F-prime values of additional heavy-atom types. You must specify the same number of entries of `mad_ha_add_f_prime_list` as you do for `mad_ha_add_f_double_prime_list` and for `mad_ha_add_list`. Only use for Phaser SAD phasing with a single dataset
- `mad_ha_add_list = None` You can specify heavy atom types in addition to the one you named in `mad_ha_type`. The heavy-atoms found in initial HySS searches will be given the type of `mad_ha_type`, and Phaser (if used for phasing) will try to find additional heavy atoms of both the type `mad_ha_type` and any listed in `mad_ha_add_list`. You must also specify the same number of `mad_ha_add_f_prime_list` entries and of `mad_ha_add_f_double_prime_list` entries. Only use for Phaser SAD phasing with a single dataset
- `n_ha_list = None` Enter a guess of number of HA sites Normally not used. Use sites in deriv instead
- `nat_der_list = None` Enter Native or a heavy-atom symbol (Pt, Se) Normally not used. Use `atom_type` in deriv instead
- `density_modificationadd_classic_denmod = Auto` You can run classic density modification with solvent flipping after any other kind of density modification. Default is False if `quick=True` and `extreme_dm=False`
`skip_classic_if_worse_fom = True` Skip results of `add_classic_denmod` if FOM gets worse during density modifications
`skip_ncs_in_add_classic = True` Skip using NCS in `add_classic_denmod` (speeds it up)
`tolerance_add_classic = 0.02` Take classic density modification if R-factor is not made worse by more than this
`fix_xyz = False` You can choose to not refine coordinates, and instead to fix them to the values found by the heavy-atom search.
`fix_xyz_after_denmod = None` When sites are found after density modification you can choose whether you want to fix the coordinates to the values found in that map.
`hl_in_resolve = False` AutoSol normally does not write out HL coefficients in the `resolve.mtz` file with density-modified phases. You can turn them on with `hl_in_resolve=True`
`mask_type = *histograms probability wang classic` Choose method for obtaining probability that a point is

in the protein vs solvent region. Default is "histograms". If you have a SAD dataset with a heavy atom such as Pt or Au then you may wish to choose "wang" because the histogram method is sensitive to very high peaks. Options are: histograms: compare local rms of map and local skew of map to values from a model map and estimate probabilities. This one is usually the best. probability: compare local rms of map to distribution for all points in this map and estimate probabilities. In a few cases this one is much better than histograms. wang: take points with highest local rms and define as protein. Classic runs classical density modification with solvent flipping.test_mask_type = None You can choose to have AutoSol test histograms/wang methods for identifying solvent region and statistical vs classical density modification based on the final density modification r-factor. mask_cycles = 5 Number of mask cycles in density modification (5 is usual for thorough density modification minor_cycles = 10 Number of minor cycles in density modification for each mask cycle (10 is usual for thorough density modification)thorough_denmod = None Choose whether you want to go for density modification (usual) or quick (speeds it up and for a terrible map is sometimes better) truncate_ha_sites_in_resolve = Auto *True False You can choose to truncate the density near heavy-atom sites at a maximum of 2.5 sigma. This is useful in cases where the heavy-atom sites are very strong, and rarely hurts in cases where they are not. The heavy-atom sites are specified with "input_ha_file" and radius is rad_maskrad_mask = None You can define the radius for calculation of the protein mask Applies only to truncate_ha_sites_in_resolve. Default is resolution of data.use_ncs_in_denmod = True This script normally uses available ncs information in density modification. Say No to skip this. See also find_ncs mask_as_mtz = False Defines how omit_output_mask_file ncs_output_mask_file and protein_output_mask_file are written out. If mask_as_mtz=False it will be a ccp4 map. If mask_as_mtz=True it will be an mtz file with map coefficients FP PHIM FOMM (all three required)protein_output_mask_file = None Name of map to be written out representing your protein (non-solvent) region. If mask_as_mtz=False the map will be a ccp4 map. If mask_as_mtz=True it will be an mtz file with map coefficients FP PHIM FOMM (all three required)ncs_output_mask_file = None Name of map to be written out representing your ncs asymmetric unit. If mask_as_mtz=False the map will be a ccp4 map. If mask_as_mtz=True it will be an mtz file with map coefficients FP PHIM FOMM (all three required)omit_output_mask_file = None Name of map to be written out representing your omit region. If mask_as_mtz=False the map will be a ccp4 map. If mask_as_mtz=True it will be an mtz file with map coefficients...

- add_classic_denmod = Auto You can run classic density modification with solvent flipping after any other kind of density modification. Default is False if quick=True and extreme_dm=False
- skip_classic_if_worse_fom = True Skip results of add_classic_denmod if FOM gets worse during density modification
- skip_ncs_in_add_classic = True Skip using NCS in add_classic_denmod (speeds it up)
- tolerance_add_classic = 0.02 Take classic density modification if R-factor is not made worse by more than this
- fix_xyz = False You can choose to not refine coordinates, and instead to fix them to the values found by the heavy-atom search.
- fix_xyz_after_denmod = None When sites are found after density modification you can choose whether you want to fix the coordinates to the values found in that map.
- hl_in_resolve = False AutoSol normally does not write out HL coefficients in the resolve.mtz file with density-modified phases. You can turn them on with hl_in_resolve=True
- mask_type = *histograms probability wang classic Choose method for obtaining probability that a point is in the protein vs solvent region. Default is "histograms". If you have a SAD dataset with a heavy atom such as Pt or Au then you may wish to choose "wang" because the histogram

method is sensitive to very high peaks. Options are: histograms: compare local rms of map and local skew of map to values from a model map and estimate probabilities. This one is usually the best. probability: compare local rms of map to distribution for all points in this map and estimate probabilities. In a few cases this one is much better than histograms. wang: take points with highest local rms and define as protein. Classic runs classical density modification with solvent flipping.

- `test_mask_type = None` You can choose to have AutoSol test histograms/wang methods for identifying solvent region and statistical vs classical density modification based on the final density modification r-factor.
- `mask_cycles = 5` Number of mask cycles in density modification (5 is usual for thorough density modification)
- `minor_cycles = 10` Number of minor cycles in density modification for each mask cycle (10 is usual for thorough density modification)
- `thorough_denmod = None` Choose whether you want to go for density modification (usual) or quick (speeds it up and for a terrible map is sometimes better)
- `truncate_ha_sites_in_resolve = Auto *True False` You can choose to truncate the density near heavy-atom sites at a maximum of 2.5 sigma. This is useful in cases where the heavy-atom sites are very strong, and rarely hurts in cases where they are not. The heavy-atom sites are specified with `"input_ha_file"` and radius is `rad_mask`
- `rad_mask = None` You can define the radius for calculation of the protein mask Applies only to `truncate_ha_sites_in_resolve`. Default is resolution of data.
- `use_ncs_in_denmod = True` This script normally uses available ncs information in density modification. Say No to skip this. See also `find_ncs`
- `mask_as_mtz = False` Defines how `omit_output_mask_file` `ncs_output_mask_file` and `protein_output_mask_file` are written out. If `mask_as_mtz=False` it will be a ccp4 map. If `mask_as_mtz=True` it will be an mtz file with map coefficients FP PHIM FOMM (all three required)
- `protein_output_mask_file = None` Name of map to be written out representing your protein (non-solvent) region. If `mask_as_mtz=False` the map will be a ccp4 map. If `mask_as_mtz=True` it will be an mtz file with map coefficients FP PHIM FOMM (all three required)
- `ncs_output_mask_file = None` Name of map to be written out representing your ncs asymmetric unit. If `mask_as_mtz=False` the map will be a ccp4 map. If `mask_as_mtz=True` it will be an mtz file with map coefficients FP PHIM FOMM (all three required)
- `omit_output_mask_file = None` Name of map to be written out representing your omit region. If `mask_as_mtz=False` the map will be a ccp4 map. If `mask_as_mtz=True` it will be an mtz file with map coefficients FP PHIM FOMM (all three required)
- `use_hl_anom_in_denmod = None` Default is False (use HL coefficients including model information in density modification) Allows you to specify that HL coefficients including only the phase information from the imaginary (anomalous difference) contribution from the anomalous scatterers are to be used in density modification. Two sets of HL coefficients are produced by Phaser. HLA HLB etc are HL coefficients including the contribution of both the real scattering and the anomalous differences. HLANomA HLANomB etc are HL coefficients including the contribution of the anomalous differences alone. These HL coefficients for anomalous differences alone are the ones that you will want to use in cases where you are bringing in model information that includes the real scattering from the model used in Phaser, such as when you are carrying out density modification with a model or refinement of a model If `use_hl_anom_in_denmod=True` then the HLANom HL coefficients from Phaser are used in density modification

- `use_hl_anom_in_denmod_with_model = None` Default is True if `input_partpdb_file` is included. (See also `use_hl_anom_in_denmod`) If `use_hl_anom_in_denmod=True` then the HLANom HL coefficients from Phaser (not including model information) are used in density modification with a model
- `precondition = False` Precondition density before modification
- `model_buildingbuild = True` Build model after density modification? `phase_improve_and_build = True` Carry out cycles of phase improvement with quick model-building followed by a full model-building step
NOTE: This is now the standard model-building approach for AutoSolsort_hetatms = False Waters are automatically named with the chain of the closest macromolecule if you set `sort_hetatms=True` This is for the final model only.
`map_to_object = None` you can supply a target position for your model with `map_to_object=my_target.pdb`. Then at the very end your molecule will be placed as close to this as possible. The center of mass of the autobuild model will be superimposed on the center of mass of `my_target.pdb` using space group symmetry, taking any match closer than 15 Å within 3 unit cells of the original position. The new file will be `overall_best_mapped.pdb`
`helices_strands_only = False` You can choose to use a quick model-building method that only builds secondary structure. At low resolution this may be both quicker and more accurate than trying to build the entire structure
`resolution_helices_strands = 3.1` Resolution to switch to `helices_strands_only`
`helices_strands_start = False` You can choose to use a quick model-building method that builds secondary structure as a way to get started...then model completion is done as usual. (Contrast with `helices_strands_only` which only does secondary structure)
`cc_helix_min = None` Minimum CC of helical density to map at low resolution when using `helices_strands_only`
`cc_strand_min = None` Minimum CC of strand density to map when using `helices_strands_only`
`build_type = *RESOLVE RESOLVE_AND_BUCCANEER` You can choose to build models with RESOLVE or with RESOLVE and BUCCANEER #and TEXTAL and how many different models to build with RESOLVE. The more you build, the more likely to get a complete model. Note that `rebuild_in_place` can only be carried out with RESOLVE model-building. For BUCCANEER model building you need CCP4 version 6.1.2 or higher and BUCCANEER version 1.3.0 or higher
`resolveParameters` specific for RESOLVE model-building
`n_cycle_build = None` Choose number of cycles (3).
`refine = True` This script normally refines the model during building. Say False to skip refinement
`ncycle_refine = 3` Choose number of refinement cycles (3)
`number_of_builds = None` Number of different solutions to build models for
`number_of_models = None` This parameter lets you choose how many initial models to build with RESOLVE within a single build cycle.
`resolution_build = 0` Enter the high-resolution limit for model-building. If 0.0, the value of resolution is used as a default.
`fit_loops = True` You can fit loops automatically if sequence alignment has been done.
`loop_cc_min = 0.4` You can specify the minimum correlation of density from a loop with the map.
`group_ca_length = 4` In resolve building you can specify how short a fragment to keep. Normally 4 or 5 residues should be the minimum.
`group_length = 2` In resolve building you can specify how many fragments must be joined to make a connected group that is kept. Normally 2 fragments should be the minimum.
`input_compare_file = None` If you are rebuilding a model or already think you know what the model should be, you can include a comparison file in rebuilding. The model is not used for anything except to write out information on coordinate differences in the output log files. NOTE: this feature does not always work correctly.
`n_random_frag = 0` In resolve building you can randomize each fragment slightly so as to generate more possibilities for tracing based on extending it.
`n_random_loop = 3` Number of randomized tries from each end for building loops If 0, then one try. If N, then N additional tries with randomization based on `rms_random_loop`.
`offsets_list = 53 7 23` You can specify an offset for the orientation of the helix and strand templates in building. This is used in generating different starting models.
`remove_outlier_segments_z_cut = 3.0` You can remove any segments that ar...
- `build = True` Build model after density modification?

- `phase_improve_and_build = True` Carry out cycles of phase improvement with quick model-building followed by a full model-building step NOTE: This is now the standard model-building approach for AutoSol
- `sort_hetatms = False` Waters are automatically named with the chain of the closest macromolecule if you set `sort_hetatms=True` This is for the final model only.
- `map_to_object = None` you can supply a target position for your model with `map_to_object=my_target.pdb`. Then at the very end your molecule will be placed as close to this as possible. The center of mass of the autobuild model will be superimposed on the center of mass of `my_target.pdb` using space group symmetry, taking any match closer than 15 Å within 3 unit cells of the original position. The new file will be `overall_best_mapped.pdb`
- `helices_strands_only = False` You can choose to use a quick model-building method that only builds secondary structure. At low resolution this may be both quicker and more accurate than trying to build the entire structure
- `resolution_helices_strands = 3.1` Resolution to switch to `helices_strands_only`
- `helices_strands_start = False` You can choose to use a quick model-building method that builds secondary structure as a way to get started...then model completion is done as usual. (Contrast with `helices_strands_only` which only does secondary structure)
- `cc_helix_min = None` Minimum CC of helical density to map at low resolution when using `helices_strands_only`
- `cc_strand_min = None` Minimum CC of strand density to map when using `helices_strands_only`
- `build_type = *RESOLVE RESOLVE_AND_BUCCANEER` You can choose to build models with RESOLVE or with RESOLVE and BUCCANEER #and TEXTAL and how many different models to build with RESOLVE. The more you build, the more likely to get a complete model. Note that `rebuild_in_place` can only be carried out with RESOLVE model-building. For BUCCANEER model building you need CCP4 version 6.1.2 or higher and BUCCANEER version 1.3.0 or higher
- `resolveParameters` specific for RESOLVE model-building
`ncycle_build = None` Choose number of cycles (3).
`refine = True` This script normally refines the model during building. Say False to skip refinement
`ncycle_refine = 3` Choose number of refinement cycles (3)
`number_of_builds = None` Number of different solutions to build models for
`number_of_models = None` This parameter lets you choose how many initial models to build with RESOLVE within a single build cycle.
`resolution_build = 0` Enter the high-resolution limit for model-building. If 0.0, the value of resolution is used as a default.
`fit_loops = True` You can fit loops automatically if sequence alignment has been done.
`loop_cc_min = 0.4` You can specify the minimum correlation of density from a loop with the map.
`group_ca_length = 4` In resolve building you can specify how short a fragment to keep. Normally 4 or 5 residues should be the minimum.
`group_length = 2` In resolve building you can specify how many fragments must be joined to make a connected group that is kept. Normally 2 fragments should be the minimum.
`input_compare_file = None` If you are rebuilding a model or already think you know what the model should be, you can include a comparison file in rebuilding. The model is not used for anything except to write out information on coordinate differences in the output log files. NOTE: this feature does not always work correctly.
`n_random_frag = 0` In resolve building you can randomize each fragment slightly so as to generate more possibilities for tracing based on extending it.
`n_random_loop = 3` Number of randomized tries from each end for building loops. If 0, then one try. If N, then N additional tries with randomization based on `rms_random_loop`.
`offsets_list = 53 7 23` You can specify an offset for the orientation of the helix and strand templates in building. This is used in generating different starting models.
`remove_outlier_segments_z_cut = 3.0` You can remove any segments that are not assigned to sequence during model-building if the mean density at

atomic positions are more than remove_outlier_segments_z_cut sd lower than the mean for the structure.

resolve_command_list = None Commands for resolve. One per line in the form: keyword value value can be optional Examples: coarse_grid resolution 200 2.0 hklin test.mtz NOTE: for command-line usage you need to enclose the whole set of commands in double quotes (") and each individual command in single quotes (') like this: resolve_command_list="'no_build' 'b_overall 23' ";

solve_command_list = None Commands for solve. One per line in the form: keyword value, where value can be optional Examples: verbose resolution 200 2.0 For specification from command_line enclose each command and value in quotes, and then use a different type of quotes to enclose all of them (same as resolve_command_list)rms_random_frag = None Rms random position change added to residues on ends of fragments when extending them If you enter a negative number, defaults will be used.

rms_random_loop = None Rms random position change added to residues on ends of loops in tries for building loops If you enter a negative number, defaults will be used. semet = None You can specify that the dataset that is used for refinement is a selenomethionine dataset, and that the model should be the SeMet version of the protein, with all SD of MET replaced with Se of MSE. By default if your heavy-atom is Se then this will be set to Trueuse_met_in_align = Auto *True False You can use the heavy-atom positions in input_ha_file as markers for Met SD positions. start_chains_list = None You can specify the starting residue number for each of the unique chains in your structure. If you use a sequence file then the unique chains are extracted and the order must match the order of your starting residue numbers. For example, if your sequence file has chains A and B (identical) and chains C and D (identical to each other, but different than A and B) then you can ent...

- n_cycle_build = None Choose number of cycles (3).
- refine = True This script normally refines the model during building. Say False to skip refinement
- ncycle_refine = 3 Choose number of refinement cycles (3)
- number_of_builds = None Number of different solutions to build models for
- number_of_models = None This parameter lets you choose how many initial models to build with RESOLVE within a single build cycle.
- resolution_build = 0 Enter the high-resolution limit for model-building. If 0.0, the value of resolution is used as a default.
- fit_loops = True You can fit loops automatically if sequence alignment has been done.
- loop_cc_min = 0.4 You can specify the minimum correlation of density from a loop with the map.
- group_ca_length = 4 In resolve building you can specify how short a fragment to keep. Normally 4 or 5 residues should be the minimum.
- group_length = 2 In resolve building you can specify how many fragments must be joined to make a connected group that is kept. Normally 2 fragments should be the minimum.
- input_compare_file = None If you are rebuilding a model or already think you know what the model should be, you can include a comparison file in rebuilding. The model is not used for anything except to write out information on coordinate differences in the output log files. NOTE: this feature does not always work correctly.
- n_random_frag = 0 In resolve building you can randomize each fragment slightly so as to generate more possibilities for tracing based on extending it.
- n_random_loop = 3 Number of randomized tries from each end for building loops If 0, then one try. If N, then N additional tries with randomization based on rms_random_loop.
- offsets_list = 53 7 23 You can specify an offset for the orientation of the helix and strand templates in building. This is used in generating different starting models.

- `remove_outlier_segments_z_cut = 3.0` You can remove any segments that are not assigned to sequence during model-building if the mean density at atomic positions are more than `remove_outlier_segments_z_cut` sd lower than the mean for the structure.
- `resolve_command_list = None` Commands for resolve. One per line in the form: keyword value value can be optional Examples: `coarse_grid resolution 200 2.0 hkl in test.mtz` NOTE: for command-line usage you need to enclose the whole set of commands in double quotes (") and each individual command in single quotes (') like this: `resolve_command_list="'no_build' 'b_overall 23' "`;
- `solve_command_list = None` Commands for solve. One per line in the form: keyword value, where value can be optional Examples: `verbose resolution 200 2.0` For specification from command_line enclose each command and value in quotes, and then use a different type of quotes to enclose all of them (same as `resolve_command_list`)
- `rms_random_frag = None` Rms random position change added to residues on ends of fragments when extending them If you enter a negative number, defaults will be used.
- `rms_random_loop = None` Rms random position change added to residues on ends of loops in tries for building loops If you enter a negative number, defaults will be used.
- `semet = None` You can specify that the dataset that is used for refinement is a selenomethionine dataset, and that the model should be the SeMet version of the protein, with all SD of MET replaced with Se of MSE. By default if your heavy-atom is Se then this will be set to True
- `use_met_in_align = Auto *True False` You can use the heavy-atom positions in `input_ha_file` as markers for Met SD positions.
- `start_chains_list = None` You can specify the starting residue number for each of the unique chains in your structure. If you use a sequence file then the unique chains are extracted and the order must match the order of your starting residue numbers. For example, if your sequence file has chains A and B (identical) and chains C and D (identical to each other, but different than A and B) then you can enter 2 numbers, the starting residues for chains A and C. NOTE: you need to specify an input sequence file for `start_chains_list` to be applied.
- `thorough_loop_fit = None` Try many conformations and accept them even if the fit is not perfect. If you say True the parameters for thorough loop fitting are: `n_random_loop=100 rms_random_loop=0.3 rho_min_main=0.5` while if you say No those for quick loop fitting are: `n_random_loop=20 rms_random_loop=0.3 rho_min_main=1.0`
- `trace_as_lig = False` You can specify that in building steps the ends of chains are to be extended using the LigandFit algorithm. This is default for nucleic acid model-building.
- `use_any_side = False` You can choose to have resolve model-building place the best-fitting side chain at each position, even if the sequence is not matched to the map.
- `trim_ends_if_needed = True` If a single loop is specified and no loop is found, try trimming back the ends.
- `loop_backup_residues = None` Number of tries removing one residue at a time from each end of existing ends of loop if no loop is found with initial gap Default is 4 (1 if quick)
- `split_by_gap = True` Split up work by gaps and run individually (normally True)
- `loop_lib = False` Use loop library to fit loops Only applicable for `chain_type=PROTEIN`
- `standard_loops = True` Use standard loop fitting
- `trace_loops = False` Use loop tracing to fit loops Only applicable for `chain_type=PROTEIN`
- `refine_trace_loops = True` Refine loops (real-space) after `trace_loops`

- `density_of_points = None` Packing density of points to consider as as possible CA atoms in `trace_loops`. Try 1.0 for a quick run, up to 5 for much more thorough run If None, try value depending on value of quick.
- `a_cut_min = None` Minimum density (relative to SD of map, normalized for solvent content) for `trace_loop` points
- `max_density_of_points = None` Maximum packing density of points to consider as as possible CA atoms in `trace_loops`.
- `cutout_model_radius = None` Radius to cut out density for `trace_loops` If None, guess based on length of loop
- `max_cutout_model_radius = 20`. Maximum value of `cutout_model_radius` to try
- `padding = 1`. Padding for cut out density in `trace_loops`
- `cut_out_density = True` Cut out density for `trace_loops`
- `max_span = 30` Maximum length of a gap to try to fill
- `max_c_ca_dist = None` Maximum C-CA distance for a connection
- `max_overlap_rmsd = 2`. Maximum rmsd for 3 residues on each end of loop-lib fit
- `max_overlap = None` Maximum number of residues from ends to start with. (1=use existing ends, 2=one in from ends etc) If None, set based on value of quick.
- `min_overlap = None` Minimum number of residues from ends to start with. (1=use existing ends, 2=one in from ends etc)
- `ncsfind_ncs = *Auto True False` The wizard normally deduces ncs information from the NCS in heavy atom sites, and then later from any NCS in chains of models that are built during model-building. The update is done each cycle in which an improved model is obtained. Say No to skip this update. If your molecule can only fit in the asymmetric unit once the `find_ncs` will be set to False unless you specifically set it to True or you set `ncs_copies=xxxx`
`ncs_copies = None` Number of copies of the molecule in the au (note: only one type of molecule allowed at present)
`ncs_refine_coord_sigma_from_rmsd = False` You can choose to use the current NCS rmsd as the value of the sigma for NCS restraints. See also `ncs_refine_coord_sigma_from_rmsd_ratio`
`ncs_refine_coord_sigma_from_rmsd_ratio = 1` You can choose to multiply the current NCS rmsd by this value before using it as the sigma for NCS restraints See also `ncs_refine_coord_sigma_from_rmsd`
`optimize_ncs = True` This script normally deduces ncs information from the NCS in chains of models that are built during iterative model-building. Optimize NCS adds a step to try and make the molecule formed by NCS as compact as possible, without losing any point-group symmetry.
`refine_with_ncs = True` This script can allow phenix.refine to automatically identify NCS and use it in refinement.
`ncs_in_refinement = *torsion cartesian None` Use `torsion_angle` refinement of NCS. Alternative is `cartesian` or `None` (None will use phenix.refine default)
- `find_ncs = *Auto True False` The wizard normally deduces ncs information from the NCS in heavy atom sites, and then later from any NCS in chains of models that are built during model-building. The update is done each cycle in which an improved model is obtained. Say No to skip this update. If your molecule can only fit in the asymmetric unit once the `find_ncs` will be set to False unless you specifically set it to True or you set `ncs_copies=xxxx`
- `ncs_copies = None` Number of copies of the molecule in the au (note: only one type of molecule allowed at present)
- `ncs_refine_coord_sigma_from_rmsd = False` You can choose to use the current NCS rmsd as the value of the sigma for NCS restraints. See also `ncs_refine_coord_sigma_from_rmsd_ratio`

- `ncs_refine_coord_sigma_from_rmsd_ratio = 1` You can choose to multiply the current NCS rmsd by this value before using it as the sigma for NCS restraints See also `ncs_refine_coord_sigma_from_rmsd`
- `optimize_ncs = True` This script normally deduces ncs information from the NCS in chains of models that are built during iterative model-building. Optimize NCS adds a step to try and make the molecule formed by NCS as compact as possible, without losing any point-group symmetry.
- `refine_with_ncs = True` This script can allow phenix.refine to automatically identify NCS and use it in refinement.
- `ncs_in_refinement = *torsion cartesian None` Use `torsion_angle` refinement of NCS. Alternative is `cartesian` or `None` (`None` will use phenix.refine default)
- `refinementrefine_b = True` You can choose whether phenix.refine is to refine individual atomic displacement parameters (B values) `refine_se_occ = True` You can choose to refine the occupancy of SE atoms in a SEMET structure (default=True). This only applies if `semet=true` `skip_clash_guard = True` Skip refinement check for atoms that clash Also skips check for side chains crossing special position (`allow_polymer_cross_special_position=True` in refinement) `correct_special_position_tolerance = None` Adjust tolerance for special position check. If 0., then check for clashes near special positions is not carried out. This sometimes allows phenix.refine to continue even if an atom is near a special position. If 1., then checks within 1 Å of special positions. If `None`, then uses phenix.refine default. (1) `use_mhlh = True` This script normally uses information from the input file (HLA HLB HLC HLD) in refinement. Say No to only refine on Fobs `generate_hl_if_missing = False` This script normally uses information from the input file (HLA HLB HLC HLD) in refinement. Say No to not generate HL coeffs from input phases. `place_waters = True` You can choose whether phenix.refine automatically places ordered solvent (waters) during the refinement process. `refinement_resolution = 0` Enter the high-resolution limit for refinement only. This high-resolution limit can be different than the high-resolution limit for other steps. The default ("None" or 0.0) is to use the overall high-resolution limit for this run (as set by resolution) `ordered_solvent_low_resolution = None` You can choose what resolution cutoff to use for placing ordered solvent in phenix.refine. If the resolution of refinement is greater than this cutoff, then no ordered solvent will be placed, even if `refinement.main.ordered_solvent=True`. `link_distance_cutoff = 3` You can specify the maximum bond distance for linking residues in phenix.refine called from the wizards. `r_free_flags_fraction = 0.1` Maximum fraction of reflections in the free R set. You can choose the maximum fraction of reflections in the free R set and the maximum number of reflections in the free R set. The number of reflections in the free R set will be up the lower of the values defined by these two parameters. `r_free_flags_max_free = 2000` Maximum number of reflections in the free R set. You can choose the maximum fraction of reflections in the free R set and the maximum number of reflections in the free R set. The number of reflections in the free R set will be up the lower of the values defined by these two parameters. `r_free_flags_use_lattice_symmetry = True` When generating `r_free_flags` you can decide whether to include lattice symmetry (good in general, necessary if there is twinning). `r_free_flags_lattice_symmetry_max_delta = 5` You can set the maximum deviation of distances in the lattice that are to be considered the same for purposes of generating a lattice-symmetry-unique set of free R flags. `allow_overlapping = None` Default is `None` (set automatically, normally `False` unless S or Se atoms are the anomalously-scattering atoms). You can allow atoms in your ligand files to overlap atoms in your protein/nucleic acid model. This overrides 'keep_pdb_atoms' Useful in early stages of model-building and refinement The ligand atoms get the altloc indicator 'L' NOTE: The ligand occupancy will be refined by default if you set `allow_overlapping=True` (because of the altloc indicator) You can turn this off with `fix_ligand_occupancy=True` `fix_ligand_occupancy = None` If `allow_overlapping=True` then ligand occupancies are refined as a group. You can turn this off with `fix_ligand_occupancy=True` NOTE: has no effect if `allow_overlapping=False` `remove_outlier_segments = True` You can remove any segments that are not assigned to sequence if their mean B values are more than `remove_outlier_segments_z_cut`

sd higher than the mean for the structure. NOTE: this is done after refinement, so the R/Rfree are no longer applicable; the remarks...

- `refine_b = True` You can choose whether phenix.refine is to refine individual atomic displacement parameters (B values)
- `refine_se_occ = True` You can choose to refine the occupancy of SE atoms in a SEMET structure (default=True). This only applies if `semet=true`
- `skip_clash_guard = True` Skip refinement check for atoms that clash Also skips check for side chains crossing special position (`allow_polymer_cross_special_position=True` in refinement)
- `correct_special_position_tolerance = None` Adjust tolerance for special position check. If 0., then check for clashes near special positions is not carried out. This sometimes allows phenix.refine to continue even if an atom is near a special position. If 1., then checks within 1 Å of special positions. If None, then uses phenix.refine default. (1)
- `use_mlhl = True` This script normally uses information from the input file (HLA HLB HLC HLD) in refinement. Say No to only refine on Fobs
- `generate_hl_if_missing = False` This script normally uses information from the input file (HLA HLB HLC HLD) in refinement. Say No to not generate HL coeffs from input phases.
- `place_waters = True` You can choose whether phenix.refine automatically places ordered solvent (waters) during the refinement process.
- `refinement_resolution = 0` Enter the high-resolution limit for refinement only. This high-resolution limit can be different than the high-resolution limit for other steps. The default ('None' or 0.0) is to use the overall high-resolution limit for this run (as set by resolution)
- `ordered_solvent_low_resolution = None` You can choose what resolution cutoff to use for placing ordered solvent in phenix.refine. If the resolution of refinement is greater than this cutoff, then no ordered solvent will be placed, even if `refinement.main.ordered_solvent=True`.
- `link_distance_cutoff = 3` You can specify the maximum bond distance for linking residues in phenix.refine called from the wizards.
- `r_free_flags_fraction = 0.1` Maximum fraction of reflections in the free R set. You can choose the maximum fraction of reflections in the free R set and the maximum number of reflections in the free R set. The number of reflections in the free R set will be up the lower of the values defined by these two parameters.
- `r_free_flags_max_free = 2000` Maximum number of reflections in the free R set. You can choose the maximum fraction of reflections in the free R set and the maximum number of reflections in the free R set. The number of reflections in the free R set will be up the lower of the values defined by these two parameters.
- `r_free_flags_use_lattice_symmetry = True` When generating `r_free_flags` you can decide whether to include lattice symmetry (good in general, necessary if there is twinning).
- `r_free_flags_lattice_symmetry_max_delta = 5` You can set the maximum deviation of distances in the lattice that are to be considered the same for purposes of generating a lattice-symmetry-unique set of free R flags.
- `allow_overlapping = None` Default is None (set automatically, normally False unless S or Se atoms are the anomalously-scattering atoms). You can allow atoms in your ligand files to overlap atoms in your protein/nucleic acid model. This overrides 'keep_pdb_atoms' Useful in early stages of model-building and refinement The ligand atoms get the altloc indicator 'L' NOTE: The ligand occupancy will be refined by default if you set `allow_overlapping=True` (because of the altloc indicator) You can turn this off with

fix_ligand_occupancy=True

- fix_ligand_occupancy = None If allow_overlapping=True then ligand occupancies are refined as a group. You can turn this off with fix_ligand_occupancy=true NOTE: has no effect if allow_overlapping=False
- remove_outlier_segments = True You can remove any segments that are not assigned to sequence if their mean B values are more than remove_outlier_segments_z_cut sd higher than the mean for the structure. NOTE: this is done after refinement, so the R/Rfree are no longer applicable; the remarks in the PDB file are removed
- twin_law = None You can specify a twin law for refinement like this: twin_law='-h,k,-l'
- use_hl_anom_in_refinement = None Default is True if input_partpdb_file is used (See also use_hl_anom_in_denmod). If use_hl_anom_in_refinement=True then the Hlanom HL coefficients from Phaser (not including model information) are used in refinement
- include_ha_in_refinement = None You can choose to include your heavy-atom sites in the model for refinement. This is a good idea if your structure includes these heavy-atom sites (i.e., for SAD or MAD structures where you are not using a native dataset). Heavy-atom sites that overlap an atom in your model will be ignored. Default is True unless the dataset is SAD/MAD with Se or S
- displaynumber_of_solutions_to_display = 1 Number of solutions to put on screen and to write out
solution_to_display = 0 Solution number of the solution to display and write out (use 0 to let the wizard display the top solution)
- number_of_solutions_to_display = 1 Number of solutions to put on screen and to write out
- solution_to_display = 0 Solution number of the solution to display and write out (use 0 to let the wizard display the top solution)
- generaldata_quality = *moderate strong weak The defaults are set for you depending on the anticipated data quality. You can choose "moderate" if you are unsure. NOTE: if best FOM of phasing is less than fom_for_extreme_dm data_quality will be automatically reset to weak.thoroughness = *quick medium thorough You can try to run quickly and see if you can get a solution ("quick") or more thoroughly to get the best possible solution ("thorough"). NOTE: if best FOM of phasing is less than fom_for_extreme_dm thoroughness will be automatically reset to thorough.nproc = 1 Normally you may want to set nproc to the number of processors on your machine so HySS can use them all. You can specify the number of processors to use (nproc) and the number of batches to divide the data into for parallel jobs. Normally you will set nproc to the number of processors available and leave nbatch alone. If you leave nbatch as None it will be set automatically, with a value depending on the Wizard. This is recommended. The value of nbatch can affect the results that you get, as the jobs are not split into exact replicates, but are rather run with different random numbers. If you want to get the same results, keep the same value of nbatch. nbatch = 1 You can specify the number of processors to use (nproc) and the number of batches to divide the data into for parallel jobs. Normally you will set nproc to the number of processors available and leave nbatch alone. If you leave nbatch as None it will be set automatically, with a value depending on the Wizard. This is recommended. The value of nbatch can affect the results that you get, as the jobs are not split into exact replicates, but are rather run with different random numbers. If you want to get the same results, keep the same value of nbatch. keep_files = overall_best* phaser_*.mtz denmod_*.mtz resolve_*.mtz solve_*.mtz ha_*.pdb dataset*.log List of files that are not to be cleaned up. wildcards permittedcoot_name = "coot" If your version of coot is called something else, then you can specify that here. i_ran_seed = 72432 Random seed (positive integer) for model-building and simulated annealing refinement raise_sorry = False You can have any failure end with a Sorry instead of simply printout to the screenbackground = True When you specify nproc=nn, you can run the

jobs in background (default if nproc is greater than 1) or foreground (default if nproc=1). If you set run_command=qsub (or otherwise submit to a batch queue), then you should set background=False, so that the batch queue can keep track of your runs. There is no need to use background=True in this case because all the runs go as controlled by your batch system. If you use run_command='sh ' (or similar, sh is default) then normally you will use background=True so that all the jobs run simultaneously.

check_wait_time = 1.0 You can specify the length of time (seconds) to wait between checking for subprocesses to end

max_wait_time = 1.0 You can specify the length of time (seconds) to wait when looking for a file. If you have a cluster where jobs do not start right away you may need a longer time to wait. The symptom of too short a wait time is 'File not found'

wait_between_submit_time = 1.0 You can specify the length of time (seconds) to wait between each job that is submitted when running sub-processes. This can be helpful on NFS-mounted systems when running with multiple processors to avoid file conflicts. The symptom of too short a wait_between_submit_time is File exists:....

cache_resolve_libs = True Use caching of resolve libraries to speed up resolve

resolve_size = 12 Size for solve/resolve ('""", "_giant", "_huge", "_extra_huge" or a number where 12=giant 18=huge)

check_run_command = False You can have the wizard check your run command at startup

run_command = "sh " When you specify nproc=nn, you can run the subprocesses as ...

- data_quality = *moderate strong weak The defaults are set for you depending on the anticipated data quality. You can choose 'moderate' if you are unsure. NOTE: if best FOM of phasing is less than fom_for_extreme_dm data_quality will be automatically reset to weak.
- thoroughness = *quick medium thorough You can try to run quickly and see if you can get a solution ('quick') or more thoroughly to get the best possible solution ('thorough'). NOTE: if best FOM of phasing is less than fom_for_extreme_dm thoroughness will be automatically reset to thorough.
- nproc = 1 Normally you may want to set nproc to the number of processors on your machine so HySS can use them all. You can specify the number of processors to use (nproc) and the number of batches to divide the data into for parallel jobs. Normally you will set nproc to the number of processors available and leave nbatch alone. If you leave nbatch as None it will be set automatically, with a value depending on the Wizard. This is recommended. The value of nbatch can affect the results that you get, as the jobs are not split into exact replicates, but are rather run with different random numbers. If you want to get the same results, keep the same value of nbatch.
- nbatch = 1 You can specify the number of processors to use (nproc) and the number of batches to divide the data into for parallel jobs. Normally you will set nproc to the number of processors available and leave nbatch alone. If you leave nbatch as None it will be set automatically, with a value depending on the Wizard. This is recommended. The value of nbatch can affect the results that you get, as the jobs are not split into exact replicates, but are rather run with different random numbers. If you want to get the same results, keep the same value of nbatch.
- keep_files = overall_best* phaser_*.mtz denmod_*.mtz resolve_*.mtz solve_*.mtz ha_*.pdb dataset*.log List of files that are not to be cleaned up. wildcards permitted
- coot_name = "coot" If your version of coot is called something else, then you can specify that here.
- i_ran_seed = 72432 Random seed (positive integer) for model-building and simulated annealing refinement
- raise_sorry = False You can have any failure end with a Sorry instead of simply printout to the screen
- background = True When you specify nproc=nn, you can run the jobs in background (default if nproc is greater than 1) or foreground (default if nproc=1). If you set run_command=qsub (or otherwise submit to

a batch queue), then you should set `background=False`, so that the batch queue can keep track of your runs. There is no need to use `background=True` in this case because all the runs go as controlled by your batch system. If you use `run_command='sh '` (or similar, `sh` is default) then normally you will use `background=True` so that all the jobs run simultaneously.

- `check_wait_time = 1.0` You can specify the length of time (seconds) to wait between checking for subprocesses to end
- `max_wait_time = 1.0` You can specify the length of time (seconds) to wait when looking for a file. If you have a cluster where jobs do not start right away you may need a longer time to wait. The symptom of too short a wait time is 'File not found'
- `wait_between_submit_time = 1.0` You can specify the length of time (seconds) to wait between each job that is submitted when running sub-processes. This can be helpful on NFS-mounted systems when running with multiple processors to avoid file conflicts. The symptom of too short a `wait_between_submit_time` is File exists:....
- `cache_resolve_libs = True` Use caching of resolve libraries to speed up resolve
- `resolve_size = 12` Size for solve/resolve ('giant', 'huge', 'extra_huge' or a number where 12=giant 18=huge)
- `check_run_command = False` You can have the wizard check your run command at startup
- `run_command = "sh "` When you specify `nproc=nn`, you can run the subprocesses as jobs in background with `sh` (default) or submit them to a queue with the command of your choice (i.e., `qsub`). If you have a multi-processor machine, use `sh`. If you have a cluster, use `qsub` or the equivalent command for your system. NOTE: If you set `run_command=qsub` (or otherwise submit to a batch queue), then you should set `background=False`, so that the batch queue can keep track of your runs. There is no need to use `background=True` in this case because all the runs go as controlled by your batch system. If `nproc` is greater than 1 and you use `run_command='sh '` (or similar, `sh` is default) then normally you will use `background=True` so that all the jobs run simultaneously.
- `queue_commands = None` You can add any commands that need to be run for your queueing system. These are written before any other commands in the file that is submitted to your queueing system. For example on a PBS system you might say: `queue_commands='#PBS -N mr_rosetta'`
`queue_commands='#PBS -j oe'` `queue_commands='#PBS -l walltime=03:00:00'`
`queue_commands='#PBS -l nodes=1:ppn=1'` NOTE: you can put in the characters '<path>' in any `queue_commands` line and this will be replaced by a string of characters based on the path to the run directory. The first character and last two characters of each part of the path will be included, separated by '_', up to 15 characters. For example 'test_autobuild/WORK_5/AutoBuild_run_1/TEMP0/RUN_1' would be represented by: 'tld_W_5_A1__TP0_1'
- `condor_universe = vanilla` The universe for condor is usually vanilla. However you might need to set it to local for your cluster
- `add_double_quotes_in_condor = True` You might need to turn on or off double quotes in condor job submission scripts. These are already default elsewhere but may interfere with condor paths.
- `condor = None` Specifies if the `group_run_command` is submitting a job to a condor cluster. Set by default to True if `group_run_command=condor_submit`, otherwise False. For condor job submission `mr_rosetta` uses a customized script with condor commands. Also uses `one_subprocess_level=True`
- `last_process_is_local = True` If true, run the last process in a group in background with `sh` as part of the job that is submitting jobs. This prevents having the job that is submitting jobs sit and wait for all the others while doing nothing

- `skip_r_factor = False` You can skip R-factor calculation if refinement is not done and `maps_only=True`
- `test_flag_value = Auto` Normally leave this at Auto (default). This parameter sets the value of the test set that is to be free. Normally phenix sets up test sets with values of 0 and 1 with 1 as the free set. The CCP4 convention is values of 0 through 19 with 0 as the free set. Either of these is recognized by default in Phenix. If you have any other convention (for example values of 0 to 19 and test set is 1) then you can specify this with `test_flag_value`.
- `skip_xtriage = False` You can bypass xtriage if you want. This will prevent you from applying anisotropy corrections, however.
- `base_path = None` You can specify the base path for files (default is current working directory)
- `temp_dir = None` Define a temporary directory
- `local_temp_directory = None` Write all temporary files to this directory and copy files specified by `keep_files` back to TEMP directory in working directory at the end of run. Used to speed up read/write operations. NOTE: Deletes all files except those specified by `keep_files` just like `clean_up=True`.
- `clean_up = None` At the end of the entire run the TEMP directories will be removed if `clean_up` is True. Also all files except for overall_best_xxx (final) files and .eff files In the AutoSol_run_xx_ AutoBuild_run_xx_ etc directory will be removed Files listed in `keep_files` will not be deleted. If you want to remove files after your run is finished use a command like `"phenix.autobuild run=1 clean_up=True"`;
- `print_citations = True` Print citations at end of run
- `solution_output_pickle_file = None` At end of run, write solutions to this file in output directory if defined
- `job_title = None` Job title in PHENIX GUI, not used on command line
- `top_output_dir = None` This is used in subprocess calls of wizards and to tell the Wizard where to look for the STOPWIZARD file.
- `wizard_directory_number = None` This is used by the GUI to define the run number for Wizards. It is the same as `desired_run_number` NOTE: this value can only be specified on the command line, as the directory number is set before parameters files are read.
- `verbose = False` Command files and other verbose output will be printed
- `extra_verbose = False` Facts and possible commands will be printed every cycle if True
- `debug = False` You can have the wizard stop with error messages about the code if you use debug. Additionally the output goes to the terminal if you specify `"debug=True"`;
- `require_nonzero = True` Require non-zero values in data columns to consider reading in.
- `remove_path_word_list = None` List of words identifying paths to remove from PATH These can be used to shorten your PATH. For example... `cns ccp4 coot` would remove all paths containing these words except those also containing phenix. Capitalization is ignored.
- `fill = False` Fill in all missing reflections to resolution `res_fill`. Applies to density modified maps. See also `filled_2fofc_maps` in autobuild.
- `res_fill = None` Resolution for filling in missing data (default = highest resolution of any datafile). Only applies to density modified maps. Default is fill to high resolution of data. Ignored if `fill=False`
- `check_only = False` Just read in and check initial parameters. Not for general use
- `run_controlcoot = None` Set coot to True and optionally `run=[run-number]` to run Coot with the current model and map for run run-number. In some wizards (AutoBuild) you can edit the model and give it back to PHENIX to use as part of the model-building process. If you just say coot then the facts for the

highest-numbered existing run will be shown. `ignore_blanks = None` `ignore_blanks` allows you to have a command-line keyword with a blank value like `"input_lig_file_list="`; `stop = None` You can stop the current wizard with `"stopwizard"`; or `"stop"`; . If you type `"phenix.autobuild run=3 stop"`; then this will stop run 3 of autobuild. `display_facts = None` Set `display_facts` to True and optionally `run=[run-number]` to display the facts for run `run-number`. If you just say `display_facts` then the facts for the highest-numbered existing run will be shown. `display_summary = None` Set `display_summary` to True and optionally `run=[run-number]` to show the summary for run `run-number`. If you just say `display_summary` then the summary for the highest-numbered existing run will be shown. `carry_on = None` Set `carry_on` to True to carry on with highest-numbered run from where you left off. `run = None` Set `run` to `n` to continue with run `n` where you left off. `copy_run = None` Set `copy_run` to `n` to copy run `n` to a new run and continue where you left off. `display_runs = None` List all runs for this wizard. `delete_runs = None` List runs to delete: 1 2 3-5 9:12 `display_labels = None` `display_labels=test.mtz` will list all the labels that identify data in `test.mtz`. You can use the label strings that are produced in AutoSol to identify which data to use from a datafile like this: `peak.data="F+ SIGF+ F- SIGF-"`; . The entire string in quotes counts here You can use the individual labels from these strings as identifiers for data columns in AutoSol or AutoBuild like this: `input_refinement_labels="FP SIGFP FreeR_flags"`; # each individual label counts `dry_run = False` Just read in and check parameter `namesparams_only = False` Just read in and return parameter defaults. Not for general use `display_all = False` Just read in and display parameter defaults

- `coot = None` Set `coot` to True and optionally `run=[run-number]` to run Coot with the current model and map for run `run-number`. In some wizards (AutoBuild) you can edit the model and give it back to PHENIX to use as part of the model-building process. If you just say `coot` then the facts for the highest-numbered existing run will be shown.
- `ignore_blanks = None` `ignore_blanks` allows you to have a command-line keyword with a blank value like `"input_lig_file_list="`;
- `stop = None` You can stop the current wizard with `"stopwizard"`; or `"stop"`; . If you type `"phenix.autobuild run=3 stop"`; then this will stop run 3 of autobuild.
- `display_facts = None` Set `display_facts` to True and optionally `run=[run-number]` to display the facts for run `run-number`. If you just say `display_facts` then the facts for the highest-numbered existing run will be shown.
- `display_summary = None` Set `display_summary` to True and optionally `run=[run-number]` to show the summary for run `run-number`. If you just say `display_summary` then the summary for the highest-numbered existing run will be shown.
- `carry_on = None` Set `carry_on` to True to carry on with highest-numbered run from where you left off.
- `run = None` Set `run` to `n` to continue with run `n` where you left off.
- `copy_run = None` Set `copy_run` to `n` to copy run `n` to a new run and continue where you left off.
- `display_runs = None` List all runs for this wizard.
- `delete_runs = None` List runs to delete: 1 2 3-5 9:12
- `display_labels = None` `display_labels=test.mtz` will list all the labels that identify data in `test.mtz`. You can use the label strings that are produced in AutoSol to identify which data to use from a datafile like this: `peak.data="F+ SIGF+ F- SIGF-"`; . The entire string in quotes counts here You can use the individual labels from these strings as identifiers for data columns in AutoSol or AutoBuild like this: `input_refinement_labels="FP SIGFP FreeR_flags"`; # each individual label counts
- `dry_run = False` Just read in and check parameter names

- `params_only = False` Just read in and return parameter defaults. Not for general use
- `display_all = False` Just read in and display parameter defaults
- `special_keywordswrite_run_directory_to_file = None` Writes the full name of a run directory to the specified file. This can be used as a call-back to tell a script where the output is going to go.
- `write_run_directory_to_file = None` Writes the full name of a run directory to the specified file. This can be used as a call-back to tell a script where the output is going to go.
- `non_user_parameters` These are obsolete parameters and parameters that the wizards use to communicate among themselves. Not normally for general use.
- `gui_output_dir = None` Used only by the GUI
- `allow_negative_f_double_prime = False` Allow a negative f-double-prime value
- `inano_list = None` Choose inano for including anomalous differences and noinano not to include them and anonly for just anomalous differences (no isomorphous differences) Not normally used. Use inano in deriv instead
- `ha_sites_file = None` Not normally used. Use sites_file for wavelength or deriv
- `expt_type = *Auto mad sir sad` Not normally used. Determined automatically from your inputs for wavelength and native/deriv. Experiment type (MAD SIR SAD) NOTE: Please treat MIR experiments as a set of SIR experiments. NOTE: The default for this keyword is Auto which means "carry out normal process to guess this keyword". If you have a single file, then it is assumed to be SAD. If you specify native.data and deriv.data it is SIR, if you specify peak.data and infl.data it is MAD. If the Wizard does not guess correctly, you can set it with this keyword.
- `wavelength_list = None` Optional wavelength of x-ray data (A) Not normally used. Use wavelength/deriv and lambda instead
- `wavelength_name_list = None` Names of wavelengths. Not normally used. Use wavelength/deriv and name instead
- `sg = None` Obsolete. Use space_group instead

Experimental phasing in the AutoSol GUI

autosol_gui.html

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Introduction

NOTE: this document is mainly an introduction for users who are not familiar with AutoSol (or experimental phasing in general), rather than a comprehensive reference for the program. The documentation for the command-line version covers the full range of options and functionality.

AutoSol is an experimental phasing pipeline that combines HySS (Hybrid Substructure Search) for finding heavy-atom sites, Phaser or SOLVE for calculating experimental phases, and RESOLVE for density modification and (optional) model-building. It will automatically try multiple solutions as necessary and determine which is most likely to give the desired result; several different measures of solution quality (discussed below) are supplied. With good data, a partial model can be obtained for an average-size protein (30kD) in under an hour, starting only from scaled reflections and a sequence file.

The examples below mostly use the p9-sad example included in the Phenix distribution. If you have not previously performed experimental phasing or used AutoSol, we recommend setting up a tutorial project in the GUI and running the full example. Because these data extend to a relatively high resolution (1.7Å), AutoSol can build a nearly complete model with R-factors well below 0.3, but to save time you may prefer to truncate the resolution at 2.5Å, which will finish in less than 15 minutes.

Video Tutorial

A tutorial video is available on the Phenix YouTube channel and covers the following topics:

- basic overview
- how to run AutoSol via the GUI
- how to interpret the output

Input files and mandatory parameters

Most users will have anomalous data from a single crystal (or merged from multiple similar crystals); although SIR and MIR phasing (with or without anomalous information) is supported, these types of experiment are typically done as a last resort. At a minimum, one reflections file containing scaled anomalous intensities or amplitudes is required. Any input format is supported, and in most cases AutoSol is run immediately after data processing. Files can be added to the input list by dragging them into the main window, or by clicking the "Browse" button and selecting them from a file dialog. The GUI will automatically determine the file type, usually assuming that a SAD or MAD experiment is being performed. In most cases the data labels for reflections will be guessed automatically (in the p9-sad example, only one set of values is available anyway).

You should always supply a sequence file (usually in FASTA format) if you have this information, although it is optional if you do not perform model-building. If more than one chain is present, multiple sequences may be specified in a single file. At this time, AutoSol (and AutoBuild) can build either protein or nucleic acid structures, but not both at once.

When you add a reflections file, additional controls will appear for specifying experiment details such as heavy atom type and number, and anomalous scattering parameters. If you are trying to phase using selenomethionine (SeMet), the number of atoms can be automatically determined from the sequence. The wavelength is required for SAD but optional for other phasing methods; anomalous coefficients (f' and f'') are optional for SAD (but highly recommended if they were measured), and required for MAD. If you did not measure these parameters at the beamline using a fluorescence scan, they can be guessed based on the wavelength by clicking the button below the file list. For MAD phasing, the experimentally determined values are more essential, and often make the difference between an uninterpretable map and an obviously correct solution. A different example (gene-5-mad, also included in the PHENIX distribution) using multiple wavelengths is shown below; the inputs are only slightly more complicated than for SAD. (In this case the wavelengths and anomalous coefficients are listed in the file README.txt in the example directory.)

Assuming that the space group and unit cell are automatically extracted from the reflection files, most of the other parameters can be left on the defaults, although in the first image above we have changed the resolution to 2.5Å. The p9-sad example contains only a single protein chain in the asymmetric unit so the number of NCS copies can be left blank, but you should enter this if you are confident of the crystal contents. We recommend leaving the model-building on, since it is the single most useful measure of whether the final map is interpretable or not. If you have an RNA or DNA crystal, you may explicitly set the chain type by clicking the "All parameters" button, select "Crystal info" from the menu, and change the drop-down menu for "Chain type".

Three other types of input file are optional but may be useful in special cases: heavy-atom sites (in PDB format) from an external program such as SHELXD, phases (in MTZ format) from an MR model or another phasing program, or a molecular replacement solution (in PDB format). The use of pre-defined sites is not discussed here as it is essentially very simple. External phases can be used to help SOLVE find heavy-atom sites in phasing methods other than SAD. The combination of MR and SAD is described at the end of this document.

Running AutoSol

Once you start AutoSol, a new tab is added to the GUI, with multiple sub-tabs displaying job status and results. The first tab prints the log output and summary of the current best statistics for the run once they are available.

The second tab displays a list of the most important output files in the job directory created by AutoSol. Buttons for viewing the results in graphics applications will be activated once the necessary files are detected. The image below is the appearance of this tab at the end of running the p9-sad example.

Xtriage

The first step in the phasing pipeline is running the data analysis program phenix.xtriage, which identifies potential problems such as twinning, translational pseudo-symmetry, or weak anomalous signal. (Xtriage can also be run by itself, starting from the main GUI or the toolbar button in AutoSol, but you do not need to do this if AutoSol has already run it for you.) The graphs and summary text from Xtriage may be viewed by clicking the button that appears in this section. We recommend that you read the documentation for the Xtriage GUI for more information about interpreting the output. If AutoSol is not successful in phasing your structure, it is especially important to look over the Xtriage results before spending more effort on the same data.

Heavy-atom search

Within a few minutes, information on heavy-atom site finding will appear in the third tab. There may be more than one solution if you are running a MAD or MIR experiment, but AutoSol will pick the one mostly likely to be correct for further analysis. The most important statistic here is the number of sites located; it will usually be less than the final number of refined sites, and the difference will be made up in the phasing step. If the number of sites is much less than expected, you may not have sufficient anomalous signal in your data, or (when not using SeMet) the heavy atom may not be bound.

Phasing and density modification

Phasing is done by Phaser for SAD experiments, SOLVE for everything else, but the reported statistics are the same, and provide the first indication whether the structure is solveable by AutoSol. If Phaser is run, additional heavy-atom sites will be added based on the log-likelihood-gradient map, and phases recalculated using the new sites. Phasing is followed by density modification in RESOLVE; these two steps combined take only a couple of minutes for an average-sized structure. Once solutions are finished with each step, you may view the maps and sites in Coot or PyMOL by clicking one of the buttons below the solution lists:

The map produce by the phasing step will usually be uninterpretable due to poor phase quality, especially in SAD experiments where the phase ambiguity can not be resolved. The images below show the experimental maps (over the entire unit cell) for the p9-sad example; first the map from Phaser, then the density-modified map from RESOLVE, in identical orientations and contour levels. Note that while the protein region is more electron-dense even in the initial SAD map, the boundary between protein and bulk solvent is impossible to determine. However, even in the map from Phaser, the heavy-atom sites should be clearly inside strong density.

The phasing and density modification steps generate several statistics; the full AutoSol manual provides a detailed explanation of these (see the section "Scoring of heavy-atom solutions"). Some approximate guidelines for their interpretation:

The Figure of Merit (FOM) is an estimate of phase quality, ranging from 0 to 1. For SAD, above 0.45 is very good, and between 0.25 and 0.45 is marginal. For MAD, these thresholds increase to 0.45 and 0.65. The Overall Score (also BAYES-CC) is simply a correlation coefficient multiplied by 100. The higher this number is, the better the solution. The R-factor for density modification is calculated by performing an inverse Fourier transform on the density-modified map to obtain $F(\text{calc})$, and comparing this to $F(\text{obs})$. This is on the same scale as a refinement R-factor, so below 0.3 is ideal; however, a low value does not guarantee a usable solution. The "skew" is a measure of the deviation of the distribution of electron density values in the raw phased map from a Gaussian; a macromolecular crystal will have a large "tail" of very high density values. A skew of 0.2 is usually very good.

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- The "skew" is a measure of the deviation of the distribution of electron density values in the raw phased map from a Gaussian; a macromolecular crystal will have a large "tail" of very high density values. A skew of 0.2 is usually very good.

Model-building

The automated model-building step is the most time-consuming stage, but by far the most useful indication of success. Building is done by RESOLVE, and alternated with refinement in phenix.refine. AutoSol will automatically build SeMet residues when appropriate, based on heavy-atom positions. If possible, the user-provided sequence will be docked into the peptide chain. A schematic of the sequence and current model coverage (including secondary structure content) will be added in a new tab once a PDB file is available. If you have the model and map open in Coot or PyMOL, double-clicking on any residue or secondary structure element will zoom to that region.

The final model is rarely complete and may contain gross errors, but a genuine solution is usually unambiguous based on visual inspection, the refinement R-factors, and the correlation coefficient.

If using the full resolution range of the p9-sad example, an R-free below 0.25 is possible; when truncating at 2.5Å, the R-free is nearly 0.35, but this is still obviously correct. An R-free between 0.4 and 0.5 is usually indicative of a solved structure, but may indicate poor resolution and/or low-quality phases. In any case, running the more comprehensive (but slower) AutoBuild wizard often leads to significant improvement. R-free above 0.5 is rarely usable; even if the phases are approximately correct, the map is not interpretable enough to build automatically.

What to do next

Important: if you are going to use the final model from AutoSol in further rounds of building and refinement, it is essential that you use the reflections in the file `exptl_fobs_phases_freeR_flags_X.mtz` (where X is the solution number), which will be listed in the Summary tab. This contains the R-free set used for refinement during building; failing to use these data will bias the R-free in future refinements. It also contains experimental phase information (Hendrickson-Lattman coefficients), which may be useful as additional restraints during refinement.

Assuming that you have a correct solution, there are several immediate options for improving the model. Validation at this stage is usually not worthwhile, as many of the automatic procedures may yield a significant improvement right away. We generally recommend running the AutoBuild wizard, which will usually drop R-free by several percent if not more. This can be started by clicking the button next to the list of output files in the Summary tab; appropriate files will be pre-loaded, including heavy-atom sites. AutoBuild may take several hours to run, but if you have a multi-core system it can be set to run in parallel.

With a very good solution, such as the p9-sad data with the full resolution range, the model from AutoSol may be good enough to jump directly to running phenix.refine. At this point, simulated annealing will probably be helpful; at moderate-to-high resolutions (2.5Å or better), you should also turn on rotamer fitting and solvent

picking.

What to do if AutoSol fails

The most common reason for experimental phasing to fail is poor data quality - this may mean radiation damage, low signal-to-noise ratio, severe incompleteness, very little anomalous signal (sometimes due to lack of high-occupancy heavy-atom sites), or lattice pathologies such as twinning or pseudo-translation. The Xtriage output may be helpful in diagnosing faults in the data.

If the data are of relatively good quality and anomalous signal is present but few or no heavy-atom sites were found, it may be helpful to supply additional input files. Heavy-atom sites identified by another program are one possibility; another is a partial molecular replacement solution, which can be used to bootstrap MR-SAD phasing. In MR-SAD, Phaser uses the log-likelihood-gradient completion algorithm to add anomalous scatterers to the molecular replacement solution. The resulting phases combine information from both sources, MR and SAD. This procedure may be successful in situations where neither MR nor SAD is sufficient by itself. To run MR-SAD, first carry out MR using either the Phaser-MR or MRage GUI; when it is finished, use the molecular replacement solution as a partial model file in the AutoSol GUI.

References

Decision-making in structure solution using Bayesian estimates of map quality: the PHENIX AutoSol wizard. T.C. Terwilliger, P.D. Adams, R.J. Read, A.J. McCoy, N.W. Moriarty, R.W. Grosse-Kunstleve, P.V. Afonine, P.H. Zwart, and L.W. Hung. *Acta Crystallogr D Biol Crystallogr* 65, 582-601 (2009).

- Decision-making in structure solution using Bayesian estimates of map quality: the PHENIX AutoSol wizard. T.C. Terwilliger, P.D. Adams, R.J. Read, A.J. McCoy, N.W. Moriarty, R.W. Grosse-Kunstleve, P.V. Afonine, P.H. Zwart, and L.W. Hung. *Acta Crystallogr D Biol Crystallogr* 65, 582-601 (2009).

Substructure search procedures for macromolecular structures. R.W. Grosse-Kunstleve, and P.D. Adams. *Acta Cryst. D* 59, 1966-1973. (2003).

- Substructure search procedures for macromolecular structures. R.W. Grosse-Kunstleve, and P.D. Adams. *Acta Cryst. D* 59, 1966-1973. (2003).

Phaser crystallographic software. A.J. McCoy, R.W. Grosse-Kunstleve, P.D. Adams, M.D. Winn, L.C. Storoni, and R.J. Read. *J Appl Crystallogr* 40, 658-674 (2007).

- Phaser crystallographic software. A.J. McCoy, R.W. Grosse-Kunstleve, P.D. Adams, M.D. Winn, L.C. Storoni, and R.J. Read. *J Appl Crystallogr* 40, 658-674 (2007).

Maximum-likelihood density modification. T.C. Terwilliger. *Acta Crystallogr D Biol Crystallogr* 56, 965-72 (2000).

- Maximum-likelihood density modification. T.C. Terwilliger. *Acta Crystallogr D Biol Crystallogr* 56, 965-72 (2000).

Hybrid Substructure Search

hyss.html

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Authors

Ralf Grosse-Kunstleve, Paul Adams, Randy Read, Gabor Bunkoczi, Tom Terwilliger

HySS overview

The HySS (Hybrid Substructure Search) submodule of the Phenix package is a highly-automated procedure for the location of anomalous scatterers in macromolecular structures. HySS starts with the automatic detection of the reflection file format and analyses all available datasets in a given reflection file to decide which of these is best suited for solving the structure. The search parameters are automatically adjusted based on the available data and the number of expected sites given by the user. The search method is a systematic multi-trial procedure employing

direct-space Patterson interpretation followed by reciprocal-space Patterson interpretation followed by dual-space direct methods or Phaser LLG completion followed by automatic comparison of the solutions and automatic termination detection.

- direct-space Patterson interpretation followed by
- reciprocal-space Patterson interpretation followed by
- dual-space direct methods or Phaser LLG completion followed by
- automatic comparison of the solutions and
- automatic termination detection.

The end result is a consensus model which is exported in a variety of file formats suitable for frequently used phasing and density modification packages.

The core search procedure is applicable to both anomalous diffraction and isomorphous replacement problems. However, currently the command line interface is limited to work with anomalous diffraction data or externally preprocessed difference data.

A new powerful and parallel Hybrid Substructure Search (HySS)

HySS has many new features since 2014. The first thing you might notice in HySS is that it can use as many processors as you have on your computer. This can make for a really quick direct methods search for your anomalously-scattering substructure.

You might notice next that for SAD data HySS now automatically tries Phaser completion to find a solution if the direct methods approach does not give a clear solution right away. Phaser completion uses the SAD likelihood function to create an LLG map that is used to find additional sites and then scores the solutions with the likelihood function. This is really great because Phaser completion in HySS can be much more powerful than direct methods in HySS. Phaser completion takes a lot longer than direct methods completion but it is now quite feasible, particularly if you have several processors on your computer.

The next thing you might notice in HySS is that it automatically tries several resolution cutoffs for the searches if the first try does not give a convincing solution. Also HySS will start out with a few Patterson seeds and then try more if that doesn't give a clear solution.

HySS now considers a solution convincing if it finds the same solution several times, starting with different initial Patterson peaks as seeds. The more sites in the solution, the fewer duplicates need to be found to have a convincing solution.

Putting all these together, the new HySS is much faster than the old HySS and it can solve substructures that the old HySS could not touch.

Graphical interface

The HySS GUI is listed in the "Experimental phasing" category of the main PHENIX GUI. Most options are shown in the main window, but only the fields highlighted below are mandatory. The data labels will be selected automatically if the reflections file contains anomalous arrays, and any symmetry information present in the file will be loaded in the unit cell and space group fields.

It may be helpful to run Xtriage first to determine an appropriate high resolution cutoff, as most datasets do not have significant anomalous signal in the highest resolution shells. The wavelength is only required if Phaser is being used for rescoring. Additional options are described below in the command-line documentation.

At the end of the run, a tab will be added showing output files and basic statistics. If you are happy with the sites, you can load them into AutoSol or Phaser directly from this window.

A full list on sites is displayed in the "Edit sites" tab. For a typical high-quality selenomethionine dataset, such as the p9-sad tutorial data used here, valid sites should have an occupancy close to 1, but for certain types of heavy-atom soaks (such as bromine) all sites may have partial occupancy. You can edit the sites by changing the occupancy or unchecking any that you wish to discard, then clicking the "Save selected" button.

Command line options

Enter phenix.hyss without arguments to obtain a list of the available command line options:

```
usage: phenix.hyss [options] reflection_file n_sites element_symbol
```

```
Example: phenix.hyss w1.sca 66 Se
```

Trying a search without automatic termination

If the HySS consensus model does not lead to an interpretable electron density map you can try the search=full option:

```
phenix.hyss your_file.sca 100 se search full
```

This disables the automatic termination detection and the run will in general take considerably longer. If the full search leads to a better consensus model please let us know because we will want to improve the automatic termination detection.

Brute-force searches

A second option in cases where HySS does not find a solution is to try is the strategy=brute-force option. The brute-force approach uses Phaser LLG completion and likelihood scoring as in a standard run of HySS with SAD data, except that instead of adding the highest set of sites from the LLG map as a group, all combinations of the top 20 or so sites are taken two at a time to form new trial solutions. These are all then used in LLG completion and scoring as well. The process takes quite a long time and you would normally use it only if you have a multiprocessor machine.

To carry out a very extensive search with brute force, you may want to adjust the parameters n_top_patt (target number of Patterson seeds to consider), n_top_patt_min (minimum number of seeds to consider, normally the search is terminated when the target number of sites is found once n_top_patt_min seeds have been examined), and n_top_llg (number of sites from LLG map to consider for each seed.)

Varying the high-resolution limit

Another possibility is to override the automatic determination of the high-resolution limit with the resolution option. In some cases the resolution limit is very critical. Truncating the high-resolution limit of the data can sometimes lead to a successful search, as more reflections with a weak anomalous signal are excluded.

Trying likelihood-based rescoring with direct methods

You can use likelihood-based rescoring along with direct methods. For this procedure, the keyword rescore=phaser-complete is recommended. This approach is less affected by suboptimal resolution cutoffs and also provides more discrimination with noisy data. Switching on the phaser-map extrapolation protocol is also worthwhile, since it increases success rate and is only a small runtime overhead compared to phaser-based rescoring.

Searches for heavy-atom cluster compounds

You can search for cluster compounds with HySS. Normally you should supply a PDB file with an example of the cluster using the keyword cluster_compound_structure=my_cluster_compound.pdb and scattering_type=XX (yes, really the two letters XX, not XX replaced by some other name). If your cluster is Ta6Br12 then you can simply put scattering_type=TX and skip the PDB file.

Notes

The `site_min_distance`, `site_min_distance_sym_equiv`, and `site_min_cross_distance` options are available to override the default minimum distance of 3.5 Angstroms between substructure sites.

The `real_space_squaring` option can be useful for large structures with high-resolution data. In this case the large number of triplets generated for the reciprocal-space direct methods procedure (i.e. the tangent formula) may lead to excessive memory allocation. By default HySS switches to real-space direct methods (i.e. E-map squaring) if it searches for more than 100 sites. If this limit is too high given the available memory use the `real_space_squaring` option. For substructures with a large number of sites it is in our experience not critical to employ reciprocal-space direct methods.

If the `molecular_weight` and `solvent_content` options are used HySS will help in determining the number of substructures sites in the unit cell, interpreting the number of sites specified on the command line as number of sites per molecule. For example:

```
phenix.hyss gere_MAD.mtz 2 se molecular_weight=8000 solvent_content=0.70
```

This is telling HySS that we have a molecule with a molecular weight of 8 kD, a crystal with an estimated solvent content of 70%, and that we expect to find 2 Se sites per molecule. The HySS output will now show the following:

```
#-----#
| Formula for calculating the number of molecules given a molecular weight. |
|-----|
| n_mol = ((1.0-solvent_content)*v_cell)/(molecular_weight*n_sym*.783) |
#-----#
Number of molecules: 6
Number of sites: 12
Values used in calculation:
  Solvent content: 0.70
  Unit cell volume: 476839
  Molecular weight: 8000.00
  Number of symmetry operators: 4
```

HySS will go on searching for 12 sites.

Auxiliary programs

phenix.emma

EMMA stands for Euclidean Model Matching which allows two sets of coordinates to be superimposed as best as possible given symmetry and origin choices. See the `phenix.emma` documentation for more details.

References

Substructure search procedures for macromolecular structures. R.W. Grosse-Kunstleve, and P.D. Adams. Acta Cryst. D59, 1966-1973. (2003).

- Substructure search procedures for macromolecular structures. R.W. Grosse-Kunstleve, and P.D. Adams. Acta Cryst. D59, 1966-1973. (2003).

Simple algorithm for a maximum-likelihood SAD function. A.J. McCoy, L.C. Storoni, and R.J. Read. Acta Crystallogr D Biol Crystallogr 60, 1220-8 (2004).

- Simple algorithm for a maximum-likelihood SAD function. A.J. McCoy, L.C. Storoni, and R.J. Read. Acta Crystallogr D Biol Crystallogr 60, 1220-8 (2004).

List of all available keywords

data = None Anomalous data (I+ and I- or F+ and F-). Any standard format is fine.data_label = None Optional label specifying which columns of anomalous data to use. Not necessary if your input file has only one set of anomalous data.n_sites = None Number of sites to search for. This number does affect the outcome when direct methods are used because the number of sites kept in each cycle of dual-space recycling is based on it. For LLG-based searches the number is less important as sites with high contributions to the likelihood are kept. For brute-force searches the search is terminated when this number of sites is found.scattering_type = None Name of the anomalously-scattering element. Default is Se .wavelength = None Wavelength for X-ray data collection. The wavelength is required for likelihood-based scoring. If the scattering type is Se then the default wavelength will be the peak for selenium, however otherwise the value must be input.space_group = None Space group (normally read from the data file)unit_cell = None Unit Cell (normally read from the data file)symmetry = None Optional file with crystal symmetry (only needed if data file) does not contain symmetry information and it is not supplied with space_group and unit_cell keywords.resolution = None High-resolution cutoff for input data. If specified, this overrides setting up a series of resolution cutoff values.low_resolution = None Low-resolution limit. Note: if set to None, anomalous differences will be truncated with a low resolution limit of 15 Åsearch = *fast full You can allow automatic termination when the same solution is found several times (search=fast) or a more exhaustive search which is terminated when all starting seeds have been tried (search=full).root = None Root for output file names (default is the name of your data file)output_dir = None Directory where your output files are to be placedstrategy = *Auto simple brute_force Strategy. Setting strategy=Auto and rescore=Auto will run the automatic procedure, varying resolution and scoring method. Setting strategy=Auto and rescore=phaser-complete will run phaser completion at varying resolutions. Setting strategy=simple and rescore=correlation will do a single run of correlation-based scoring at one resolution. Setting strategy=brute_force will use the brute-force completion approach. If you set strategy to simple or brute_force then you will need to set the parameter rescore to specify whether to use correlation- or likelihood-based scoring.rescore = *Auto correlation phaser-refine phaser-complete Rescore function. Correlation_coefficient, Phaser_refinement, and Phaser_substructure_completion are choices of approach for scoring of individual solutions. Phaser_refinement (rescore=phaser-refine) will use likelihood scoring without addition of new sites. Phaser_substructure_completion (rescore=phaser-complete) will use likelihood scoring and add additional sites as well. The automatic procedure (strategy=Auto) for substructure determination will use correlation scoring for the direct methods completion and phaser-complete scoring for likelihood completion. If you specify phaser-refine or phaser-complete or use strategy Auto then you will need to specify the wavelength that was used for data collection.llgc_sigma = 5. Log-likelihood gradient map sigma cutoff. Default value is 5. If your mean anomalous difference is very low you might want to try lowering this parameter to as low as 2.nproc = None Number of processors to use.phaser_resolution = None Resolution for phaser rescoring and extrapolation. If not set, same as overall resolution.phaser_low_resolution = None Low resolution for phaser rescoring and extrapolation. If not set all low-resolution data will be used.rms_cutoff = 3.5 Anomalous differences larger than rms_cutoff times the rms will be ignoredsigma_cutoff = 1 Anomalous differences smaller than sigma_cutoff times sigma will be ignoredverbose = False pdb_only = False Suppress all output except fo...

- data = None Anomalous data (I+ and I- or F+ and F-). Any standard format is fine.
- data_label = None Optional label specifying which columns of anomalous data to use. Not necessary if your input file has only one set of anomalous data.
- n_sites = None Number of sites to search for. This number does affect the outcome when direct methods are used because the number of sites kept in each cycle of dual-space recycling is based on it. For LLG-based searches the number is less important as sites with high contributions to the likelihood are kept. For brute-force searches the search is terminated when this number of sites is found.

- `scattering_type` = None Name of the anomalously-scattering element. Default is Se .
- `wavelength` = None Wavelength for X-ray data collection. The wavelength is required for likelihood-based scoring. If the scattering type is Se then the default wavelength will be the peak for selenium, however otherwise the value must be input.
- `space_group` = None Space group (normally read from the data file)
- `unit_cell` = None Unit Cell (normally read from the data file)
- `symmetry` = None Optional file with crystal symmetry (only needed if data file) does not contain symmetry information and it is not supplied with `space_group` and `unit_cell` keywords.
- `resolution` = None High-resolution cutoff for input data. If specified, this overrides setting up a series of resolution cutoff values.
- `low_resolution` = None Low-resolution limit. Note: if set to None, anomalous differences will be truncated with a low resolution limit of 15 Å
- `search` = *fast full You can allow automatic termination when the same solution is found several times (`search=fast`) or a more exhaustive search which is terminated when all starting seeds have been tried (`search=full`).
- `root` = None Root for output file names (default is the name of your data file)
- `output_dir` = None Directory where your output files are to be placed
- `strategy` = *Auto simple brute_force Strategy. Setting `strategy=Auto` and `rescore=Auto` will run the automatic procedure, varying resolution and scoring method. Setting `strategy=Auto` and `rescore=phaser-complete` will run phaser completion at varying resolutions. Setting `strategy=simple` and `rescore=correlation` will do a single run of correlation-based scoring at one resolution. Setting `strategy=brute_force` will use the brute-force completion approach. If you set `strategy` to simple or `brute_force` then you will need to set the parameter `rescore` to specify whether to use correlation- or likelihood-based scoring.
- `rescore` = *Auto correlation phaser-refine phaser-complete Rescore function. `Correlation_coefficient`, `Phaser_refinement`, and `Phaser_substructure_completion` are choices of approach for scoring of individual solutions. `Phaser_refinement` (`rescore=phaser-refine`) will use likelihood scoring without addition of new sites. `Phaser_substructure_completion` (`rescore=phaser-complete`) will use likelihood scoring and add additional sites as well. The automatic procedure (`strategy=Auto`) for substructure determination will use correlation scoring for the direct methods completion and `phaser-complete` scoring for likelihood completion. If you specify `phaser-refine` or `phaser-complete` or use `strategy Auto` then you will need to specify the wavelength that was used for data collection.
- `llgc_sigma` = 5. Log-likelihood gradient map sigma cutoff. Default value is 5. If your mean anomalous difference is very low you might want to try lowering this parameter to as low as 2.
- `nproc` = None Number of processors to use.
- `phaser_resolution` = None Resolution for phaser rescoring and extrapolation. If not set, same as overall resolution.
- `phaser_low_resolution` = None Low resolution for phaser rescoring and extrapolation. If not set all low-resolution data will be used.
- `rms_cutoff` = 3.5 Anomalous differences larger than `rms_cutoff` times the rms will be ignored
- `sigma_cutoff` = 1 Anomalous differences smaller than `sima_cutoff` times sigma will be ignored
- `verbose` = False
- `pdb_only` = False Suppress all output except for pdb file output

- `cluster_termination = False` Terminate by looking for bimodal distribution in phaser rescoring. Does not apply to correlation rescoring
- `max_view = 10` Number of solutions to show when run in sub-processes
- `minimum_reflections_for_phaser = 300` Minimum reflections to run phaser rescoring
- `cluster_compound_structure = None` PDB file containing the structure of the cluster compound
- `job_title = None` Job title in PHENIX GUI, not used on command line
- `auto_controllowest_high_resolution_to_try = None` `highest_high_resolution_to_try = None` In automatic mode a range of resolutions is tested. This parameter sets the highest high-resolution limit to try.
`try.default_high_resolution_to_try = None` In automatic mode a range of resolutions is tested. This parameter sets the first high-resolution limit to try.
`direct_methods = True` Run direct methods in automatic procedure
`complete_direct_methods = False` Run Phaser completion on direct methods solutions in automatic procedure. With this option you can try to complete direct methods solutions with likelihood-based methods.
`phaser_completion = True` Run Phaser completion on Patterson seeds in automatic procedure
`try_multiple_resolutions = True` In automatic mode several high-resolution cutoffs will be tried.
`try_full_resolution = False` In automatic mode if no solutions are found using other cutoffs, the full resolution of the data is tested.
`max_multiple = 5` First try `default_seeds_to_use` or `seeds_to_use`, then multiples of that many seeds. Does not apply to brute_force search
`max_groups = 5` In automatic mode a set of solutions to be scored is broken up into up to `max_groups`, then each group is split into `nproc` processors and scored. This parameter is used to limit the number of groups.
`seeds_to_use = None` You can set the number of seeds (usually Patterson solutions) to use in automatic mode.
`starting_seed = None` You can specify which seed to start with (skipping prior seeds)
`default_seeds_to_use = 20` Default number of seeds (usually Patterson solutions) to use in automatic mode.
`solutions_to_save = 20` If you are running multiprocessing it may be necessary to make `solutions_to_save` small (e.g. 10 or 20 rather than 1000)
`max_tries = None` Used internally by automated procedures
`write_output_model = True` Write output file
- `lowest_high_resolution_to_try = None`
- `highest_high_resolution_to_try = None` In automatic mode a range of resolutions is tested. This parameter sets the highest high-resolution limit to try.
- `default_high_resolution_to_try = None` In automatic mode a range of resolutions is tested. This parameter sets the first high-resolution limit to try.
- `direct_methods = True` Run direct methods in automatic procedure
- `complete_direct_methods = False` Run Phaser completion on direct methods solutions in automatic procedure. With this option you can try to complete direct methods solutions with likelihood-based methods.
- `phaser_completion = True` Run Phaser completion on Patterson seeds in automatic procedure
- `try_multiple_resolutions = True` In automatic mode several high-resolution cutoffs will be tried.
- `try_full_resolution = False` In automatic mode if no solutions are found using other cutoffs, the full resolution of the data is tested.
- `max_multiple = 5` First try `default_seeds_to_use` or `seeds_to_use`, then multiples of that many seeds. Does not apply to brute_force search
- `max_groups = 5` In automatic mode a set of solutions to be scored is broken up into up to `max_groups`, then each group is split into `nproc` processors and scored. This parameter is used to limit the number of groups.

- `seeds_to_use = None` You can set the number of seeds (usually Patterson solutions) to use in automatic mode.
- `starting_seed = None` You can specify which seed to start with (skipping prior seeds)
- `default_seeds_to_use = 20` Default number of seeds (usually Patterson solutions) to use in automatic mode.
- `solutions_to_save = 20` If you are running multiprocessing it may be necessary to make `solutions_to_save` small (e.g. 10 or 20 rather than 1000)
- `max_tries = None` Used internally by automated procedures
- `write_output_model = True` Write output file
- `brute_force_completion` `n_top_patt = 99` `n_top_patt` top-scoring Patterson solutions will be used as seeds in exhaustive completion. `n_top_patt_min = 10` Do not terminate brute-force completion unless at least `n_top_patt_min` Patterson solutions have been examined. You can set this to a large number (100) to get an exhaustive search. `n_top_llg = 20` `n_top_llg` LLG map peaks will be used at a time in exhaustive completion. You can set this to a larger number such as 30 to have more tries. The number of tries per seed is about half the square of `n_top_llg` (`n_top_llg=20` requires 190 tries). `min_sigma_in_brute_force = 1`. In brute force completion the top `n_top_llg` map peaks are used as potential new sites, provided their heights are at least `min_sigma_in_brute_force` times the rms of the LLG map. `n_top_iter = 5` `n_top_iter` top groups will be carried forward in exhaustive completion. `n_llg_add_at_once = 2` All possible groups of `n_llg_add_at_once` of the `n_top_llg` LLG peaks will be added to each seed in each round. As the addition is exhaustive, `n_llg_add_at_once` more than 3 may take a really long time. `n_iter_max = 3` Number of iterations of exhaustive completion. `include_base_scatterers = True` Include fragments used to run LLG map in new fragments. `score_additions_with_completion = True` Score compound fragments (patterson + additional sites) using phaser-completion (`llgc_sigma` not set to 1000). `select_matching = None` Select solutions matching at least `n_select_matching` or `n_select_matching_patt` sites in comparison model, or at least `fraction_select_matching`. Not normally used. Choose `n_llg_add_at_once+2` (patterson+add). `fraction_select_matching = 0.67` Fraction of sites required to match with `select_matching`. `n_select_matching = 3` Number of sites required to match with `select_matching`. `n_select_matching_patt = 1` Number of Patterson sites required to match with `select_matching`. `comparison_tolerance = 0` Print solution matching comparison if within N of correct. `n_print = 20` Number of solutions to list
- `n_top_patt = 99` `n_top_patt` top-scoring Patterson solutions will be used as seeds in exhaustive completion.
- `n_top_patt_min = 10` Do not terminate brute-force completion unless at least `n_top_patt_min` Patterson solutions have been examined. You can set this to a large number (100) to get an exhaustive search.
- `n_top_llg = 20` `n_top_llg` LLG map peaks will be used at a time in exhaustive completion. You can set this to a larger number such as 30 to have more tries. The number of tries per seed is about half the square of `n_top_llg` (`n_top_llg=20` requires 190 tries).
- `min_sigma_in_brute_force = 1`. In brute force completion the top `n_top_llg` map peaks are used as potential new sites, provided their heights are at least `min_sigma_in_brute_force` times the rms of the LLG map.
- `n_top_iter = 5` `n_top_iter` top groups will be carried forward in exhaustive completion
- `n_llg_add_at_once = 2` All possible groups of `n_llg_add_at_once` of the `n_top_llg` LLG peaks will be added to each seed in each round. As the addition is exhaustive, `n_llg_add_at_once` more than 3 may take a really long time.

- `n_iter_max = 3` Number of iterations of exhaustive completion
- `include_base_scatterers = True` Include fragments used to run LLG map in new fragments
- `score_additions_with_completion = True` Score compound fragments (patterson + additional sites) using phaser-completion (llgc_sigma not set to 1000).
- `select_matching = None` Select solutions matching at least `n_select_matching` or `n_select_matching_patt` sites in comparison model, or at least `fraction_select_matching`. Not normally used. Choose `n_llg_add_at_once+2` (patterson+add)
- `fraction_select_matching = 0.67` Fraction of sites required to match with `select_matching`
- `n_select_matching = 3` Number of sites required to match with `select_matching`
- `n_select_matching_patt = 1` Number of Patterson sites required to match with `select_matching`
- `comparison_tolerance = 0` Print solution matching comparison if within N of correct
- `n_print = 20` Number of solutions to list
- `terminationmatches_must_not_share_patterson_vector = True` If search=fast then if the same solution is found several times the search will be terminated. If this flag is set (default) then the solution must be found several times starting with different Patterson vectors.
- `n_top_models_to_compare = None` Number of top models that must share sites to terminate (set automatically).
- `matches_must_not_share_patterson_vector = True` If search=fast then if the same solution is found several times the search will be terminated. If this flag is set (default) then the solution must be found several times starting with different Patterson vectors.
- `n_top_models_to_compare = None` Number of top models that must share sites to terminate (set automatically).
- `parameter_estimationsolvent_content = 0.55` Solvent content (default: 0.55). Only used to estimate number of sites if not given
- `molecular_weight = None` Molecular weight in asymmetric unit. Only used to estimate number of sites if not given
- `solvent_content = 0.55` Solvent content (default: 0.55). Only used to estimate number of sites if not given
- `molecular_weight = None` Molecular weight in asymmetric unit. Only used to estimate number of sites if not given
- `search_controlrandom_seed = None` Seed for random number generators
- `site_min_distance = 3.5` Minimum distance between substructure sites (default: 3.5)
- `site_min_distance_sym_equiv = None` Minimum distance between symmetrically-equivalent substructure sites (overrides `--site_min_distance`)
- `site_min_cross_distance = None` Minimum distance between substructure sites not related by symmetry (overrides `--site_min_distance`)
- `skip_consensus = False` Skip calculation of consensus structure
- `random_seed = None` Seed for random number generator
- `site_min_distance = 3.5` Minimum distance between substructure sites (default: 3.5)
- `site_min_distance_sym_equiv = None` Minimum distance between symmetrically-equivalent substructure sites (overrides `--site_min_distance`)
- `site_min_cross_distance = None` Minimum distance between substructure sites not related by symmetry (overrides `--site_min_distance`)
- `skip_consensus = False` Skip calculation of consensus structure

- `direct_methodsreal_space_squaring = False` Use real space squaring (as opposed to the tangent formula) `extrapolation = *fast_nv1995 phaser-map` Extrapolation function for direct methods. You can use a standard map (`fast_nv1995`) or an LLG map (`phaser-map`)
- `real_space_squaring = False` Use real space squaring (as opposed to the tangent formula)
- `extrapolation = *fast_nv1995 phaser-map` Extrapolation function for direct methods. You can use a standard map (`fast_nv1995`) or an LLG map (`phaser-map`)
- `fragment_searchminimum_fragments_to_consider = 0` Minimum fragments per Patterson vector. This is the number of trial solutions to the Patterson for each Patterson cross-vector. `n_patterson_vectors = None` Number of Patterson vectors to consider. The number of trial solutions will be approximately `n_patterson_vectors` times `minimum_fragments_to_consider`. `input_emma_model_list = None` Input model or models to be used as starting points. Each MODEL group in a file is treated separately. `input_add_model = None` Atoms to add to `input_model` `keep_seed_sites = False` Keep seed sites (never omit them) `n_shake = 0` Number of random variants of each fragment `rms_shake = 1`. RMS for random variants `score_initial_solutions = False` Score 2-site Patterson solutions and sort them. This can be used if you want to take all the 2-site solutions, complete them all with likelihood-based completion, and write out the top one or all of them (with `dump_all_fragments`). You can also use it to take all the 2-site solutions and simply refine them. You can use this with `dump_all_fragments`. `dump_all_fragments = False` Dump fragments to `fragment_x.pdb`. Use with `score_initial_solutions`. `dump_all_models = False` Dump models to `dump_xxx_yy.pdb` `xxx=patterson yy=trans`. Differs from `dump_all_fragments` in that models are dumped throughout the substructure completion process. `score_only = False` Just score input models. Used in automated mode internally.
- `minimum_fragments_to_consider = 0` Minimum fragments per Patterson vector. This is the number of trial solutions to the Patterson for each Patterson cross-vector.
- `n_patterson_vectors = None` Number of Patterson vectors to consider. The number of trial solutions will be approximately `n_patterson_vectors` times `minimum_fragments_to_consider`.
- `input_emma_model_list = None` Input model or models to be used as starting points. Each MODEL group in a file is treated separately.
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- `dump_all_fragments = False` Dump fragments to `fragment_x.pdb`. Use with `score_initial_solutions`.
- `dump_all_models = False` Dump models to `dump_xxx_yy.pdb` `xxx=patterson yy=trans`. Differs from `dump_all_fragments` in that models are dumped throughout the substructure completion process.
- `score_only = False` Just score input models. Used in automated mode internally.
- `comparison_filescomparison_emma_model = None` Comparison model or models to be compared with all solutions found `comparison_emma_model_tolerance = 1.5` Comparison model tolerance
- `comparison_emma_model = None` Comparison model or models to be compared with all solutions found

- `comparison_emma_model_tolerance = 1.5` Comparison model tolerance
- `chunkn = 1` Group inputs into chunks of size `n`. This is part of the procedure for multiprocessing
`i = 0` i'th input solution in each chunk will be used. `i` goes from 0 to `n-1`. This is part of the procedure for multiprocessing
- `n = 1` Group inputs into chunks of size `n`. This is part of the procedure for multiprocessing
- `i = 0` i'th input solution in each chunk will be used. `i` goes from 0 to `n-1`. This is part of the procedure for multiprocessing